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power transfer prototype
for electric vehicles, with square-shaped coils*

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بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ
" قَالُوا سُبْحَانَكَ لَا عِلْمَ لَنَا إِلَّا مَا عَلَّمْتَنَا إِنَّكَ أَنْتَ الْعَلِيمُ الْحَكِيمُ "

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Abstract

In this project, a practical prototype of a resonant inductive wireless charging system for electric vehicles is implemented. The system delivers a DC output power of 823-watt when the coils are aligned with a 15 cm air gap between the 45 x 45 cm square transmitter coil and the 45 x 45 cm square receiver coil. The system operates at a resonant frequency of 83.5 kHz which is in accordance with the SAE J2954 international standard, achieving an efficiency of 77.5%.

The system is simulated using ANSYS Maxwell 3D and ANSYS Simpler software to study the magnetic-field distribution for different coil geometries such as circular and square coils at various misalignment positions as well as the effect of ferrite magnets, and to determine the system parameters.

The simulation results are compared with practical values and verified using mathematical equations and MATLAB/Simulink software to ensure the accuracy of the simulation.

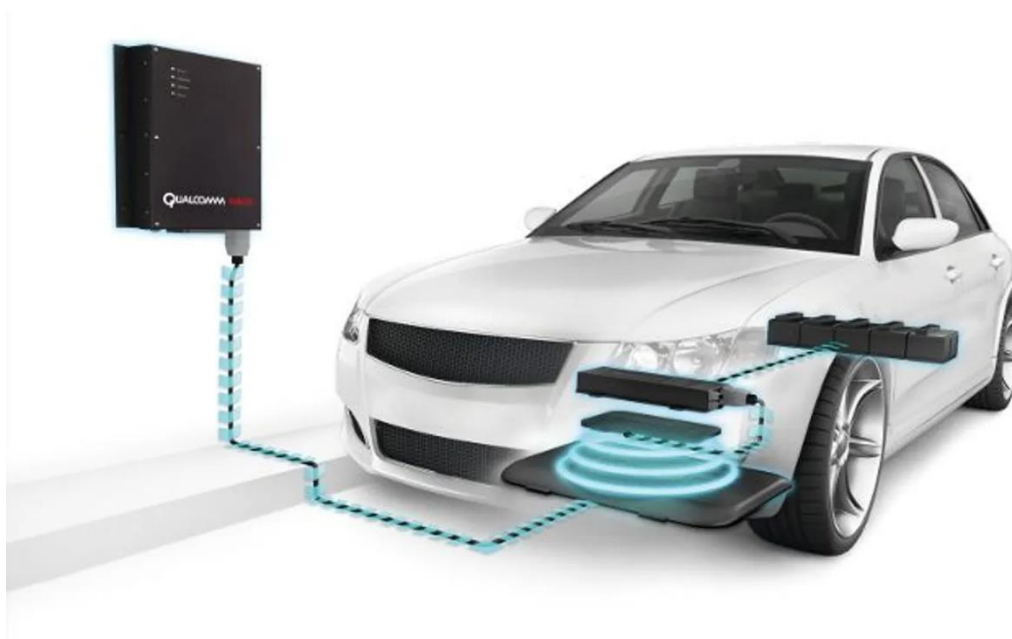


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List of Abbreviations

AC	Alternating Current
BMS	Battery Management System
CC	Constant Current
CF	Crest Factor
CL	Capacitor Inductor
CV	Constant Voltage
CWPT	Capacitive Wireless Power Transfer
DC	Direct Current
DD	Double-D
EVs	Electric Vehicles
FEA	Finite Element Analysis
FEMs	Finite Element Models
GCR	Ground Clearance Range
GSP	Ground Side Pad
HF	High Frequency
HFI	High Frequency Inverter
HV	High Voltage
IWPT	Inductive Wireless Power Transfer
KVL	Kirchhoff's Voltage Law
LC	Inductor Capacitor
LCC	Inductor Capacitor Capacitor
MEPT	Maximum Efficiency and Power Tracking
MGWPT	Magnetic Gear Wireless Power Transfer
MPP	Maximum Power Point
NPPs	Non-Polarized Pads
PF	Power Factor
PMs	Permanent Magnets
PP	Parallel-Parallel
PPs	Polarized pads
PS	Parallel-Series
QD	Quad-D
RIWPT	Resonant Inductive Wireless Power Transfer
SAE	Society of Automotive Engineering

SP	Series-Parallel
SS	Series-Series
WPT	Wireless Power Transfer

List of Acronyms

C_F	The filtering capacitance before the load
C_r	The capacitance of the receiving circuit
C_t	The capacitance of the transmitting circuit
D_i	The inner diameter of each turn
D_o	The outer diameter of each turn
D_s	The diameter of each strand
f	The resonant frequency
H	A constant correlating to the number of strands, as specified in the datasheet of Litz wire
h	The gap distance between the two coils
I_{in}	The input RMS supply current
I_L	The output load DC current
I_r	The RMS receiving current
I_t	The RMS transmitting current
K	The coupling coefficient
L_r	The self-inductance of the receiving coil
L_t	The self-inductance of the transmitting coil
l_{Litz}	The length of the Litz wire
l_r	The outer length of the receiving coil
l_{r_i}	The inner length of the receiving coil
l_t	The outer length of the transmitting coil
l_{t_i}	The inner length of the transmitting coil
M	The mutual inductance
N	The number of strands
P_{AC}	The RMS power delivered to the equivalent AC resistive load
P_{bat}	The battery power
P_{in}	The input active power
P_L	The output load DC power
Q_{in}	The input reactive power
R_{AC}	The AC resistance of the Litz wire which represents both the skin and proximity effects
R_{DC_l}	The DC resistance of the Litz wire per length
R_L	The DC resistive load
R_{Leq}	The equivalent AC resistive load

R_r	The DC resistance of the receiving coil
R_t	The DC resistance of the transmitting coil
r_r	The half of side length of the receiving coil
r_t	The half of side length of the transmitting coil
S_i	Surface copper area of each turn, without the insulation
S_{in}	The input apparent power
S_o	Surface area of each turn, considering the insulation
T_r	No. of turns of the receiving coil
T_t	No. of turns of the transmitting coil
u	The square side length of each coil
V_1	The output HF AC voltage
V_2	The AC voltage across the equivalent AC resistive load
V_{1_o}	The output HF AC voltage at resonance
V_{2_o}	The AC voltage across the equivalent AC resistive load at resonance
V_{bat}	Battery Voltage
V_{DC}	DC output voltage from the DC supply (Input DC voltage to HFI)
V_{in}	The input RMS supply voltage
V_L	The output load DC voltage
W	The resistance ratio of a single strand when isolated
w_r	The outer width of the receiving coil
w_{r_i}	The inner width of the receiving coil
w_t	The outer width of the transmitting coil
w_{t_i}	The inner width of the transmitting coil
η_{s-s}	The efficiency while using SS compensation
μ_o	The permeability of the vacuum
ρ_{cu}	The resistivity of the copper
ω	The angular frequency
ω_o	The resonant angular frequency

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1 Chapter (1): Project Introduction

1.1 Introduction

With the world's increasing interest in finding solutions to environmental issues and climate change, the need has emerged to replace traditional vehicles that run by fossil fuels with clean EVs that have been recently developed and spread widely to reduce environmental pollution[\[1,2,3\]](#). Recently, electrification has been considered as a significant revolution within the transportation sector, aiming to improve efficiency, promote environmental sustainability and enhance safety [\[4\]](#). The adoption of EVs contributes to decreasing greenhouse gas emissions, reducing dependence on fossil fuels, improving driving satisfaction and lower operational costs [\[4\]](#). The challenges facing the widespread adoption of EVs include concerns about charging infrastructure availability, range limits and long duration of the recharging [\[4\]](#).

WPT offers numerous advantages, including automatic operation without the need for driver intervention, maintenance-free operation, enhanced safety and convenience in challenging conditions such as snow, rain, and dust [\[4\]](#).

EV charging is classified into onboard and offboard charging. In onboard charging, the charging and conversion system is inside the vehicle and the charging speed is slow because it is limited by the size of the vehicle, while in offboard charging, the charging and conversion system is outside the vehicle, usually at charging stations and is connected directly to the vehicle battery and the charging speed is faster. The charging systems developed from slow charging (level 1) which is used in homes to fast charging (level 2) which is used in public places and then to faster DC charging (level 3) [\[2\]](#).

EV charging systems have developed to become wireless, which relies on the transfer of energy through a magnetic field between transmitter and receiver as it does not require cables like traditional charging. The WPT process is autonomous, which has made it attractive to charge self-driving EVs as the vehicle can charge itself automatically without connecting the charging cable. It can also be developed to charge vehicles while moving wirelessly, which helps increase the range that the vehicle can travel before the battery runs out, thus saving time and making the charging process more convenient [\[1,2,5\]](#).

1.2 Methods of WPT

I) CWPT: This type is characterized by its low cost, as it does not require a ferrite magnet, and it also features simple power transfer [2]. In this type, capacitors are used to transfer power, as shown in Figure 1-1. An AC voltage is connected to an H-bridge inverter, which in turn produces a high-frequency alternating voltage. Then, some inductors are connected in series with the capacitors to reduce the impedance on both sides of the transmitter and receiver. This type is used in low-power applications such as phone chargers and rotating machinery [6].

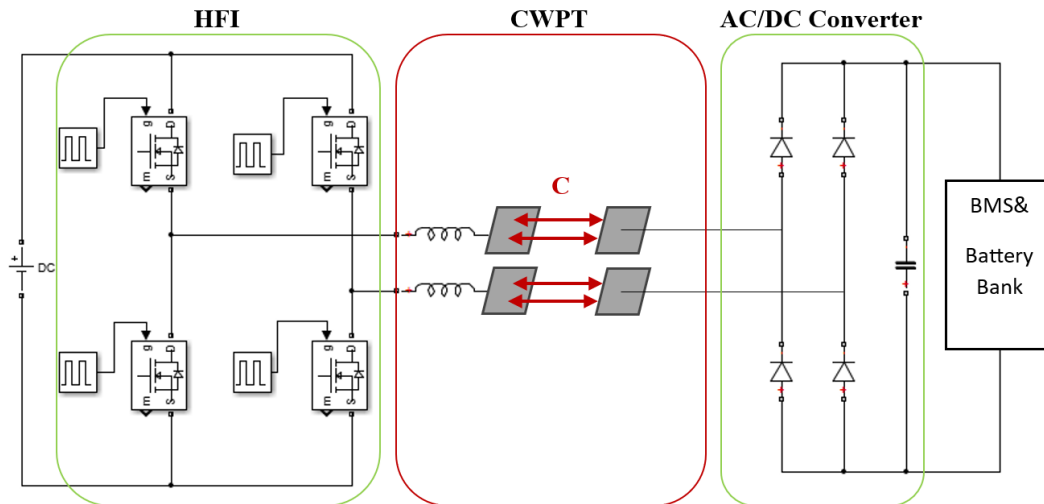


Figure 1-1: CWPT

II) MGWPT: In this type, two synchronous PMs are placed next to each other. An Alternating Current (AC) voltage is connected to the transmitter winding, which generates a mechanical torque on the transmitter magnet, causing it to rotate. This causes the receiver magnet to rotate. It is clear that the transmitter magnet acts as a generator, while the receiver magnet receives energy from the transmitter magnet. Through a Battery Management System (BMS), this energy is delivered to the battery [6], this process is shown in Figure 1-2.

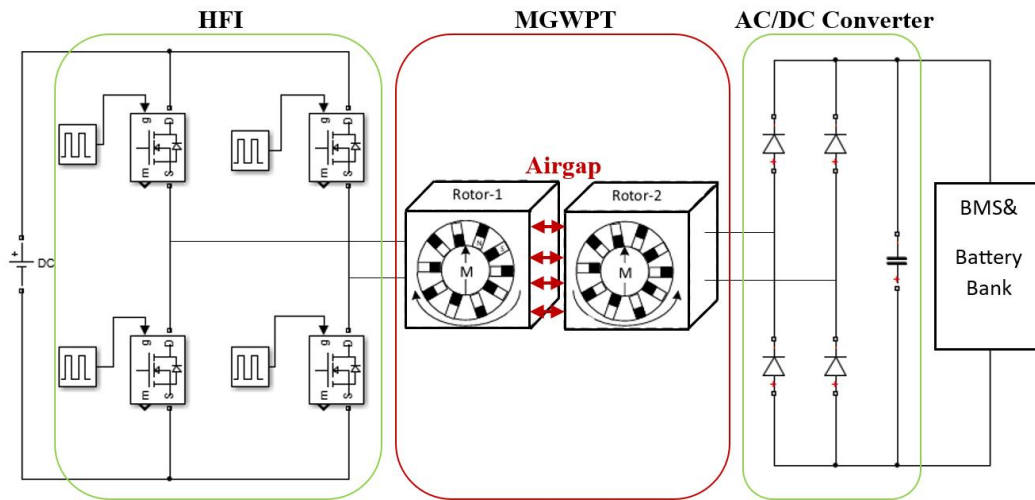


Figure 1-2: MGWPT

III) IWPT: In this type, A HF AC voltage is connected to the transmitter coil, generating a field that transfers energy from the transmitter coil to the receiver coil. The transferred energy is used to charge the battery [2], as shown in Figure 1-3.

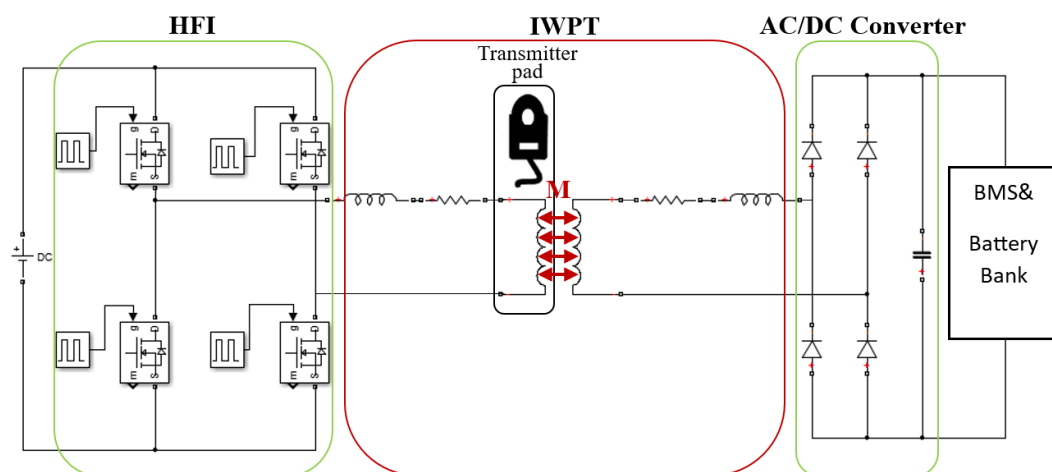


Figure 1-3: IWPT

IV) RIWPT: This type is similar to IWPT, but it is more advanced. It uses magnetically resonant coupled coils to transfer energy. A HF AC voltage is connected to the transmitter coil, generating a magnetic field that transfers energy. The receiver coil receives this energy, which is then used to charge the battery. The difference between this type and IWPT is that this type includes compensation circuits responsible for creating resonance, reducing losses and improving power factors [6], this process is shown in Figure 1-4. We will use this type in our project.

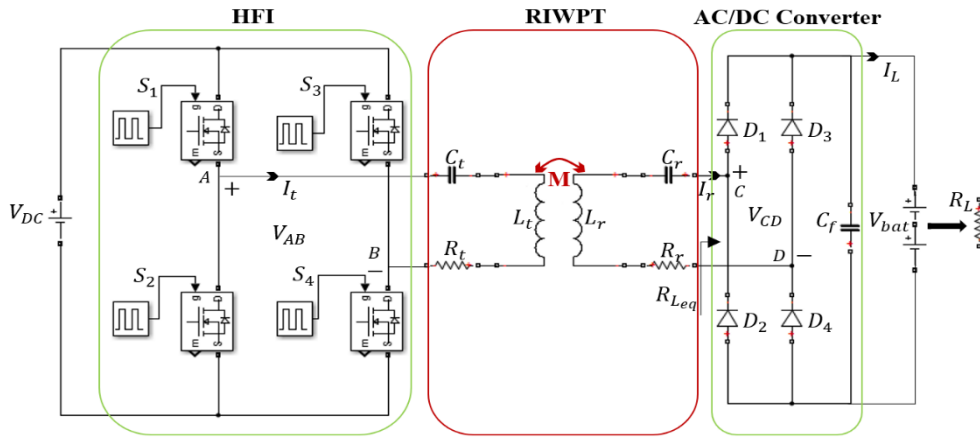


Figure 1-4: RIWPT

The schematic block diagram of WPT system for EVs is shown in Figure 1-5. The AC voltage from the grid is converted to DC voltage through AC/DC converter and then to high frequency AC voltage through DC/AC inverter. To improve efficiency, Transmitting and receiving compensation capacitors are added. The power is transferred through magnetic field from the transmitter to the receiver both with ferrite magnets to improve flux distribution and reduce leakage fluxes. The transmitted power is converted to DC for charging the battery through AC/DC converter. The transferred power decreases in case of misalignment between the transmitter and the receiver [1,6].

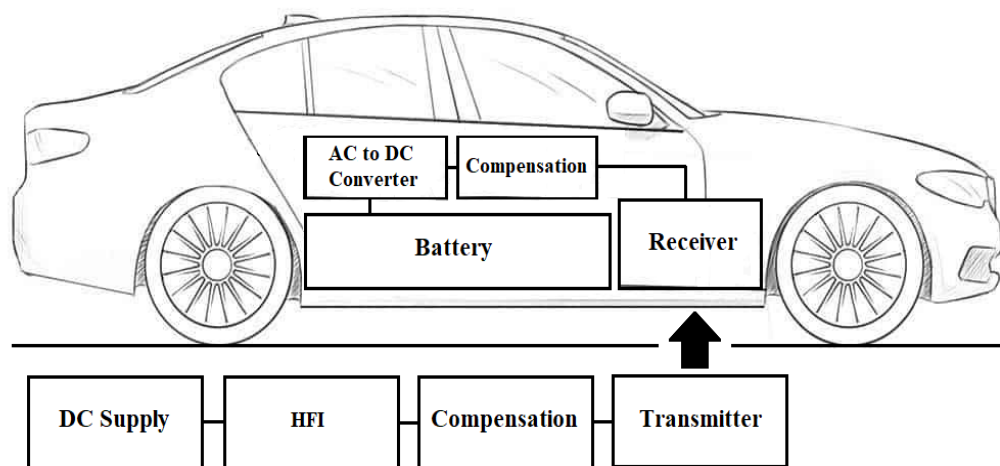


Figure 1-5: Schematic block diagram of WPT system for EVs

1.3 Compensation circuit

A major problem is efficiency in WPT, as cable charging is more efficient than WPT. The transmitter and receiver coils in WPT have high leakage inductance, requiring a large transmitter's current, which leads to transmission losses and reduced efficiency. To solve this problem, compensation capacitors are used [\[1\]](#).

Advantages of the compensation circuit include its ability to operate in CV or CC modes, and it also improves the coupling coefficient (**K**) and quality factor (**Q**) [\[7\]](#). The main disadvantages are the high current and voltage values associated with the resonance components [\[5\]](#).

To implement a compensation circuit, capacitors are added next to the transmitter and the receiver coils in specific configurations. There are four main types:

SS: A series capacitor is added to both transmitter and receiver coils.

SP: A series capacitor is added to the transmitter coil, and a parallel capacitor is added to the receiver coil.

PS: A parallel capacitor is added to the transmitter coil, and a series capacitor is added to the receiver coil.

PP: A parallel capacitor is added to the transmitter and receiver coils [\[2\]](#).

Figure 1-6. illustrates the compensation topologies. A capacitor is placed next to the transmitter coil to reduce reactive power, and another capacitor is placed next to the receiver coil to improve power transfer efficiency [\[6\]](#).

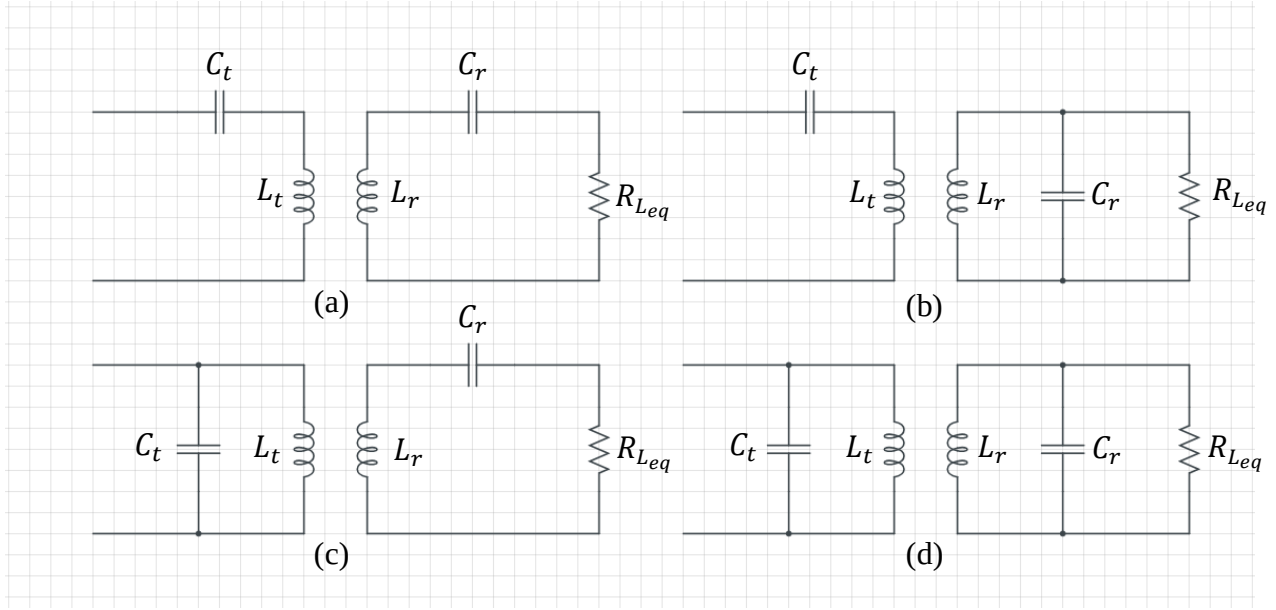


Figure 1-6: Compensation topology

a. SS b. SP c. PS d. PP

Each type of compensation circuit has its advantages and disadvantages: PP and PS are safe because the transmitter coil does not operate without the receiver coil. However, these types suffer from reduced efficiency with misalignment. SP is highly beneficial because the transmitter capacitor value does not depend on the mutual inductance but on the load [6]. SS is one of the most common types and is ideal for EV charging because it has an improved power transfer, the transmitter capacitor value does not depend on the mutual inductance, and the resonant frequency does not depend on the load or the mutual coupling [2]. Therefore, we will use this type in our project.

Each compensation circuit topology differs based on its specific characteristics, including advantages, limitations, and parameters of compensation. Table 1-1 [2,8] and table 1-2 [5,6] illustrate these differences.

Table 1-1: Specific characteristics of compensation circuit topology

Features	SS	SP	PS	PP
Suitability for EV application	High	High	Moderate	Moderate
Frequency tolerance on efficiency	Low	High	Low	High
Power transfer capability	High	High	Low	Low
The sensitivity of the power factor over a distance	Low	Low	Moderate	Moderate
Alignment tolerance	High	High	Moderate	Low
The impedance at resonant state	Low	Low	High	High

Table 1-2: Parameters of compensation

Features	SS	SP	PS	PP
Transmitter capacitor	$\frac{1}{\omega_0^2 L_t}$	$\frac{1}{\omega_0^2 (L_t - \frac{M^2}{L_r})}$	$\frac{1}{\omega_0^2 (L_t - \frac{\omega_0^2 \cdot M^4}{L_t \cdot R_L})}$	$\frac{1}{\omega_0^2 \left(\left(L_t - \frac{M^2}{L_r} \right) + \frac{\frac{M^4}{L_r^4} \cdot R_L^2}{\omega_0^2 \cdot \left(L_t - \frac{M^2}{L_r} \right)} \right)}$
Receiver capacitor	$\frac{1}{\omega_0^2 L_r}$	$\frac{1}{\omega_0^2 L_r}$	$\frac{1}{\omega_0^2 L_r}$	$\frac{1}{\omega_0^2 L_r}$
Load	$\frac{\omega_0 L_r}{Q_r}$	$\omega_0 L_r Q_r$	$\frac{\omega_0 L_r}{Q_r}$	$\omega_0 L_r Q_r$

1.4 Coil design

There are two types of coils based on main parts such as coil and protection layers:

NPPs: consist of only one coil transmits and receives the flux in the vertical direction only like traditional forms (circular, square, rectangular and hexagonal). Circular coil is the most common shape because the eddy current is as little as possible due to absence of any sharp edges. We used the square coil because it allows horizontal misalignment compared to the circular, reduces leakage flux and has higher power transfer capability. One of the disadvantages of the square coil is that it has sharp edges which lead to eddy current that increases the impedance, so it is not suitable in high power applications [\[6\]](#).

PPs: Consist of several coils, transmit and receive flux in the vertical and the horizontal directions, this is the least common type because it has some disadvantages such as complex circuit design, slow switching speed and high manufacturing costs. There are several shapes of it (Double D – Bi-polar - Double D Quadrature - Quad D Quadrature) [\[6\]](#), table 1-3 illustrates these shapes.

Table 1-3: Several shapes of PPs

Shapes	Features
Double-D	It consists of two square or rectangular coils and generates flux in one direction only so the leakage flux at the edges decreases [6].
Bi-Polar	It consists of several coils of the same size but the coupling coefficient decreases if there is misalignment [6].
Double-D quadrature	It consists of more than one circular or square coil with a high coupling coefficient, its efficiency is very high, and it is very suitable in the receiver side because it attracts the magnetic flux in horizontal and vertical directions [6].
Quad-D Quadrature	It improves power transmission efficiency as it reduces energy loss during transmission. Its design reduces the effect of misalignment due to its multiple polarizations. It provides a wider charging area without the need for precise alignment.

Coil shapes are illustrated in Figure1-7.

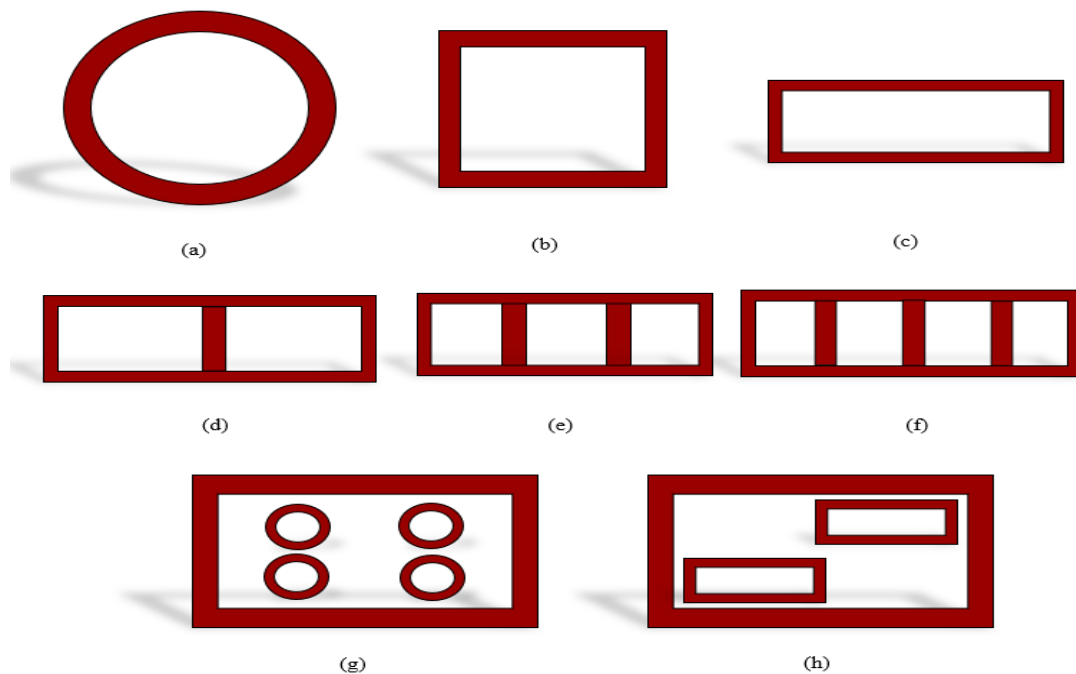


Figure 1-7: Coil shapes

- (a) Circular (b) Square (c) Rectangular (d) Double D
 (e) Bi-Polar (f) DD Quadrature (g) QD Quadrature (h) Extended DD

1.5 Ferrite magnets

Ferrite magnets are used in transmitting and receiving coils in WPT systems to improve flux distribution. The high permeability of the ferrite magnets minimizes the losses at high ranges of frequencies. The low electrical conductivity of the ferrite magnets aids in minimizing eddy currents [4].

When choosing ferrite magnets, several factors are taken into consideration including shape, size, frequency and cost. Simple ferrite shapes such as circular, square, rectangular and striated blocks have been used in WPT systems for EVs.

Although other ferrite shapes such as U-core, Double U, T-core and E-core give higher coupling coefficient, they are not suitable because of their height as EVs have limited height from the ground, as shown in Figure 1-8 [6].

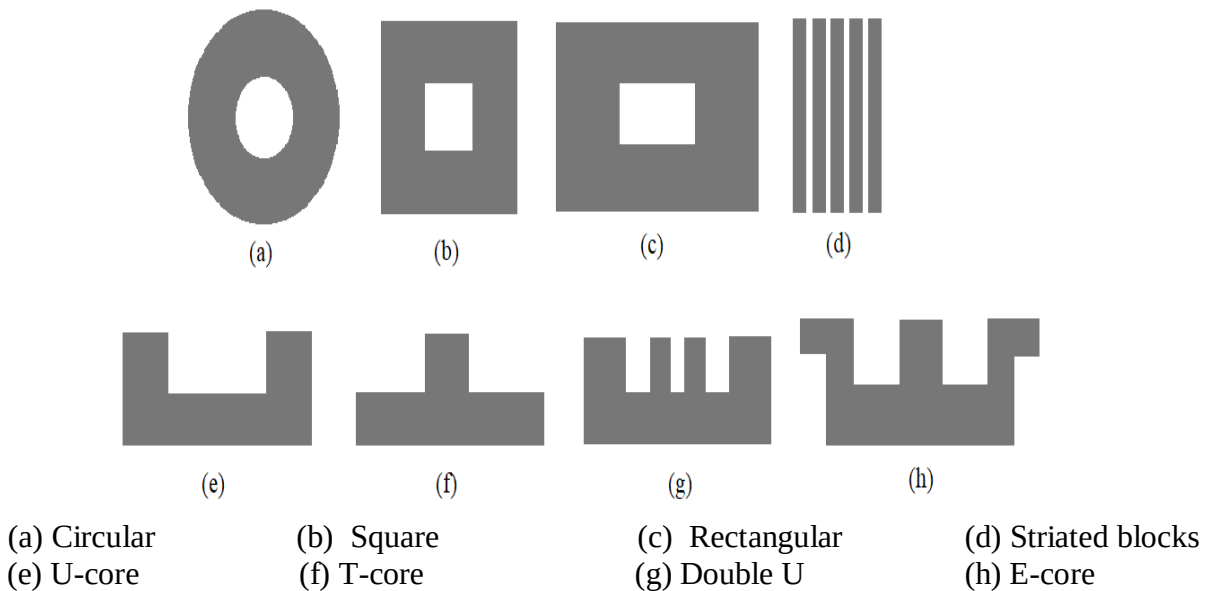


Figure 1-8: Ferrite shapes

Table 1-4 shows the characteristics of Mn-Zn large size ferrite for high power (PC40) used in our prototype.

Table 1-4: The characteristics of Mn-Zn large size ferrite for high power (PC40) used in our prototype

Ferrite specifications	
Initial permeability (μ_i)	2300
Curie temperature	>200 °C
Core loss	At 100 KHz $P_{cv}=420 \text{ kW/m}^3$
Electrical resistivity	6.5 Ωm
Saturation magnetic flux density	At T=23°C $B_s=500 \text{ mT}$ At T=100°C $B_s=380 \text{ mT}$
Remanent flux density	At T=23°C $B_r=125 \text{ mT}$

A thin aluminum layer is placed beneath the ferrites at a small separation distance. This plate acts as a passive shielding, designed to reduce the leakage electromagnetic fields [\[4\]](#).

1.6 Contribution

We conducted a detailed study on the issue of misalignment by examining 52 different misalignment positions of the secondary coil to understand their effect on the coupling coefficient and mutual inductance, and consequently, on system efficiency. The misalignment cases investigated can be categorized into the following types:

1. With uniform air gap:

- Air gap
- Horizontal misalignment
- Rotation around the Z-axis

2. With non-uniform air gap:

- Rotation around the X-axis or Y-axis

3. With multiple misalignment (hybrid):

- Horizontal misalignment with rotation around the Z-axis
 - Horizontal misalignment in (-x, -y) direction with rotation around the Z-axis
 - Horizontal misalignment in (-x, -y) direction with a non-uniform air gap along the Z-axis and rotation around the Y-axis
 - Horizontal misalignment in (-x, -y) direction, combined with a non-uniform air gap along the Z-axis and rotation around both the Y-axis and Z-axis
-

We introduce by MATLAB software about ten script files which are related to another ten script function files, in order to compare between the calculated values of many parameters such as the self-inductances, the mutual inductance, the coupling coefficient, ...etc. obtained by MATLAB command windows, with the values obtained by ANSYS Maxwell 3D or ANSYS Simpler or MATLAB Simulink, also with the practical values.

We concluded a relation Eq. (22) to obtain our equivalent AC resistive load. This is by equaling two equations of the transmitting RMS current with each other. The first equation is Eq. (18), and the second equation is Eq. (21) which is from Eq. (20) of the RMS power. At Eq. (22) all parameters are included, where P_{AC} is assumed, except R_{Leq} .

2 Chapter (2): Literature Survey

A prototype using a rectangular-shaped coil was created with a SS compensation circuit at a frequency of 85 kHz. They then carried out a comparison of different misalignment cases using ANSYS Maxwell 3D software. One of the most significant challenges they encountered was the misalignment issues [\[1\]](#).

This paper designs a WPT level 3/Z2-class with a circular transmitting with both circular and DD receiver. The efficiency in the case of circular receiver is 95.1% but it is reduced to 87.1% in maximum misalignment case. While for a DD receiver the efficiency is 91.6% and it is decreased to 89.5% which is higher compared to the circular receiver in maximum misalignment case [\[4\]](#).

A comparison was made between SS and SP compensation topologies. The effect of mutual inductance, load and operating resonant frequency on them was studied to determine the suitable topology for WPT systems for EVs. Studies have shown that SS topology is better in the case of lower loads, while SP is preferred in the case of higher loads. Also, a 500-watt WPT system was designed using ANSYS Maxwell simulation to study SS and SP compensation topologies. It was noted that at higher gaps between the transmitter and the receiver, the efficiency of SP was better than of SS. However, the SS does not depend on the mutual inductance or coupling coefficient, therefore the SS compensation topology is suitable for WPT systems for EVs [\[7,9\]](#).

A prototype using a circular coil was created with SS compensation circuit and discovered a method to improve the charging performance, thereby maximizing the charging efficiency [\[8\]](#).

A wireless charger for EVs was developed based on a high-frequency power converter to charge a lithium-ion battery at an air gap of 10 cm, achieving an efficiency of 85%. They used transmitter and receiver coils in a circular shape with 10 turns, and the distance between each turn was 1 mm. They also chose a compensation circuit of the SS type and used a ferrite magnet board to increase both the self and mutual inductances. A DC voltage was used as the main power source, which was then converted to AC at a frequency of 80 kHz using a single-phase inverter, providing 3.2 kW power on a 4-ohm load [10].

It discussed one of the main challenges in WPT, which is the misalignment issue. To address this, they created 8 scenarios for misalignment variations for a 1 kW prototype to solve the misalignment issues at the MPP. In each scenario, the output power value varied according to the specific misalignment pattern of that scenario. The coil used in all scenarios was of the circular shaped coil, with SS-type compensation [11].

A 1,000-watt prototype was created at a frequency of 85 kHz with an air gap of 15 cm [12].

A MATLAB code was developed to calculate the parameters. Using these parameters, they performed a simulation at ANSYS Maxwell 3D, where they designed a rectangular coil with a SS compensation circuit at a frequency of 17,702 Hz with an air gap of 20 cm. The output power was 15 kW with an efficiency of 75.38%. The dimensions were 100×60 cm and 60×60 cm respectively. They also studied different misalignment cases, and the efficiency varied depending on the type of misalignment used in the simulation. They analyzed the flux density distribution using ANSYS Maxwell 3D and observed that the magnetic field exposure for the human

body remained within international safety standards. When comparing MATLAB and ANSYS Maxwell 3D, it was found that the efficiency value in MATLAB differed from that in ANSYS Maxwell 3D, because MATLAB doesn't account for losses [13].

A 300-W solar energy powered SS compensated WPT system that requires fewer passive components is built over a transfer distance of 10 cm with an 88.5 % system efficiency. The proposed design has a ferrite-less structure and efficient converters. The performance of the system is evaluated under conditions of z-axes, horizontal and diagonal misalignments. A 10 cm horizontal misalignment results in a 4.34 % reduction in the output power, while the output power decreases by only 3.34 % with a diagonal misalignment. So, the system is robust under these variations since there is no significant effect on the overall performance of system [14].

The impact of tolerance on the overall performance of the WPT system was studied and measured. The focus of this study was on the test systems and coil structures outlined in SAE J2954, with particular emphasis on tolerances related to the magnetic properties of materials and the values of the compensation circuits. The main objective of this research is to analyze the effect of uncertainties on the overall performance of the WPT system, highlighting which components can have tolerances and which cannot. This sheds light on the importance of component selection. In this study, the coil used had a rectangular shape, and an SS-type compensation circuit was implemented [15].

A mathematical optimization method was discovered and applied to develop a prototype using a square coil with an LCC-S compensation circuit and a power rating of 11 KW, in compliance with the SAE J2954

standard. The primary objective of the research was to ensure that the overall system efficiency did not fall below 95%. To validate the results before implementing the prototype, they conducted simulations using MATLAB Simulink [\[16\]](#).

Two transmitter coils of same dimensions were used, connecting their inner and outer terminals together, which improves the performance of WPT system and its sensitivity to misalignment issues. A simulation was conducted using ANSYS Maxwell 3D to calculate the mutual and self-inductances of the system. Simulation results were verified using LTspice. A prototype was developed using two transmitter coils and a receiver coil, all in a circular shape. Additionally, an LCC-LCC compensation circuit was implemented. The results showed that the system achieved a power output of 1.95 kW with an efficiency of 93.07% at an air gap of 15 cm [\[17\]](#).

A system that integrates MEPT with an LCC-S compensation circuit was developed, which dynamically adjusts its parameters and criteria to enhance efficiency and overall WPT system performance, achieving optimal operation, especially in cases of misalignment. They conducted simulations using ANSYS Maxwell 3D and MATLAB Simulink to validate the results, which showed that, compared to other compensation circuits, this system improves efficiency by 4.67% and ensures dynamic stability. A 3.3 kW prototype was built using a DD coil with a resonant frequency of 85 kHz [\[18\]](#).

Table 2-1: Comparative analysis of the proposed system and existing literature

Ref.	f (KHz)	Compensation type	Coils size (cm)	Coils shape	Air gap (mm)	Power (KW)	Efficiency (%)
[1]	85	SS	42x42, 32x32	Rectangular	60	4	96.89
[4]	79.5	LC-CL	75x50, 35x32	DD	210	11.1	95.1
[7]	85	SS-SP	10 outer diameter	Circular	—	0.5	—
[8]	85	SS	50 outer diameter	Circular	150	2	90.2
[10]	80	SS	10 outer diameter	Circular	100	3.2	85
[12]	85	SS	44 outer diameter	Circular	150	1	93.8
[13]	17.702	SS	60x100, 60x60	Rectangular	200	15	75.38
[14]	85	SS	40x40, 35x35	Square	100	0.3	88.5
[17]	85	LCC-LCC	—	Dual	150	1.95	93.07
[18]	85	LCC-Series	—	DD	100	3.3	93.12
Proposed	83.3	SS	45x45	Square	150	1	Calculated 98.21 Practical 77.5

3 Chapter (3): Methodology

3.1 Equations

In the scaling of the windings, the outer and inner widths and lengths of the transmitting and the receiving windings are denoted as w_t , w_{ti} , w_r , w_{ri} , l_t , l_{ti} , l_r and l_{ri} respectively as shown in Figure 3-1.

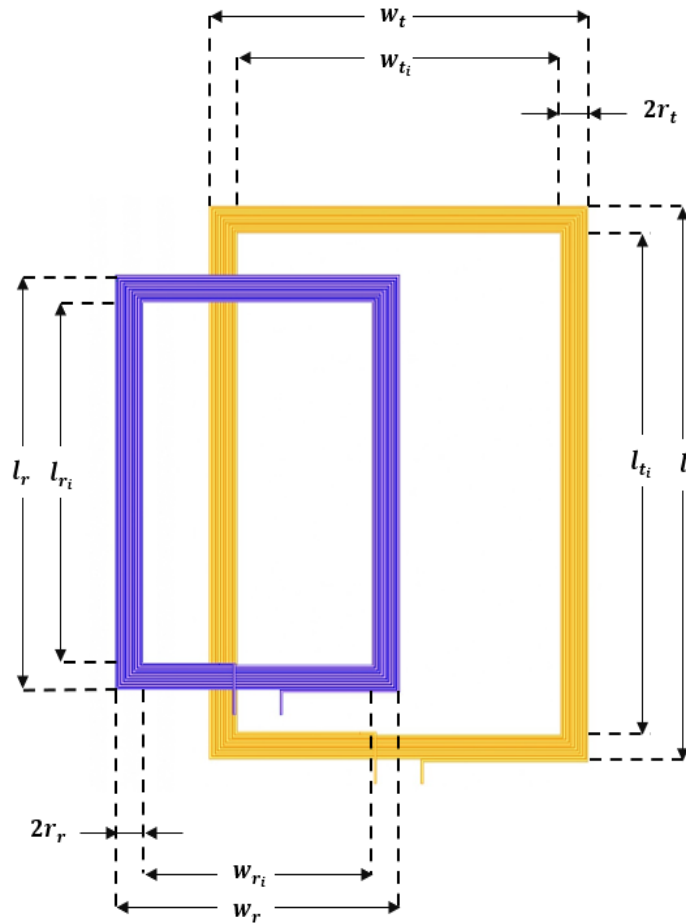


Figure 3-1: Windings geometry

By following equations given by Sallán et al., the single side lengths ($2r_t$ and $2r_r$) created by the windings are given by Eq. (1) and Eq. (2) respectively [13].

$$r_t = T_t \sqrt{\frac{S_o}{\pi}} \quad (1)$$

$$r_r = T_r \sqrt{\frac{S_o}{\pi}} \quad (2)$$

Where,

T_t is the no. of turns of the transmitting coil.

S_o is the surface area of each turn, considering the insulation.

T_r is the no. of turns of the receiving coil.

For considering both the skin and proximity effects, the AC resistance of the Litz wire (R_{AC}) is given by Eq. (3) [4].

$$R_{AC} = R_{DC_l} \left[W + H \left(\frac{N D_s}{D_o} \right)^2 \left(\frac{D_s \sqrt{f}}{10.44} \right)^4 \right] \quad (3)$$

Where,

R_{DC_l} is the DC resistance of the Litz wire per length.

W represents the resistance ratio of a single strand when isolated.

H is a constant correlating to a number of strands (N), as specified in the datasheet.

N is the number of strands.

D_s is the diameter of a strand.

D_o is the outer diameter of each turn.

f is the resonant frequency.

But since we use Litz wire, the skin and proximity effects are neglected. The DC transmitting and receiving windings resistances (R_t and R_r) are given by Eq. (4) and Eq. (5) [13].

$$R_t = \rho_{cu} T_t \frac{2(w_t + l_t)}{S_i} \quad (4)$$

$$R_r = \rho_{cu} T_r \frac{2(w_r + l_r)}{S_i} \quad (5)$$

Where,

ρ_{cu} is the resistivity of the copper.

T_t is the number of turns of the transmitting coil.

w_t is the outer width of the transmitting coil.

l_t is the outer length of the transmitting coil.

S_i is the surface copper area of each turn, without the insulation.

T_r is the number of turns of the receiving coil.

w_r is the outer width of the receiving coil.

l_r is the outer length of the receiving coil.

The transmitting and the receiving windings self-inductances are given by Eq. (6) and Eq. (7) [13].

$$L_t = \frac{\mu_o}{\pi} \times T_t^2 [a_t + b_t - 2c_t + 0.25(w_t + l_t)] \quad (6)$$

Where,

μ_o is the vacuum permeability.

T_t is the number of turns of the transmitting coil.

w_t is the outer width of the transmitting coil.

l_t is the outer length of the transmitting coil.

$$a_t = w_t \times \ln \frac{2w_t l_t}{r_t(w_t + \sqrt{w_t^2 + l_t^2})}$$

$$b_t = l_t \times \ln \frac{2w_t l_t}{r_t(l_t + \sqrt{w_t^2 + l_t^2})}$$

$$c_t = w_t + l_t - \sqrt{w_t^2 + l_t^2}$$

$$L_r = \frac{\mu_0}{\pi} \times T_r^2 [a_r + b_r - 2c_r + 0.25(w_r + l_r)] \quad (7)$$

Where,

μ_0 is the vacuum permeability.

T_r is the number of turns of the receiving coil.

w_r is the outer width of the receiving coil.

l_r is the outer length of the receiving coil.

$$a_r = w_r \times \ln \frac{2w_r l_r}{r_r(w_r + \sqrt{w_r^2 + l_r^2})}$$

$$b_r = l_r \times \ln \frac{2w_r l_r}{r_r(l_r + \sqrt{w_r^2 + l_r^2})}$$

$$c_r = w_r + l_r - \sqrt{w_r^2 + l_r^2}$$

The mutual inductance can be approximated using the following expression derived from Neumann's formula, that is, valid for square coils that are perfectly aligned [\[19\]](#).

$$M = \frac{2\mu_0}{\pi} \times T_t T_r \left[u \ln \left(\frac{u \sqrt{h^2 + u^2 + h^2 + u^2}}{hu + h \sqrt{h^2 + 2u^2}} \right) + h - 2\sqrt{h^2 + u^2} + \sqrt{h^2 + 2u^2} \right]$$

(8)

Where,

μ_0 is the vacuum permeability.

T_t is the number of turns of the transmitting coil.

T_r is the number of turns of the receiving coil.

u is the square side length of each coil (equals $w_t = l_t = w_r = l_r$, for square coils).

h is the gap distance between the two coils.

The degree of coupling between the transmitting and receiving coils is represented by the magnetic coupling coefficient K , which is directly related to mutual inductance. This can be defined by Eq. (9) [1].

$$K = \frac{M}{\sqrt{L_t \times L_r}} \quad (9)$$

Where,

M is the mutual inductance.

L_t is the self-inductance of the transmitting coil.

L_r is the self-inductance of the receiving coil.

Wireless Power Transfer (WPT) technology is less efficient than conductive charging methods while transferring and receiving power, which remains its main limitation. Most recent research on WPT focuses on developing resonant circuit topologies and optimizing power electronic converter control strategies, whether on the charger side or the load side.

For efficient power transfer, a large transmitter current is required. However, this high current reduces the system's efficiency because it causes losses in the system. To improve efficiency, reactive elements can be compensated using capacitors, leaving only resistive elements in the system, which helps minimize losses. Various compensation topologies have been explored, each with their own benefits and drawbacks. The choice of topology depends on the specific requirements of the application.

For electric vehicle (EV) charging, the series-series compensation topology is the most suitable, as it can be designed independently of coupling and load variations. At resonance, the following relation can be satisfied as shown in Eq. (10). For WPT, the SAE J2954 frequency range of 79 to 90 KHz should be used. The 85 KHz is the nominal frequency.

The capacitances of the SS compensations of the transmitting and the receiving circuits can be calculated by Eq. (10) and Eq. (11).

$$C_t = \frac{1}{4 \pi^2 f^2 L_t} \quad (10)$$

$$C_r = \frac{1}{4 \pi^2 f^2 L_r} \quad (11)$$

Where,

f is the resonant frequency.

L_t is the self-inductance of the transmitting coil.

L_r is the self-inductance of the receiving coil.

The resonant angular frequency can be calculated by Eq. (12).

$$\omega_o = \frac{1}{\sqrt{L_t \times C_t}} = \frac{1}{\sqrt{L_r \times C_r}} \quad (12)$$

Where,

L_t is the self-inductance of the transmitting coil.

C_t is the capacitance of the transmitting circuit.

L_r is the self-inductance of the receiving coil.

C_r is the capacitance of the receiving circuit.

The input DC voltage to HFI (V_{DC}), is converted into HF AC voltage. The electromagnetic field effect transfers AC power to the receiving coil. The transmitting and receiving coils act as magnetic couplers between the power source and the load. The AC power from the receiving coil is then rectified, with an output capacitor filtering any ripples as shown in Figure 3-2.

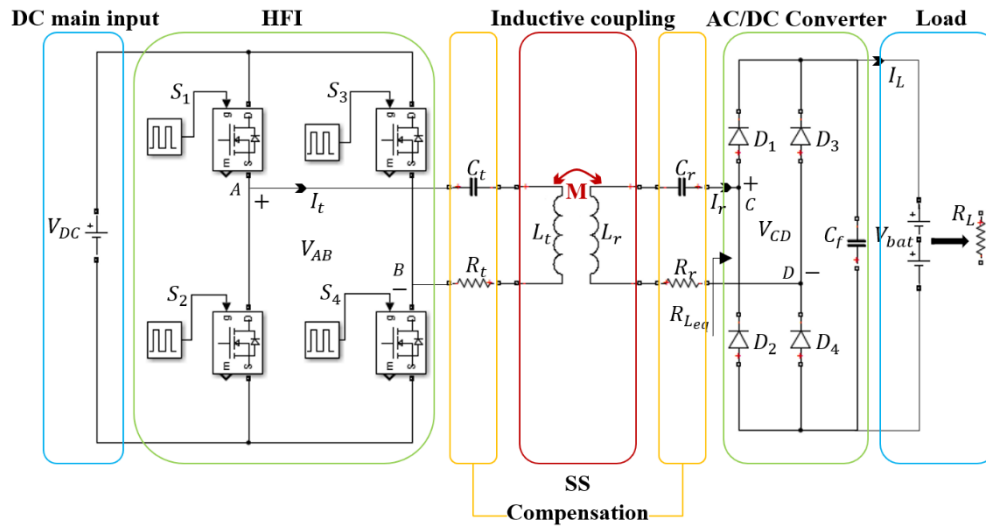


Figure 3-2: The SS compensation topology with AC/DC converter

The receiver side rectifier, filter and battery load are modeled as an equivalent AC resistive load (R_{Leq}) based on battery power (P_{bat}) and voltage (V_{bat}) [1,12], as shown in Figure 3-3. This can be defined by Eq. (13)

$$R_{Leq} = \frac{8}{\pi^2} \times \frac{V_{bat}^2}{P_{bat}} \quad (13)$$

Where,

V_{bat} is the battery voltage.

P_{bat} is the battery power.

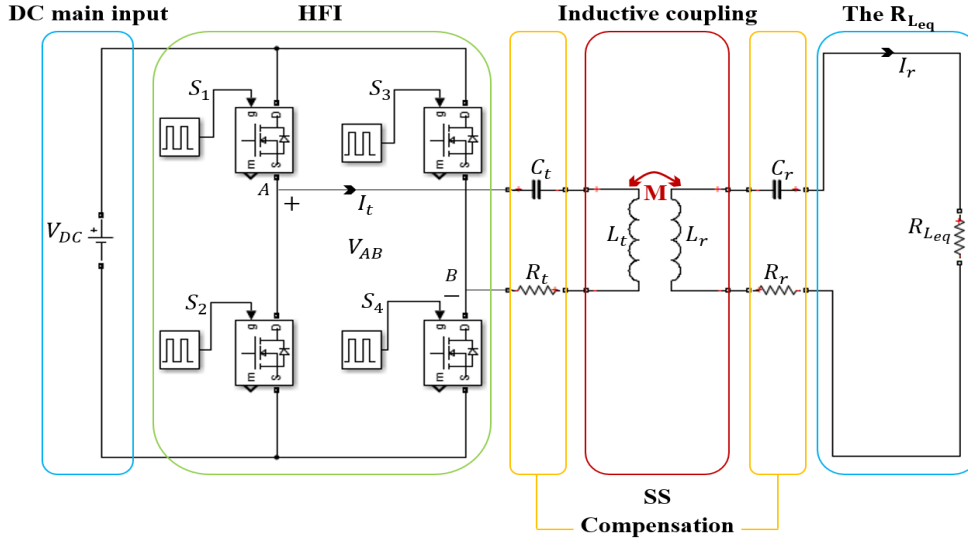


Figure 3-3: The SS compensation topology with R_{Leq} load

The effect of mutual inductance on the electrical circuit containing the transmitting coil can be represented by a dependent voltage source, which is a function of the current flowing through the receiving coil. By applying Kirchhoff's Voltage Law (**KVL**) to the transmitting coil circuit, the output high-frequency (HF) AC voltage is given by Eq. (14) [1].

$$V_1 = \left(\frac{1}{j\omega C_t} + j\omega L_t + R_t \right) I_t - j\omega M I_r \quad (14)$$

Where,

ω is the angular frequency.

C_t is the capacitance of the transmitting circuit.

L_t is the self-inductance of the transmitting coil.

R_t is the DC resistance of the transmitting coil.

I_t is the RMS transmitting current.

M is the mutual inductance.

I_r is the RMS receiving current.

Similarly, the effect of mutual inductance on the electrical circuit containing the receiving coil can be represented by a dependent voltage source, which depends on the current flowing through the transmitting coil. By applying **KVL** to the receiving coil circuit, the AC voltage across the equivalent AC resistive load is given by Eq. (15) [1].

$$V_2 = \left(\frac{1}{j\omega C_r} + j\omega L_r + R_r \right) I_r - j\omega M I_t \quad (15)$$

Where,

ω is the angular frequency.

C_r is the capacitance of the receiving circuit.

L_r is the self-inductance of the receiving coil.

R_r is the DC resistance of the receiving coil.

I_r is the RMS receiving current.

M is the mutual inductance.

I_t is the RMS transmitting current.

At resonance, V_{1_o} and V_{2_o} can be calculated by Eq. (16) and Eq. (17) [1].

$$V_{1_o} = I_t R_t - j\omega_o M I_r \quad (16)$$

$$V_{2_o} = I_r R_r - j\omega_o M I_t \quad (17)$$

Where,

I_t is the RMS transmitting current.

R_t is the DC resistance of the transmitting coil.

ω_o is the resonant angular frequency.

M is the mutual inductance.

I_r is the RMS receiving current.

R_r is the DC resistance of the receiving coil.

The expressions for RMS current flowing through the transmitting and receiving coils at resonance are given by Eq. (18) and Eq. (19) [12].

$$I_t = \frac{V_{1o}(R_{Leq} + R_r)}{R_t(R_{Leq} + R_r) + (\omega_o M)^2} \quad (18)$$

$$I_r = \frac{-j\omega_o M I_t}{R_{Leq} + R_r} \quad (19)$$

Where,

V_{1o} is the output HF AC voltage at resonance.

R_{Leq} is the equivalent AC resistive load.

R_r is the DC resistance of the receiving coil.

R_t is the DC resistance of the transmitting coil.

ω_o is the resonant angular frequency.

M is the mutual inductance.

At resonance, the RMS power delivered to the equivalent AC resistive load is given by Eq. (20) [19].

$$P_{AC} = \frac{(\omega_o M I_t)^2}{R_{Leq} + R_r} \quad (20)$$

Where,

ω_o is the resonant angular frequency.

M is the mutual inductance.

I_t is the RMS transmitting current.

R_{Leq} is the equivalent AC resistive load.

R_r is the DC resistance of the receiving coil.

From the previous equation, we get the transmitting current as given by Eq. (21).

$$I_t = \sqrt{\frac{P_{AC}(R_{Leq} + R_r)}{(\omega_o M)^2}} \quad (21)$$

Where,

R_{Leq} is the equivalent AC resistive load.

R_r is the DC resistance of the receiving coil.

ω_o is the resonant angular frequency.

M is the mutual inductance.

By equaling Eq. (18) with Eq. (21), we get the following relation (22) in which all parameters are included, where P_{AC} is assumed, except our equivalent AC resistive load.

$$\frac{P_{AC}}{(\omega_o M)^2} = \frac{V_{1o}^2 (R_{Leq} + R_r)}{[R_t(R_{Leq} + R_r) + (\omega_o M)^2]^2} \quad (22)$$

When using SS compensation, the efficiency η_{s-s} can be given by Eq. (23) [1,8].

$$\eta_{s-s} = \frac{(\omega_o M)^2 R_{Leq}}{R_t(R_{Leq} + R_r)^2 + (\omega_o M)^2 (R_{Leq} + R_r)} = \frac{R_{Leq}}{R_{Leq} + R_r + \left(\frac{R_t(R_{Leq} + R_r)^2}{(\omega_o M)^2}\right)} \quad (23)$$

Where,

ω_o is the resonant angular frequency.

M is the mutual inductance.

R_{Leq} is the equivalent AC resistive load.

R_t is the DC resistance of the transmitting coil.

R_r is the DC resistance of the receiving coil.

3.2 Simulations

3.2.1 Simulation with ANSYS Maxwell 3D

To simulate this application, we first used ANSYS Maxwell 3D to design the two coils. ANSYS Maxwell 3D uses FEA to analyze the magnetic field emitted from the transmitter to the receiver and provides the most accurate results as much as possible since it has golden value for applications where it is impossible to obtain accurate analytical solutions. FEA can calculate the values of self-inductance, mutual inductance, coupling coefficient, and magnetic flux as numerical and graphical results.

Based on the previously mentioned assumptions, we were able to calculate the outer and inner diameters of the coil. Consequently, we determined the number of turns and thus could estimate the length of the wire that is used to wind the transmitter and receiver coils. We used these parameters in the design to achieve the required power value with the desired efficiency.

Table 3-1 shows SAE j2954 recommendations concerning Z- class requirements and transmission power levels. We designed an ANSYS Maxwell 3D model for the transmitter and receiver coils in a square shape with the specifications shown in table 3-2 , and the spacing between two coils equals 15 cm.

Table 3-1: SAE j2954 recommendations concerning Z- class requirements and transmission power levels

GCR (cm)			Actual h			Power levels for light-duty EVs (kVA)			
Z1-class	Z2-class	Z3-class	Above ground	Flush ground	Under ground	WPT1	WPT2	WPT3	WPT4
10-15	14-21	17-25	>GCR	=GCR	<GCR	3.7	7.7	11.1	22

Table 3-2: The specifications of ANSYS Maxwell 3D model

Specifications	Transmitter	Receiver
Length of wire	20 m	20 m
Diameter of wire	3.25 mm	3.25 mm
Dimension of coils	450 mm × 450 mm	450 mm × 450 mm
Number of turns	10 turns	10 turns
Region medium	Air	
Material of coils	Copper	Copper
Current excitation	10 A	10 A
Terminal length	700 mm	700 mm

Figure 3-4 shows the ANSYS Maxwell 3D model for the WPT system. For minimizing computational time and effort for FEA, while striking a suitable balance between accuracy and model complexity, the coil windings may be represented in the configuration as one block of copper, each representing a single turn [4].

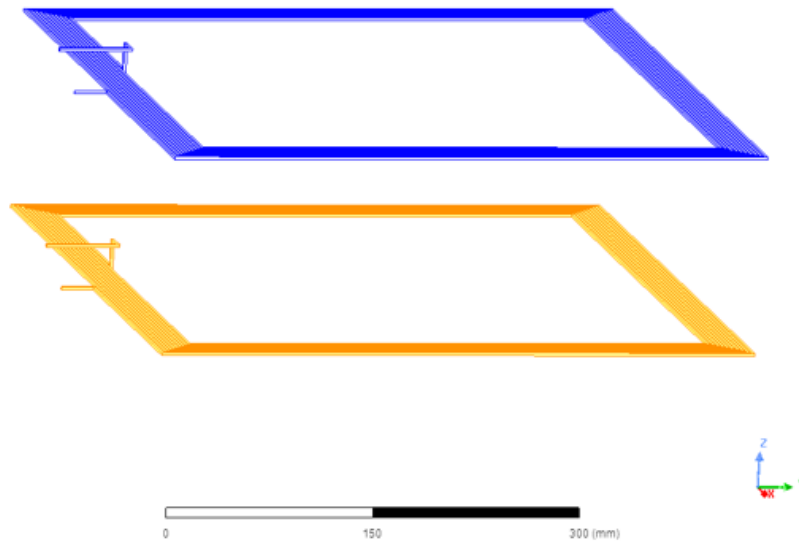


Figure 3-4: The ANSYS Maxwell 3D model for the WPT system

With this design, we have calculated the values of self-inductance, mutual inductance, and the coupling coefficient both numerically and graphically.

What sets ANSYS Maxwell 3D Electronics Desktop apart from other simulation programs is the ability to design the transmitter and receiver coils in ANSYS Maxwell 3D. This allows us to calculate values such as mutual inductance, self-inductance, coupling coefficient, and magnetic flux.

Then, this design can be used in the electric circuit in ANSYS Simplorer and two model are simulate Jointly to ensure accurate results in various electrical calculations necessary for wireless charging of electric vehicles Similar to reality results.

Figure 3-5 illustrates the electrical circuit for ANSYS Simplorer, and the components used in its design to create a WPT system.

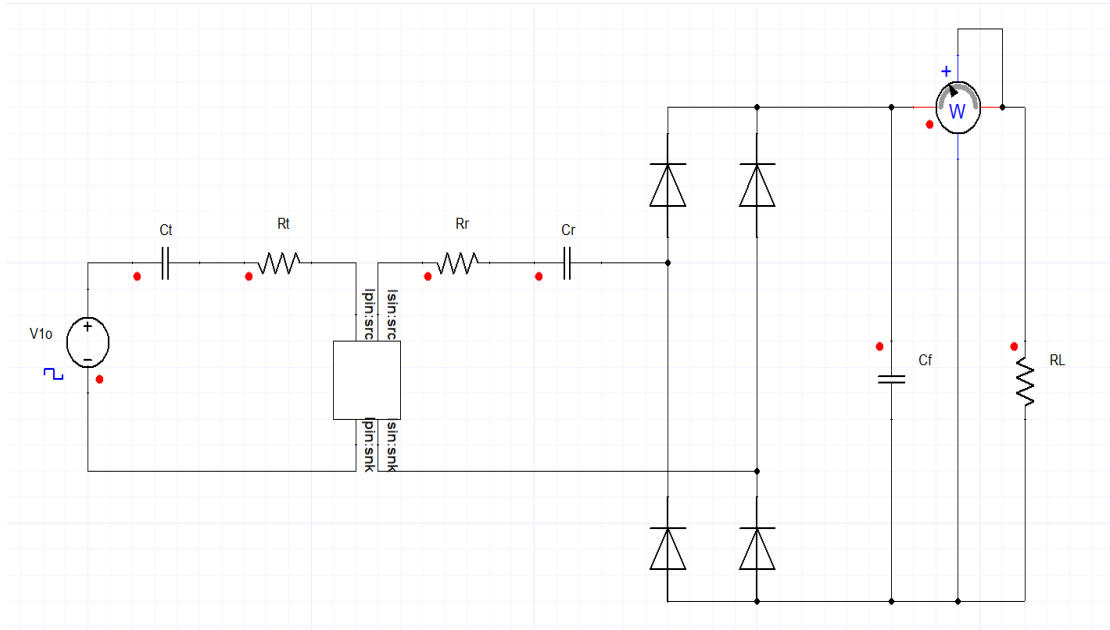
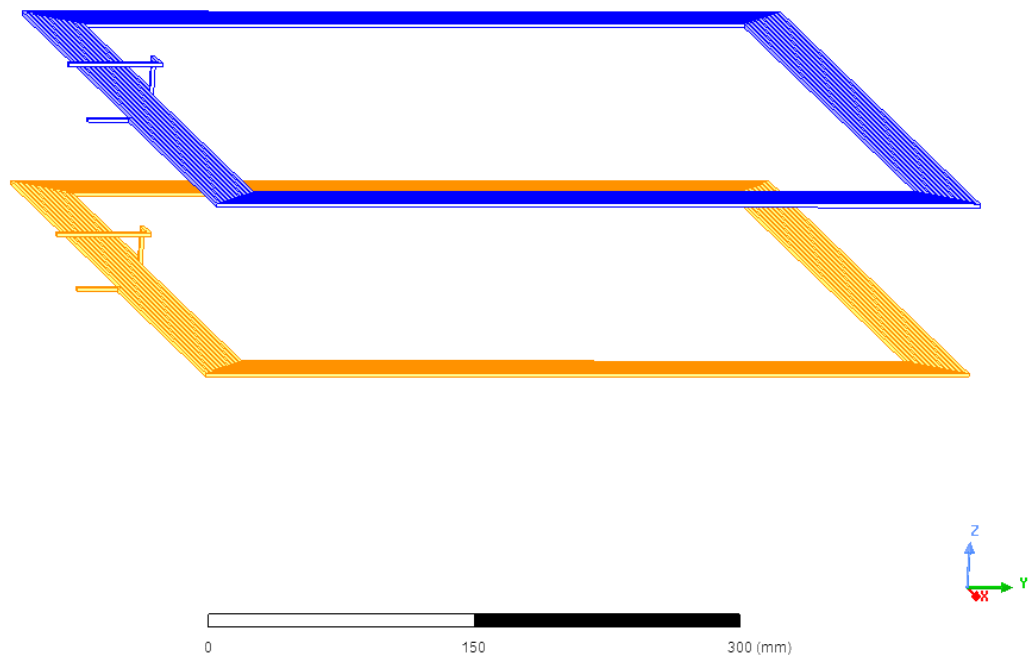


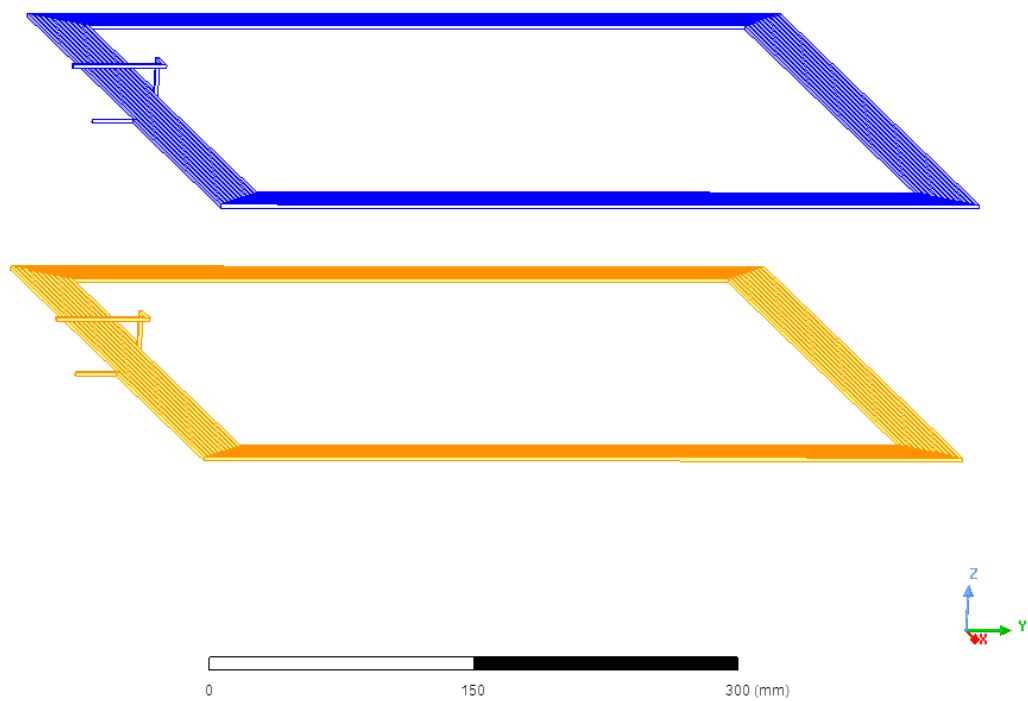
Figure 3-5: ANSYS Simpler RIWPT system

The issue of misalignment is considered one of the biggest problems facing wireless charging for electric cars, if not the biggest. For this reason, in this research paper, we conducted a detailed study on all the different misalignment scenarios that could occur in reality to understand their impact on transmission efficiency. This way, we can take them into account and try to find solutions or reduce their effects.

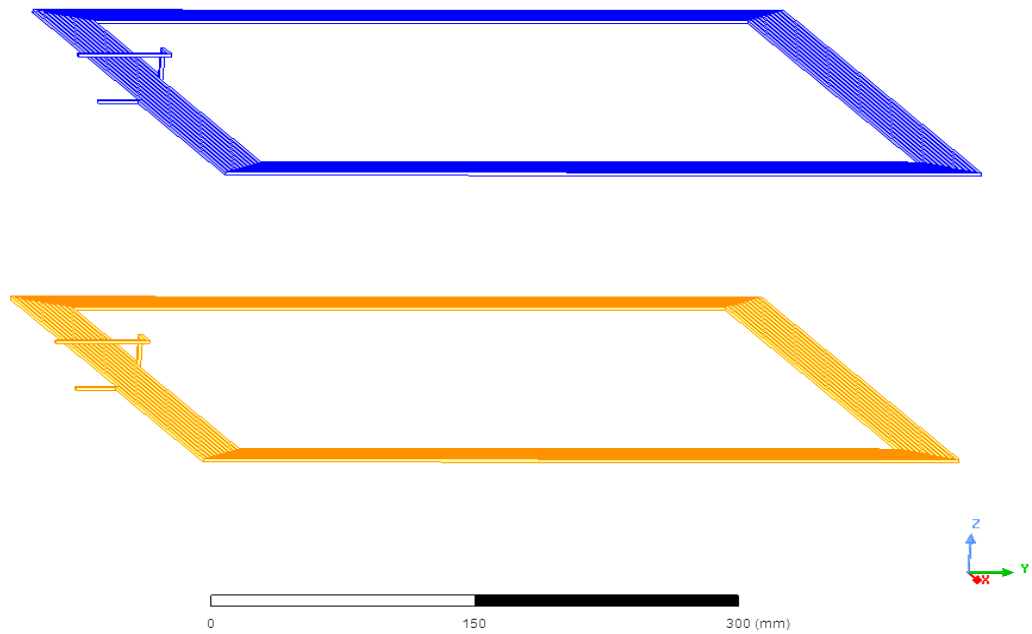
First, we studied different air gap scenarios between the transmitter and the receiver, where the vertical distance between the transmitter and receiver increases by raising the receiver upwards, as shown in Figure 3-6.



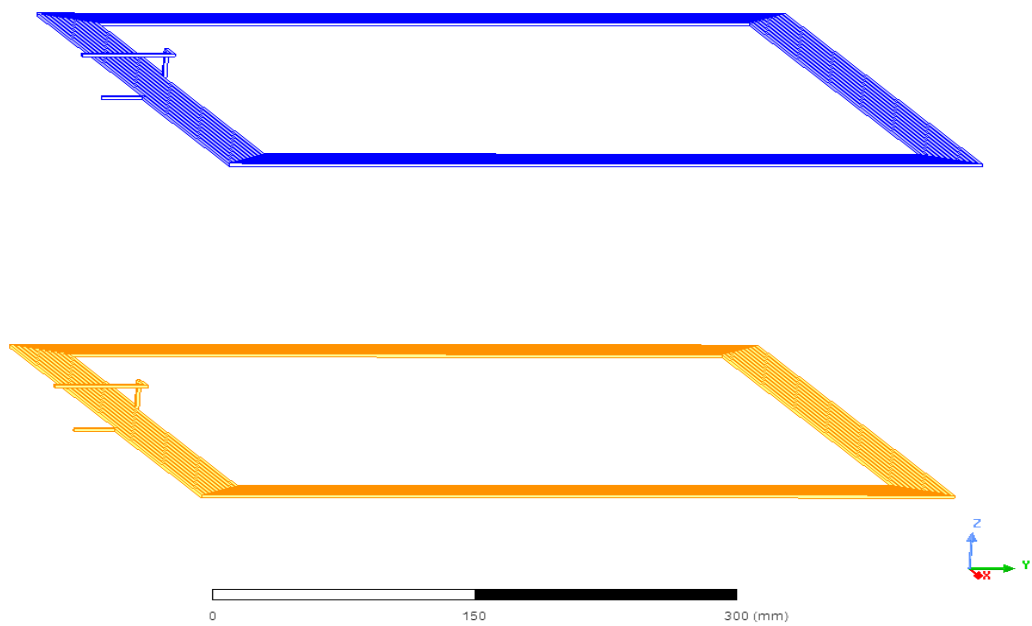
(a)



(b)



(c)



(d)

Figure 3-6: The different air gaps between the transmitter and receiver coils

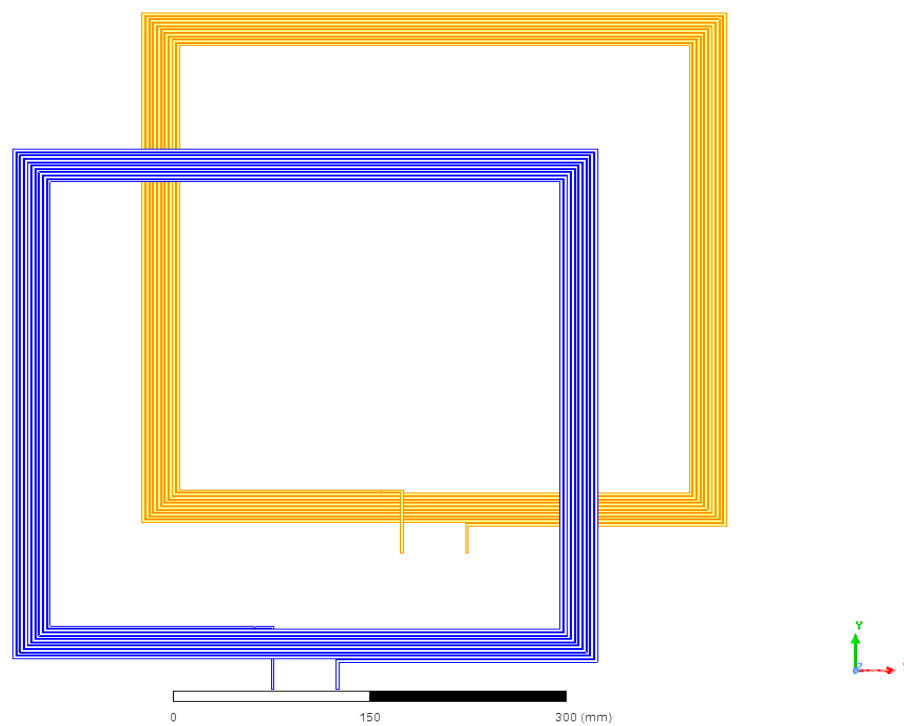
(a) 10 cm (b) 15 cm (c) 20 cm (d) 25 cm

Secondly, we studied different horizontal misalignment scenarios, as shown in table 3-3.

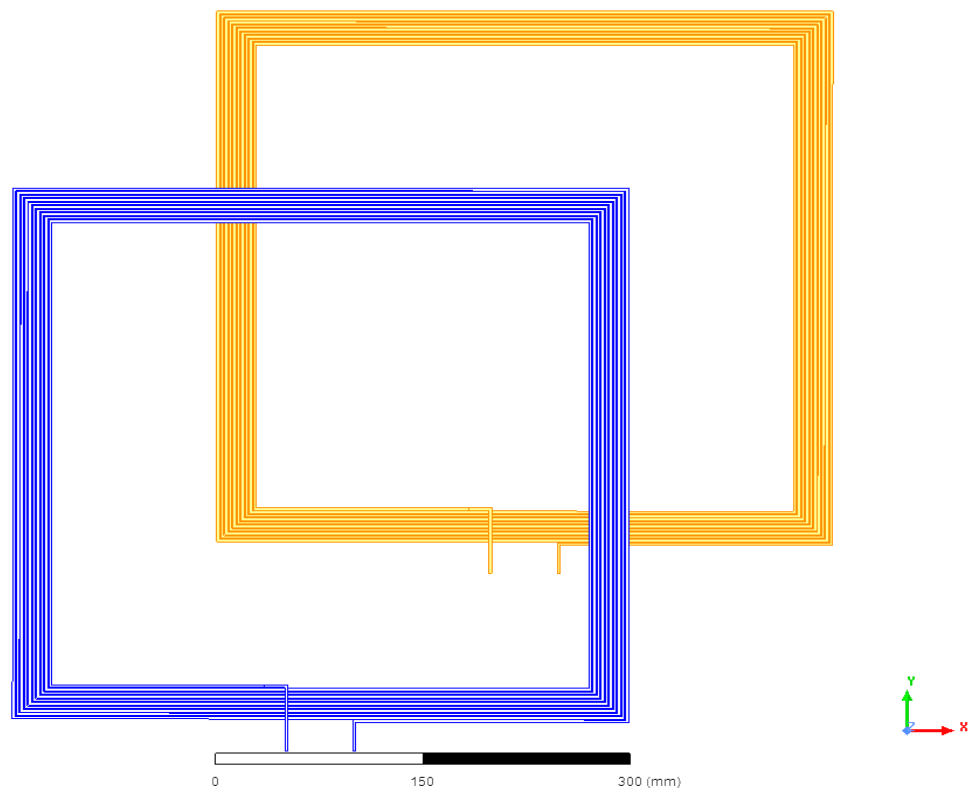
Table 3-3: Misalignment data in different positions for each scenario

Misalignment data in different positions for each scenario				
No.	Case	a	b	c
1	$(-x, -y)$ (cm)	$(-10, -12)$	$(-15, -15)$	—
2	$-x$ (cm)	-10	-15	-30
3	$-y$ (cm)	-12	-15	-30
4	$(-x, +y)$ (cm)	$(-10, +12)$	$(-15, +15)$	—

Here, the receiver coil moves parallel to the transmitter coil in different positions for each scenario, as shown in Figure 3-7.

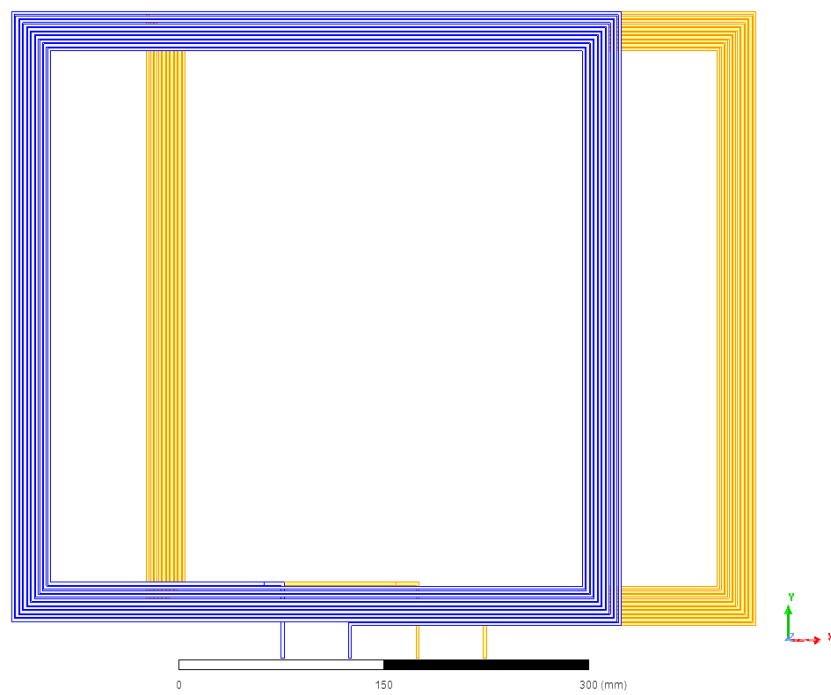


(a)

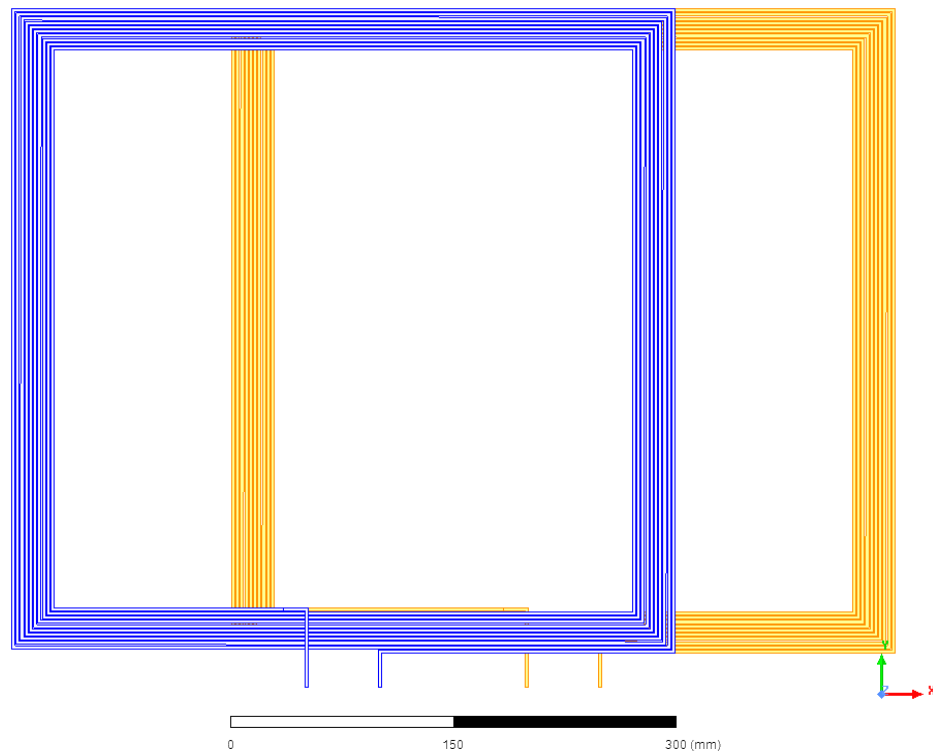


(b)

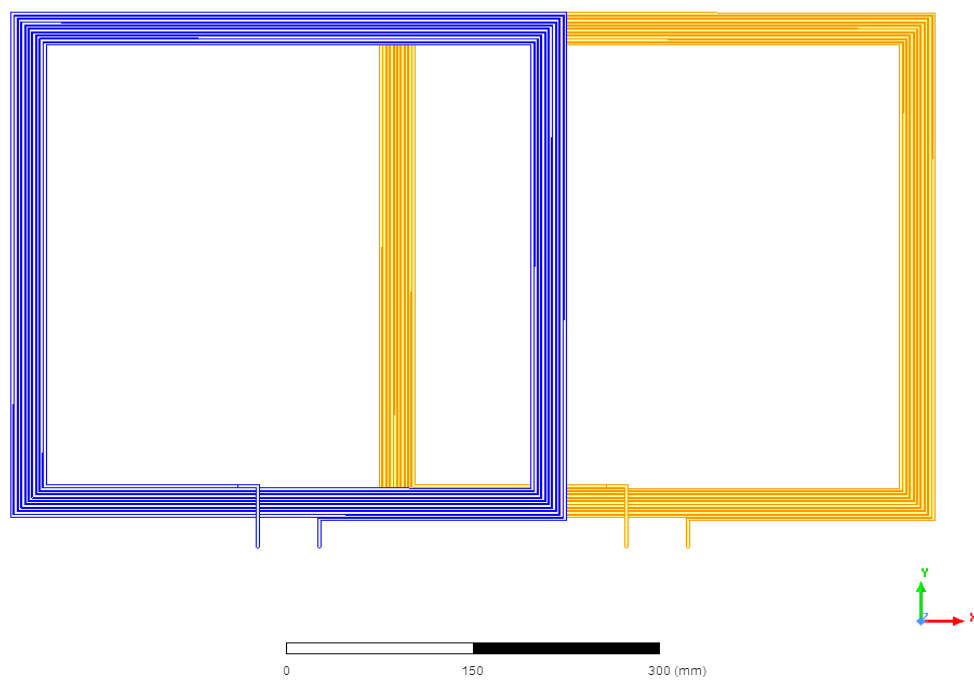
1. (-x, -y) horizontal misalignment



(a)

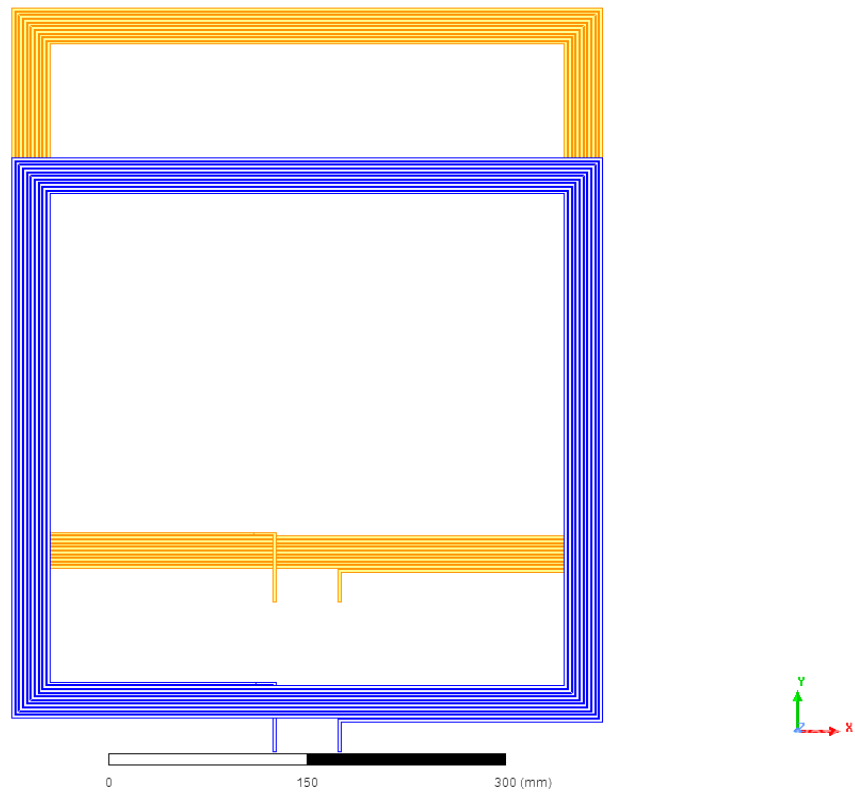


(b)

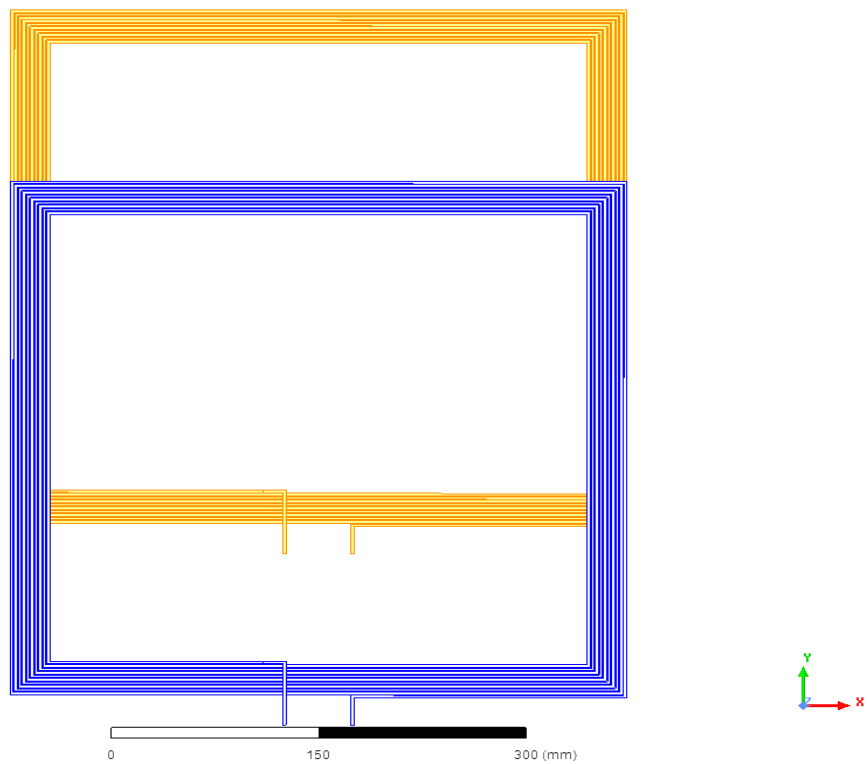


(c)

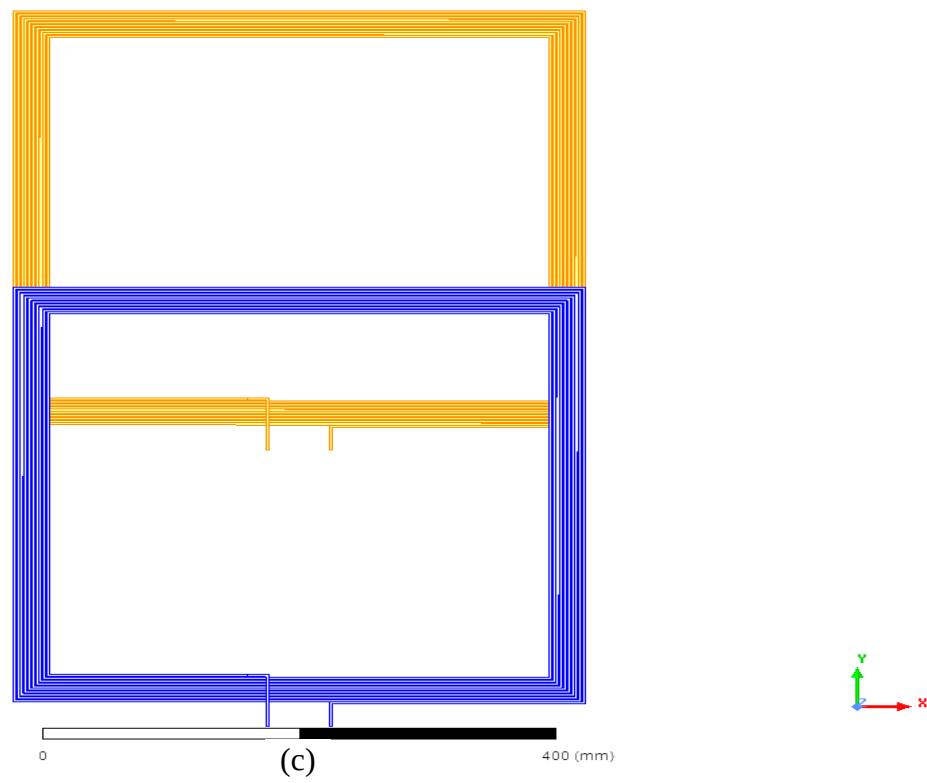
2. -x horizontal misalignment



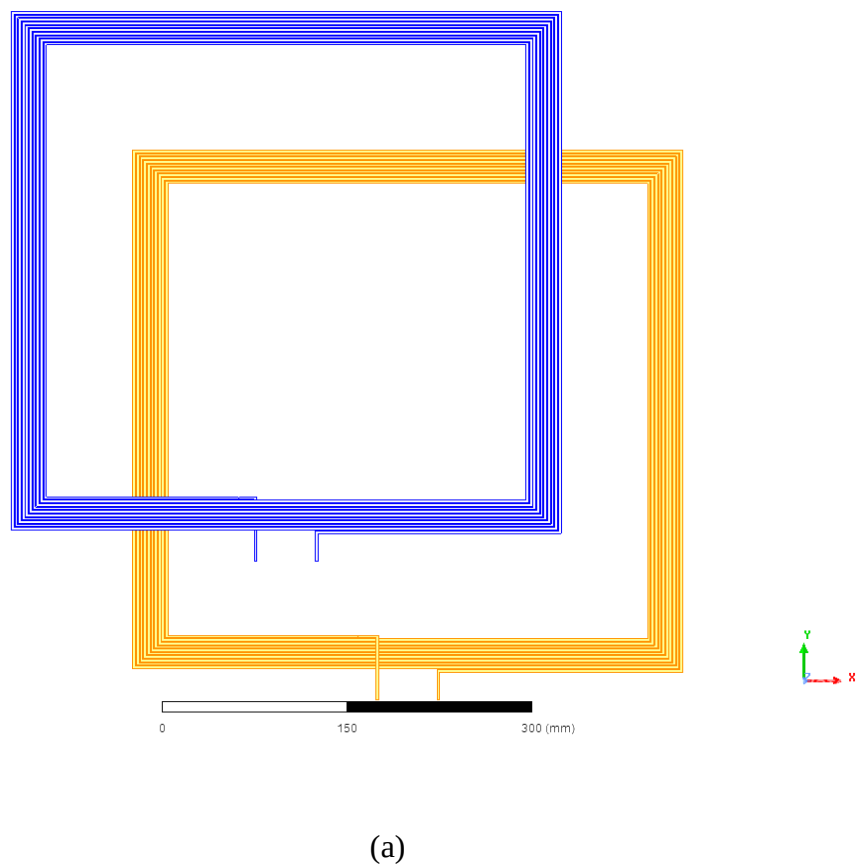
(a)

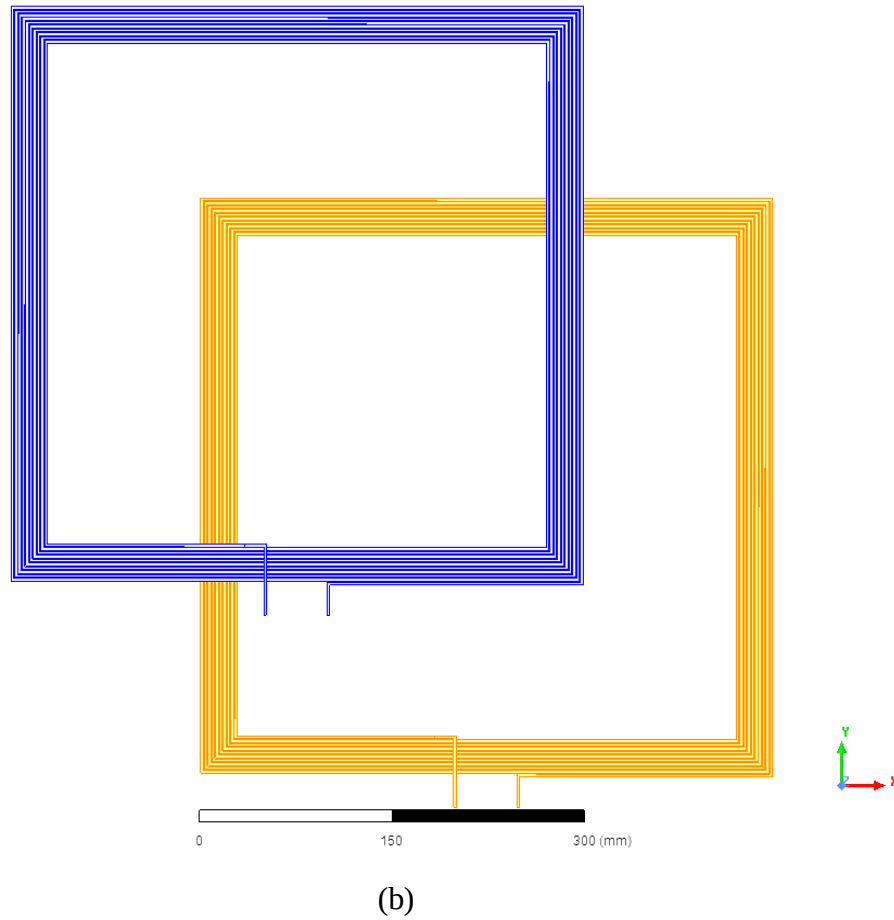


(b)



3. -y horizontal misalignment



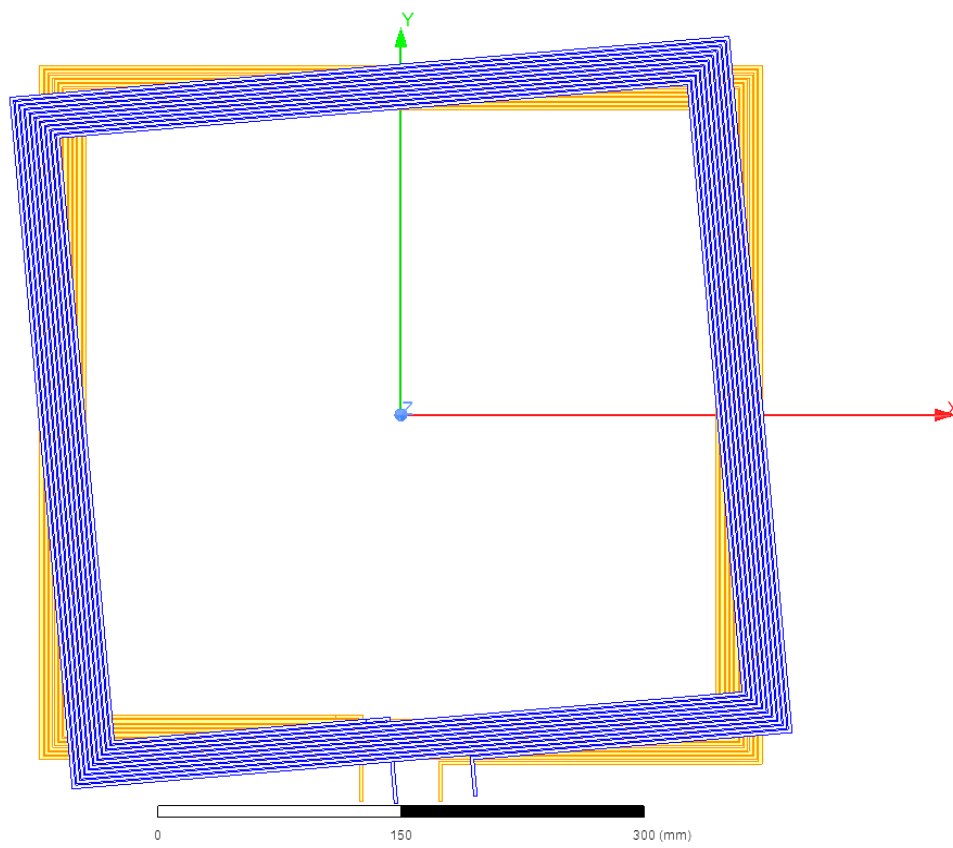


4. $(-x, +y)$ horizontal misalignment

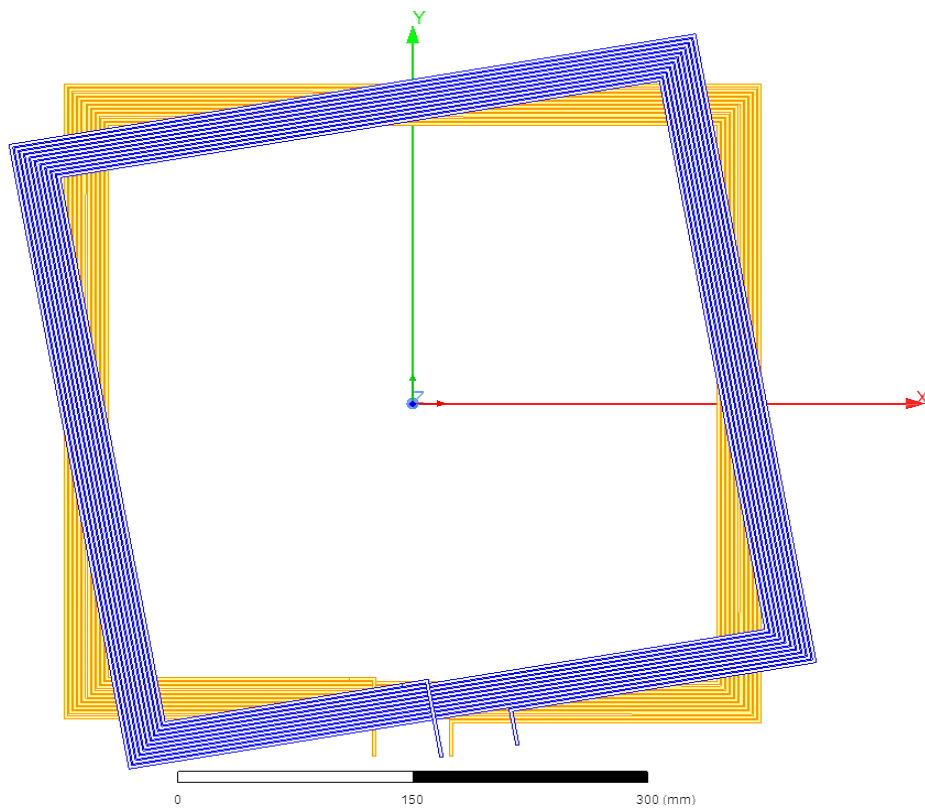
Figure 3-7: The different positions of horizontal misalignment for each scenario

1. $(-x, -y)$ 2. $-x$ 3. $-y$ 4. $(-x, +y)$

Thirdly, we studied one of the most critical misalignment cases that may frequently occur during wireless charging: the rotation around the Z-axis. In this situation, the receiver coil is parallel to the transmitter coil, but an angle exists between them concerning the Z-axis. The study was conducted for various angles to comprehensively understand this type of misalignment, as illustrated in Figure 3-8.



(a)



(b)

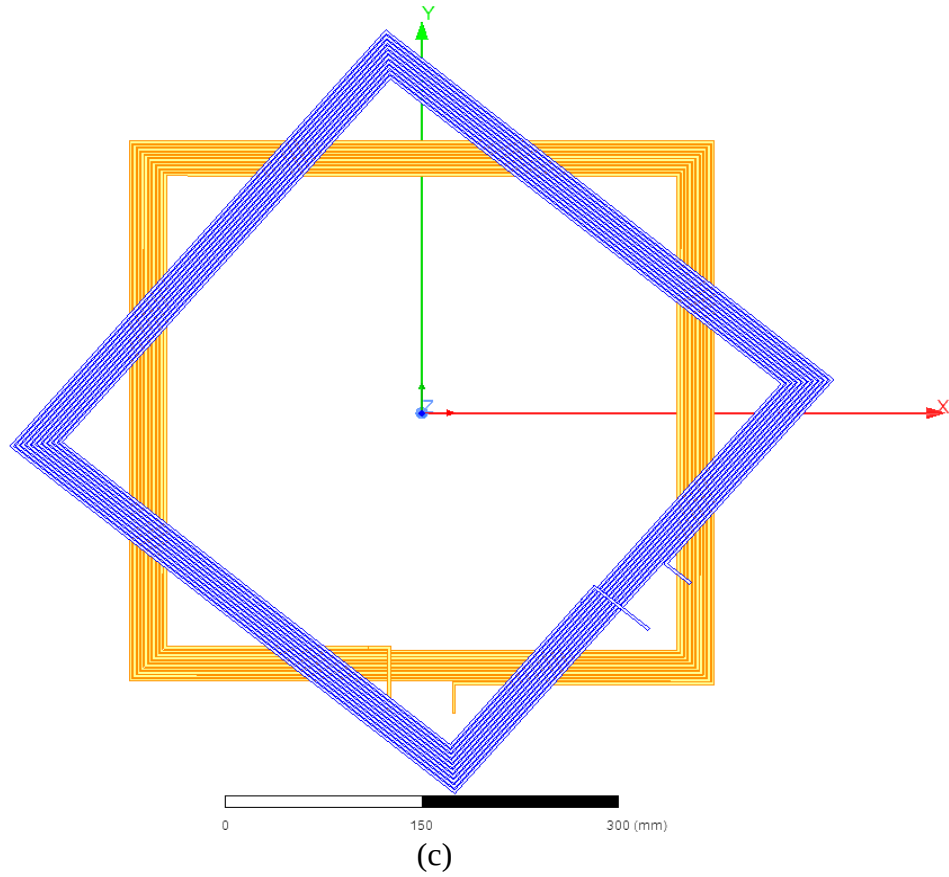
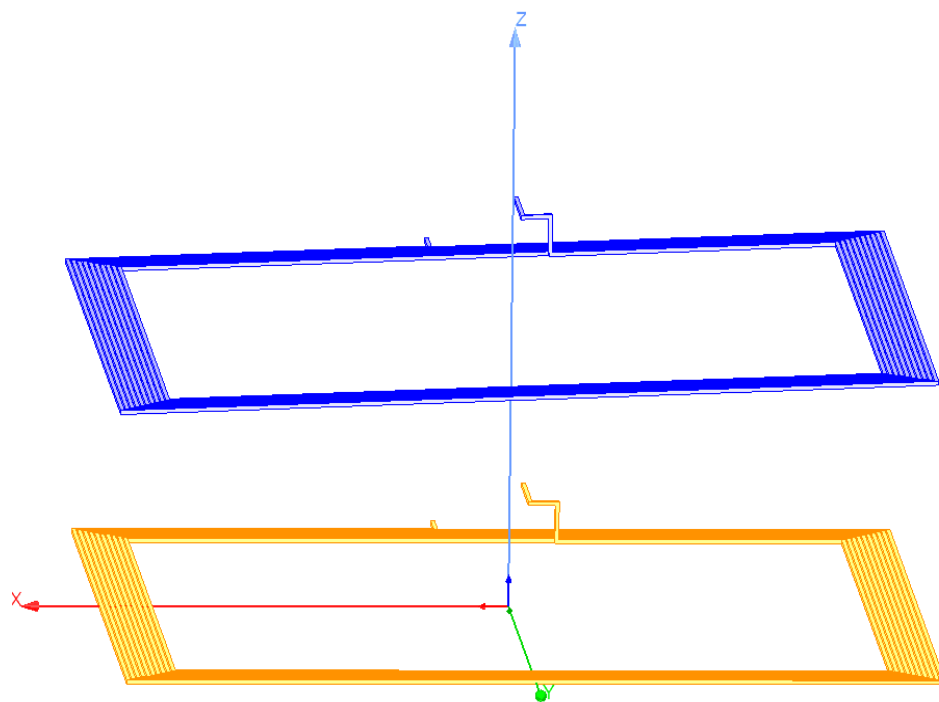


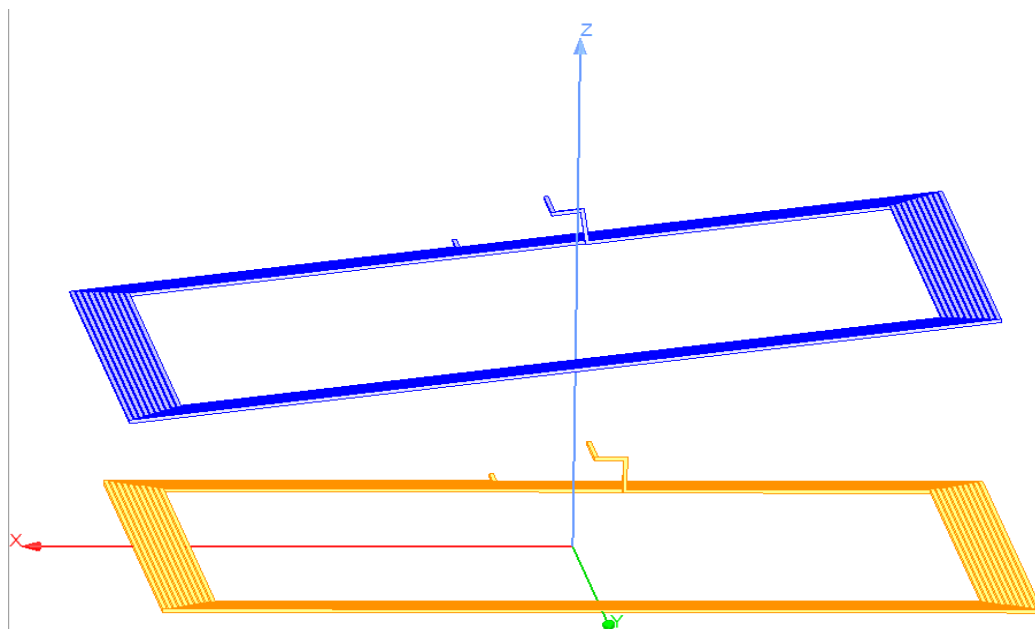
Figure 3-8: Angles exist between the transmitting and receiving coil concerning the Z-axis

(a) 5° (b) 10° (c) 50°

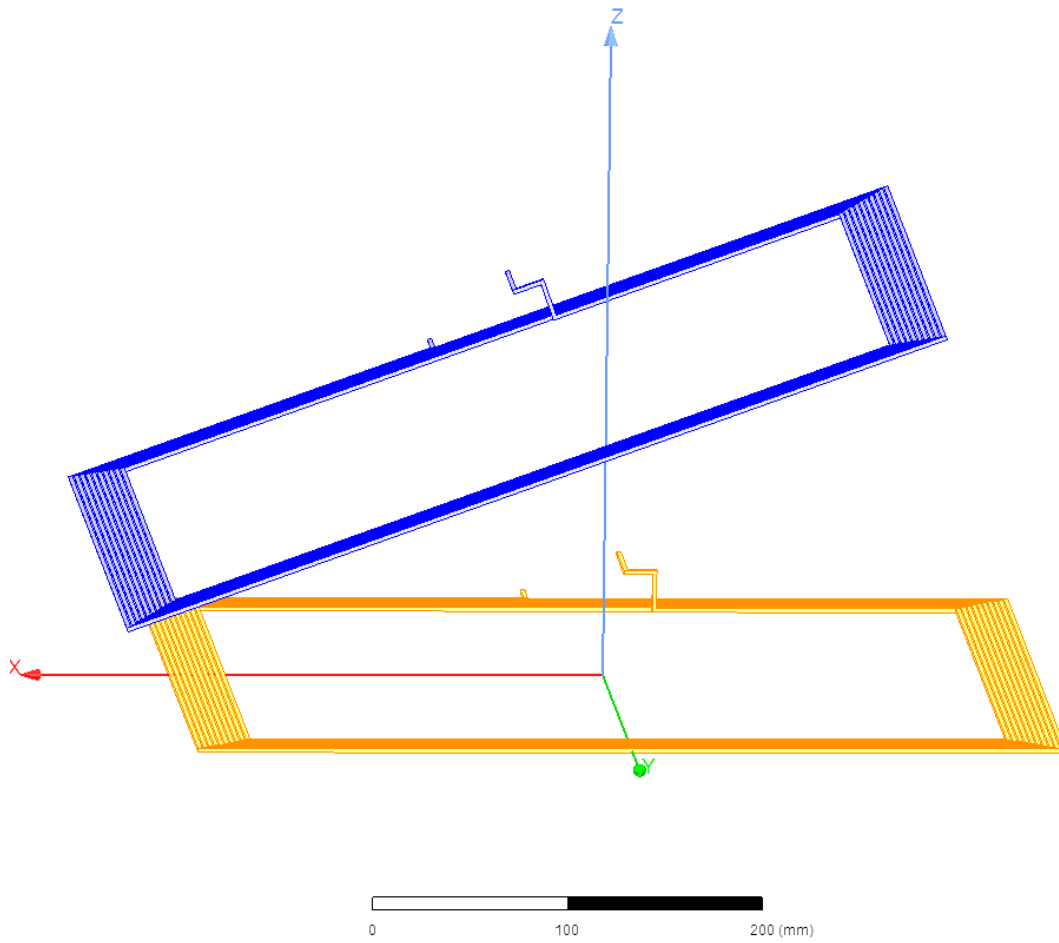
In all the previous cases, the transmitting coil was parallel to the receiver coil. Due to the possibility of one side of the receiver coil moving downward, we conducted a fourth study on the non-uniform airgap. This study was performed by rotating the receiver around the Y-axis. To understand the impact of this type of error, we analyzed it at different angles as shown in Figure 3-9.



(a)



(b)



(c)

Figure 3-9: A non-uniform air gap in the Z-axis direction with rotation around the Y- axis

(a) 5° (b) 8° (c) 20°

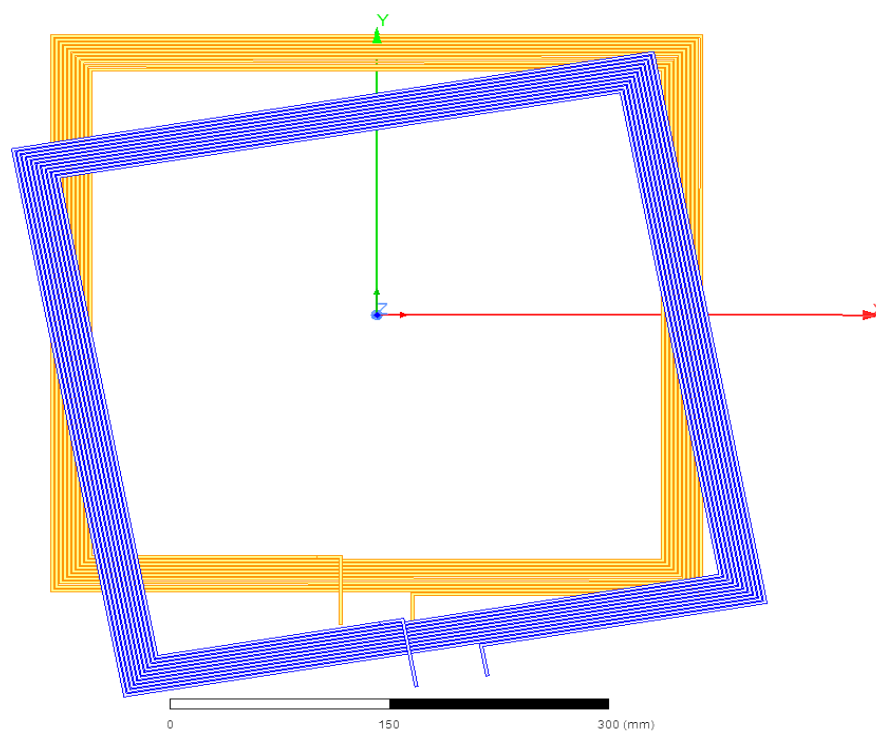
Due to the likelihood of multiple misalignment errors occurring during the wireless charging of electric vehicles, we conducted an extensive study focusing on this crucial aspect.

The first multiple misalignment error is horizontal misalignment with rotation around the Z-axis, as shown in table 3-4.

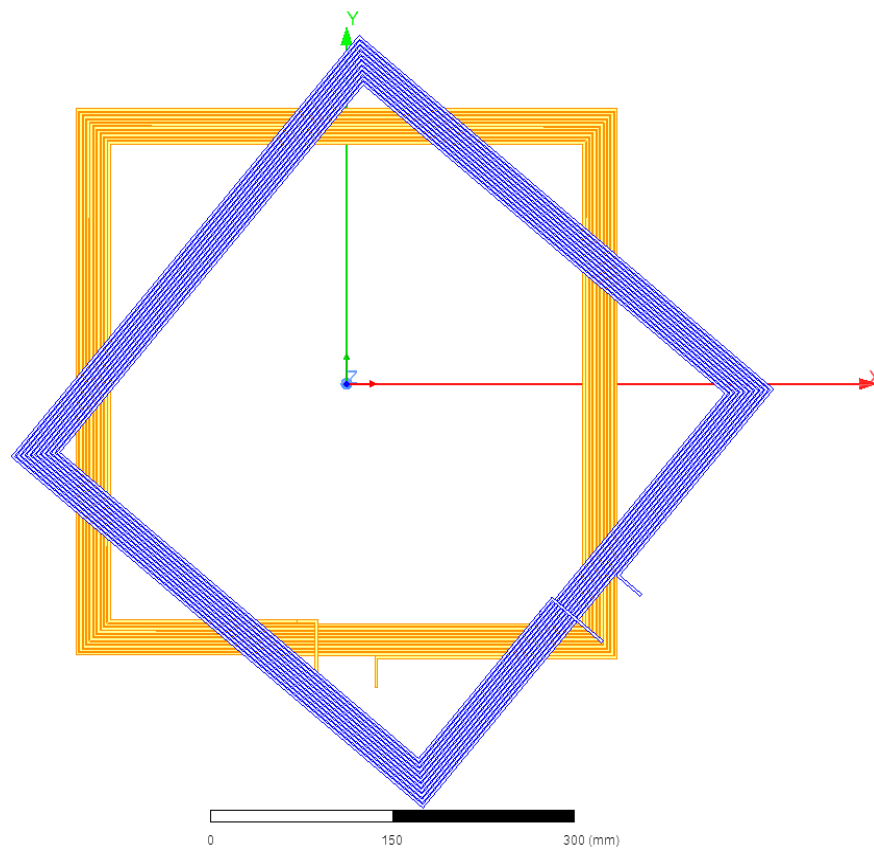
Table 3-4: Multiple misalignment error data in case of horizontal misalignment with rotation around the Z-axis

Case No.	-y (cm)	D°
1	-5	10
2	-5	50
3	-15	10
4	-15	50
5	-30	10
6	-30	50

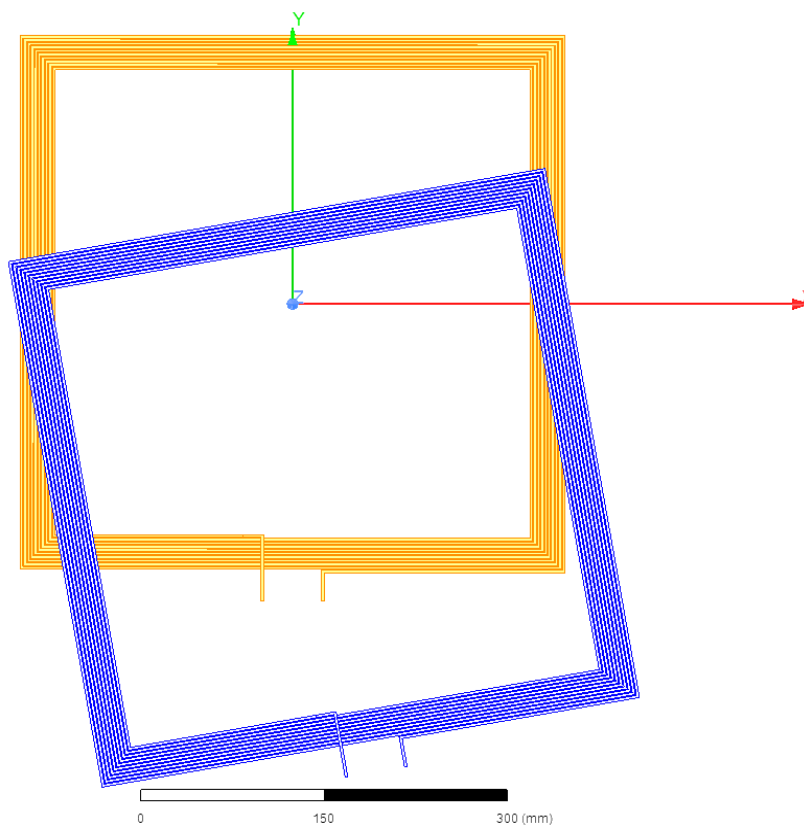
Figure 3-10 shows multiple misalignment error positions since the receiver coil moves parallel to the transmitter coil in different positions with an angle existing between them concerning the Z-axis.



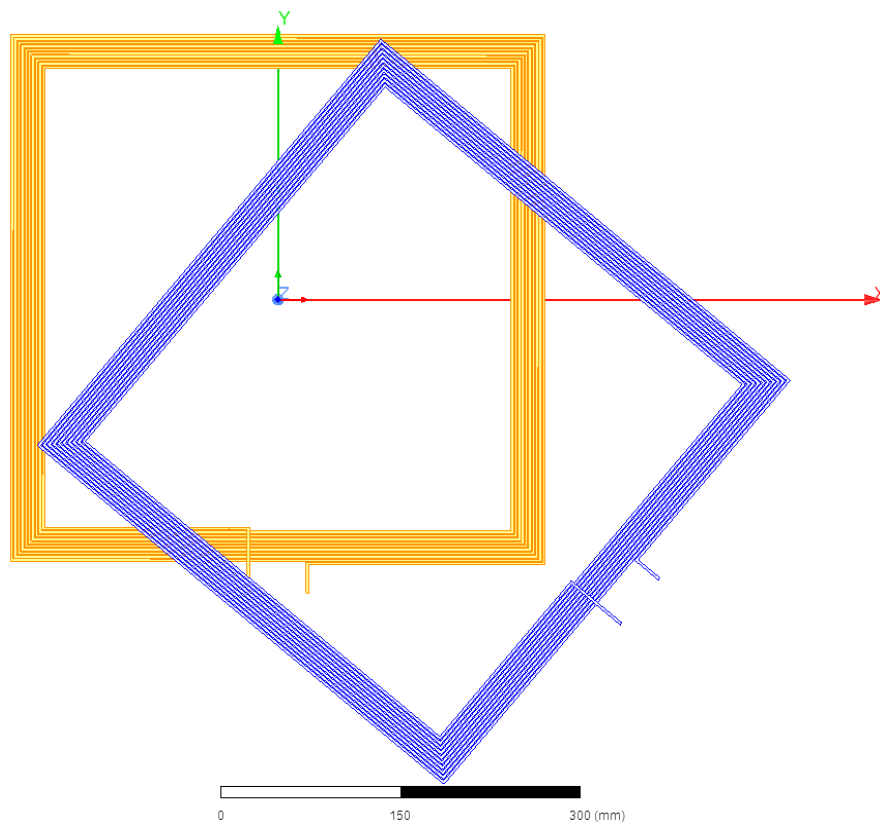
Case 1



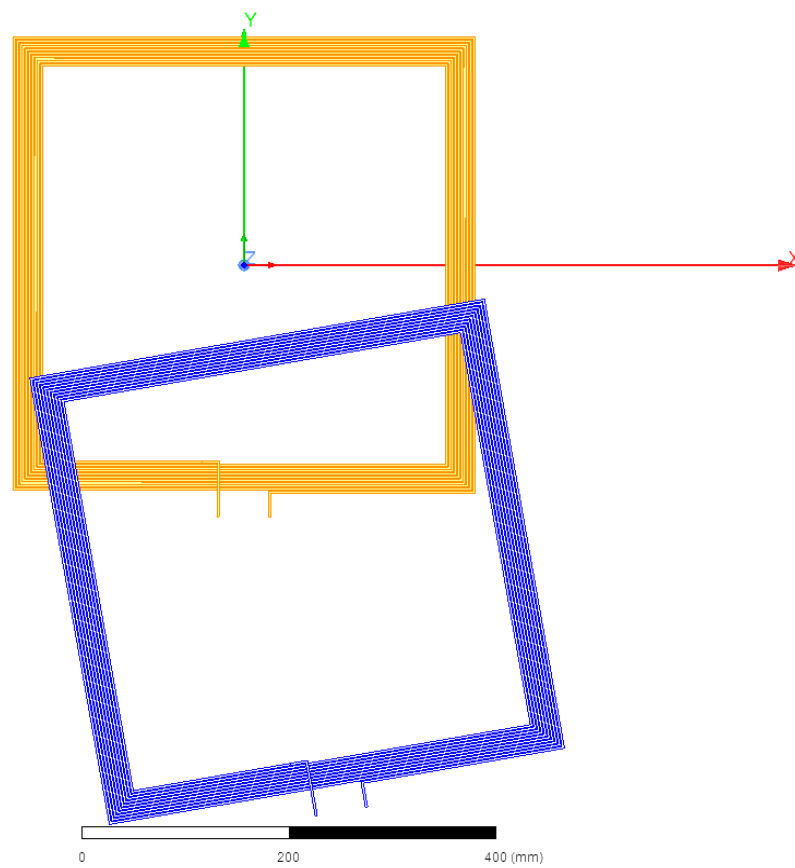
Case 2



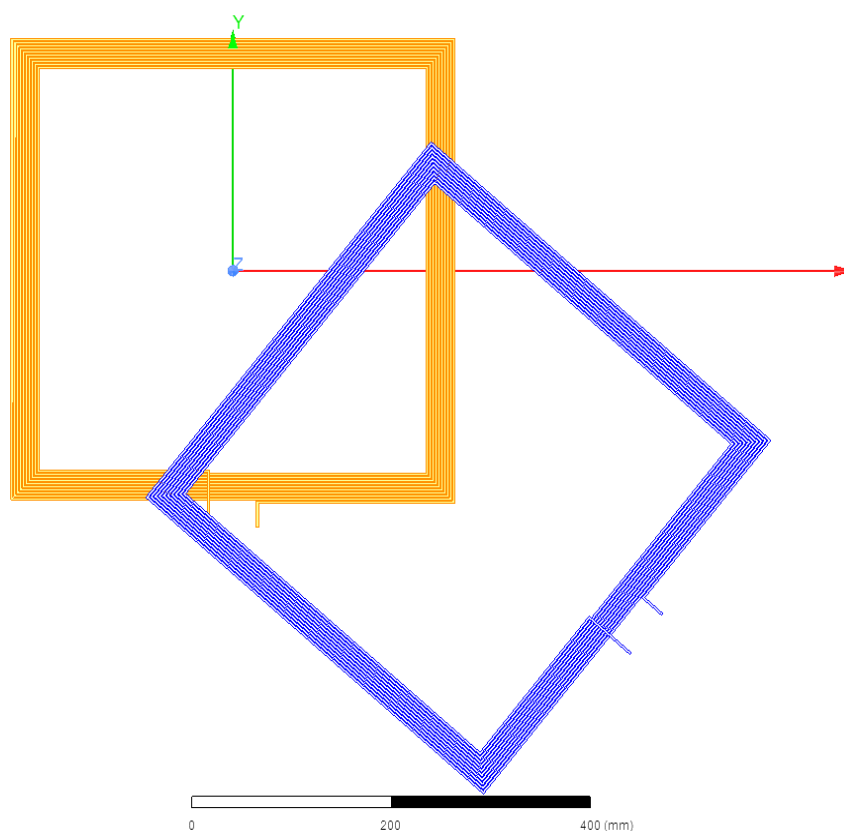
Case 3



Case 4



Case 5



Case 6

Figure 3-10: Multiple misalignment error positions in case of horizontal misalignment with rotation around the Z-axis

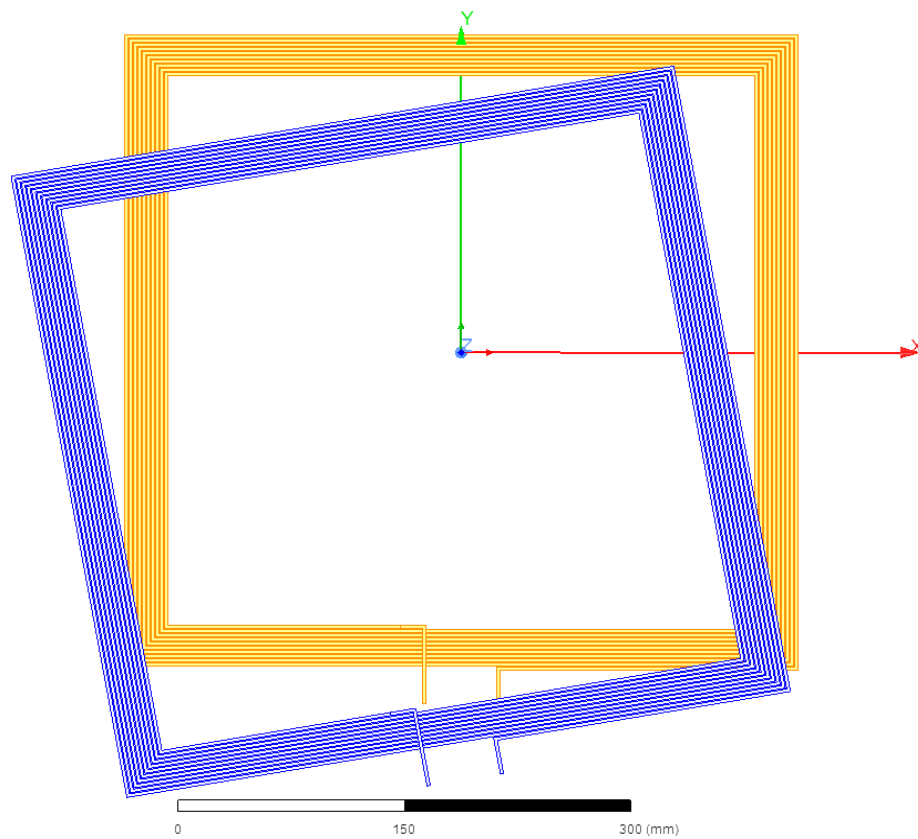
Case 1) -5 cm, 10°	Case 2) -5cm ,50°	Case 3) -15 cm ,10°
Case 4) -15 cm, 50°	Case 5) -30 cm, 10°	Case 6) -30 cm, 50°

The second multiple misalignment error is horizontal misalignment in (-x, -y) with rotation around the Z-axis as shown in table 3-5.

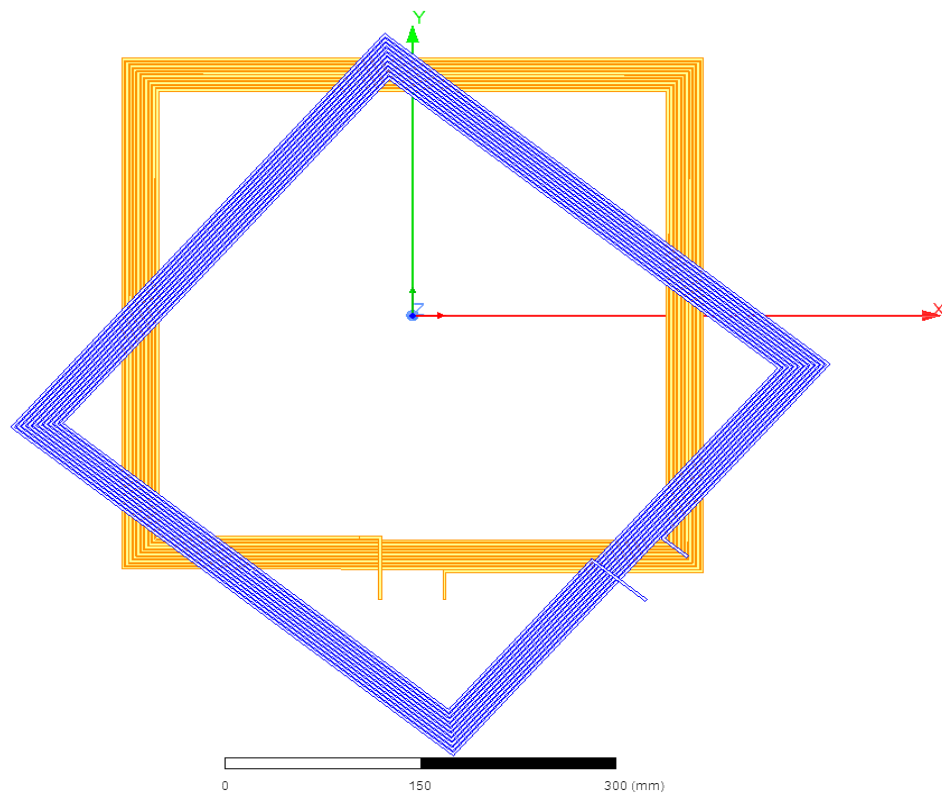
Table 3-5: Multiple misalignment error data in case of horizontal misalignment in (-x, -y) with rotation around the Z-axis

Case No.	(-x, -y) (cm)	D°
1	(-5, -5)	10
2	(-5, -5)	50
3	(-15, -15)	10
4	(-15, -15)	50

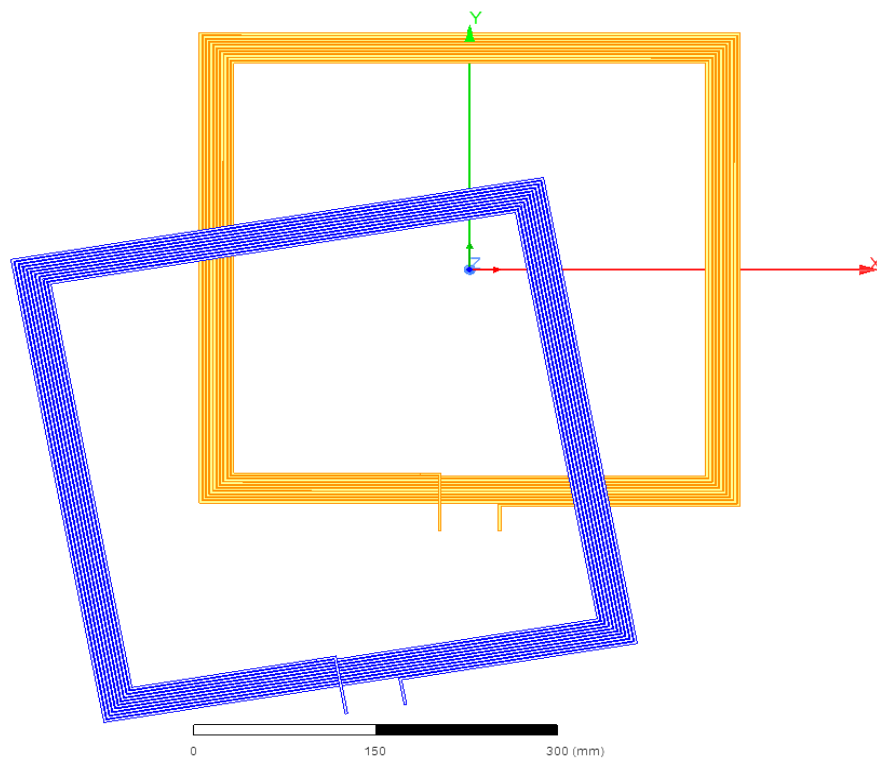
Figure 3-11 shows multiple misalignment error positions since the receiver coil moves parallel to the transmitter coil in different positions in (-x, -y) with an angle existing between them concerning the Z-axis.



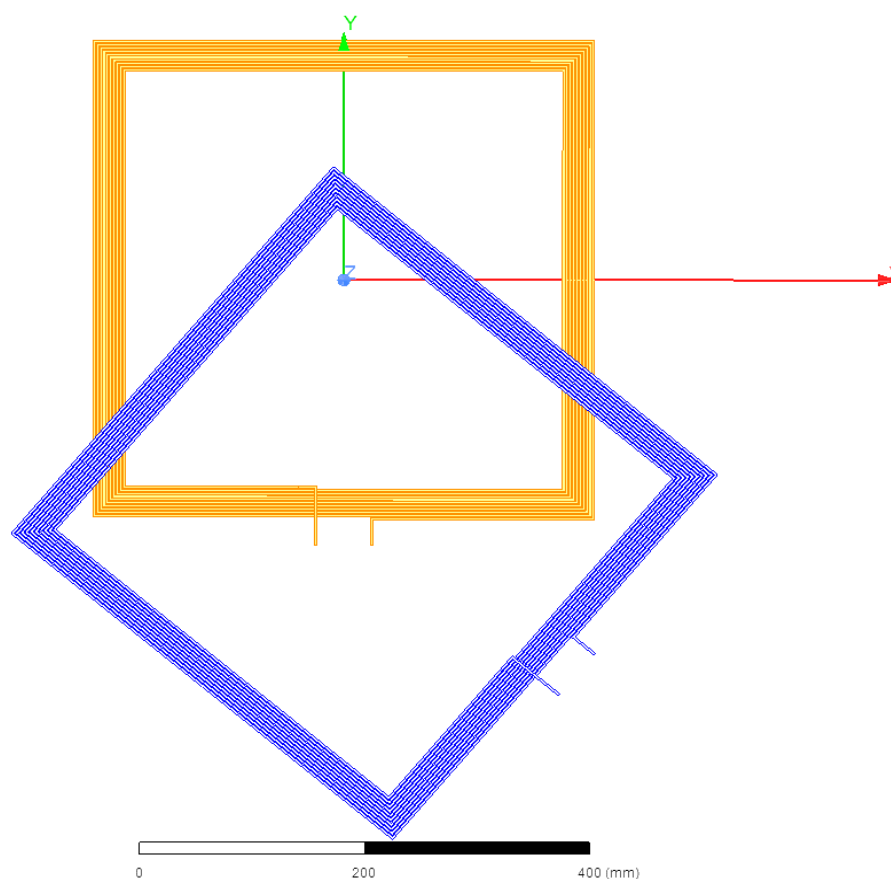
Case 1



Case 2



Case 3



Case 4

Figure 3-11: Multiple misalignment error positions in case of horizontal misalignment in $(-x, -y)$ with rotation around the Z-axis

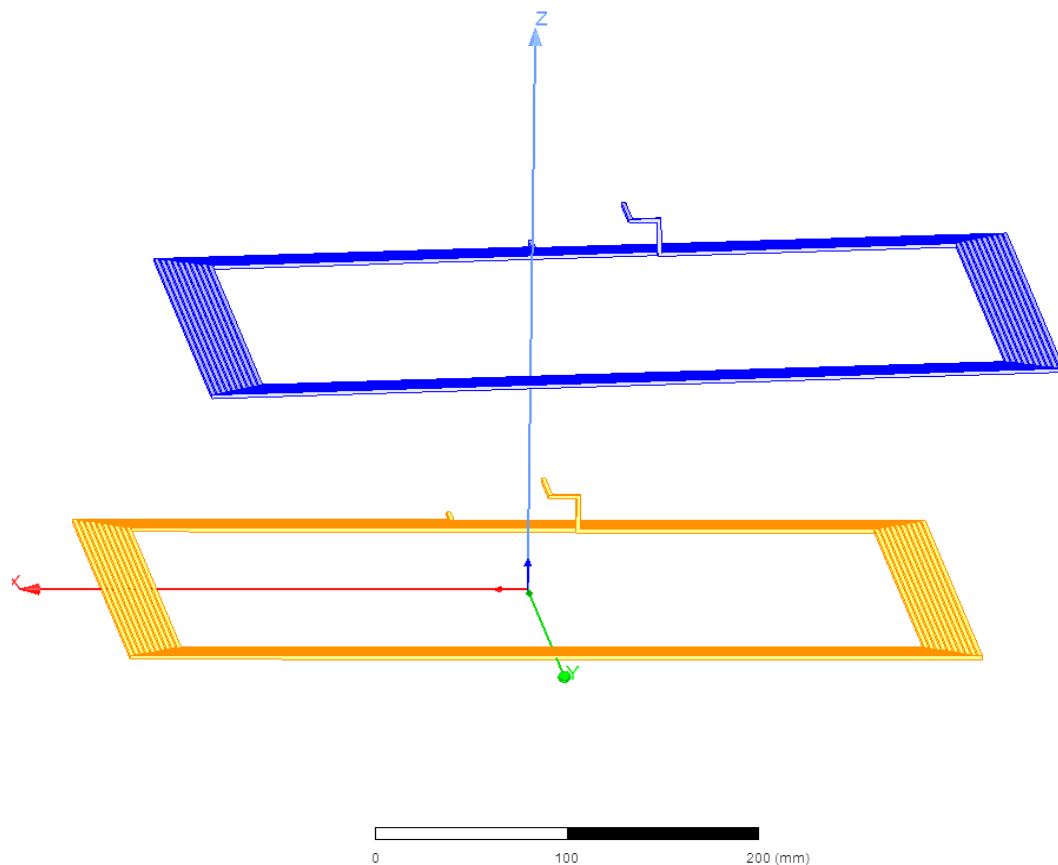
Case 1) $(-5, -5)$ cm, 10°	Case 2) $(-5, -5)$ cm, 50°
Case 3) $(-15, -15)$ cm, 10°	Case 4) $(-15, -15)$ cm, 50°

The third multiple misalignment error is horizontal misalignment in $(-x, -y)$ with a non-uniform air gap in the Z-axis direction with rotation around the Y- axis as shown in table 3-6.

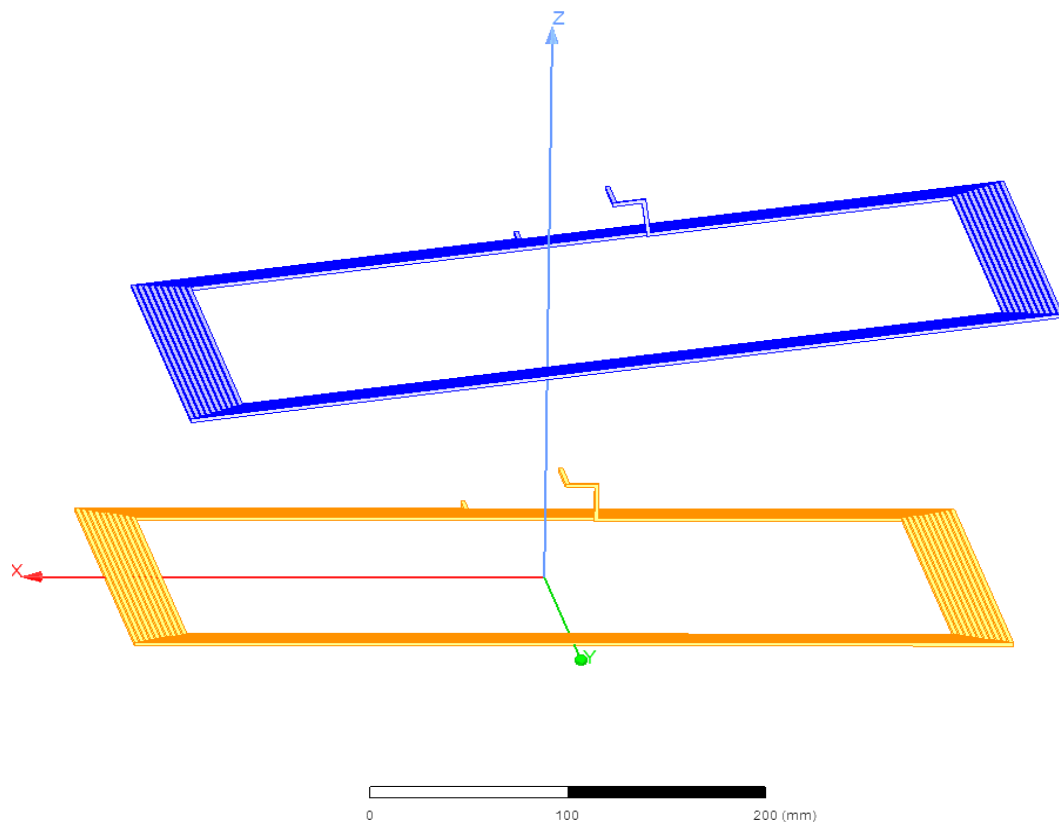
Table 3-6: Multiple misalignment error data in case of horizontal misalignment in (-x, -y) with a non-uniform air gap in the Z-axis direction with rotation around the Y- axis

Case No.	Non-uniform	Horizontal (cm)
1	5°	(-5, -5)
2	8°	(-5, -5)
3	5°	(-15, -15)
4	8°	(-15, -15)

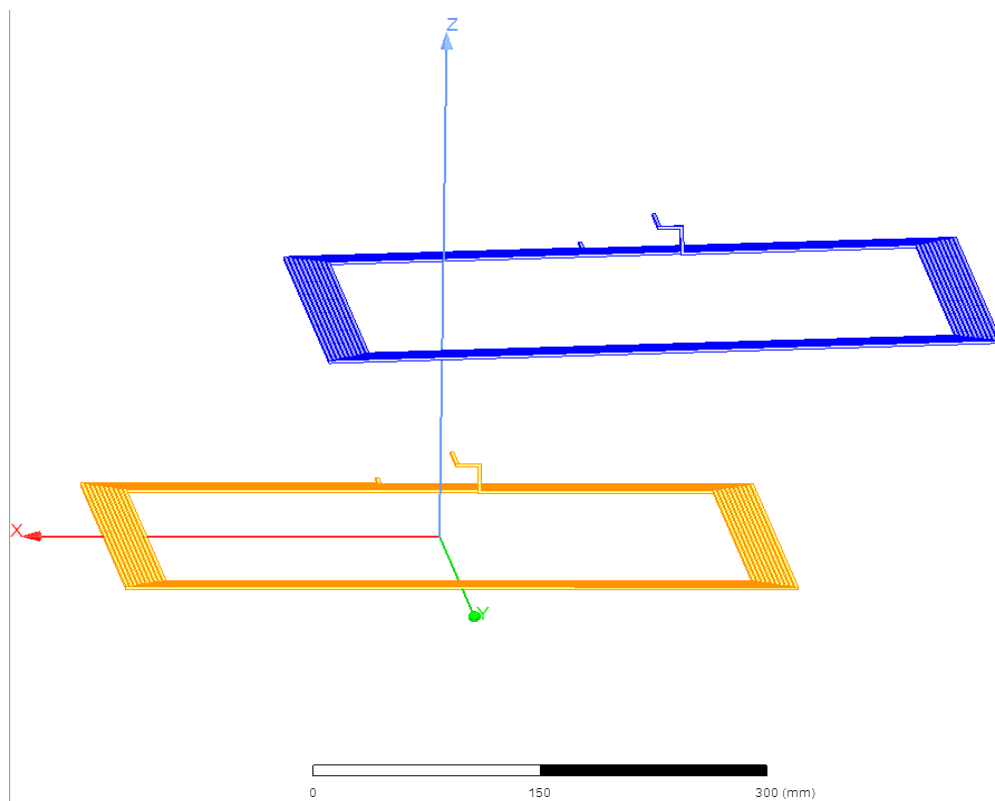
Figure 3-12 shows multiple misalignment error positions since one side of the receiver coil moving downward with an angle that exists between them concerning the Y-axis, also it moves in different positions in (-x, -y).



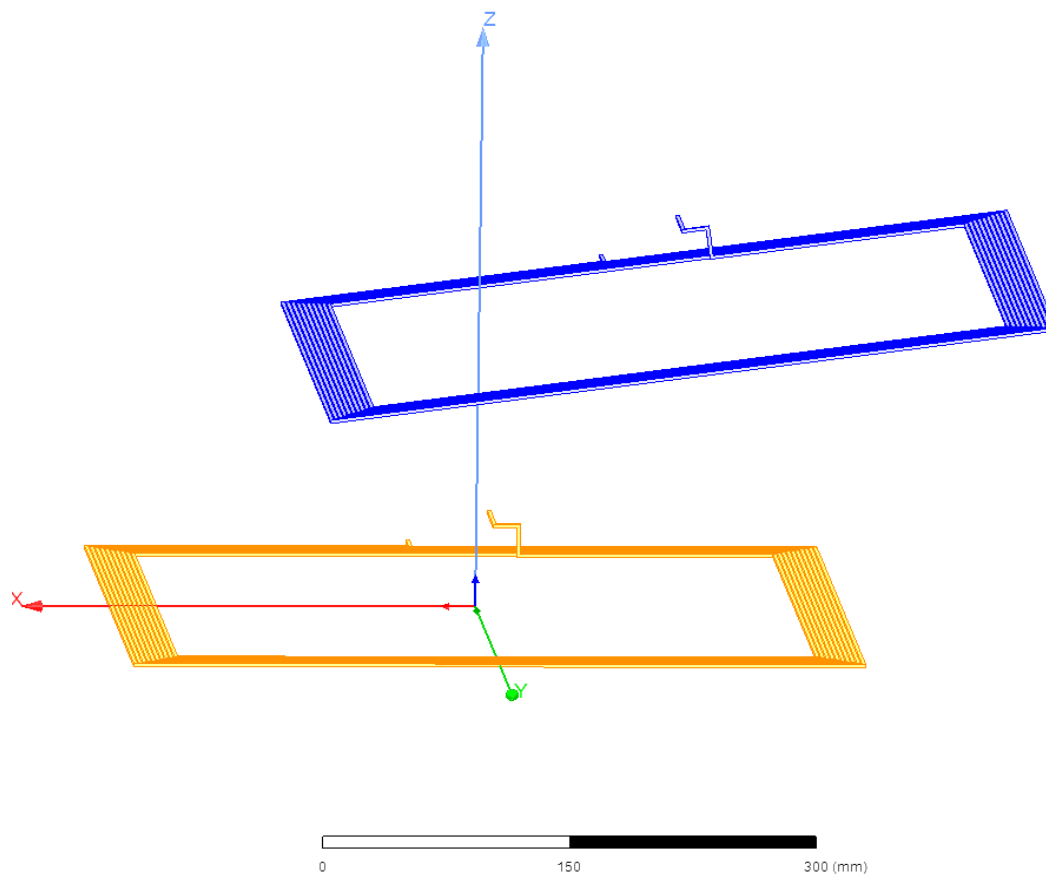
Case 1



Case 2



Case 3



Case 4

Figure 3-12: Multiple misalignment error positions in case of horizontal misalignment in $(-x, -y)$ with a non-uniform air gap in the Z-axis direction with rotation around the Y-axis

Case 1) $(-5, -5)$ cm, 5°
Case 3) $(-15, -15)$ cm, 5°

Case 2) $(-5, -5)$ cm, 8°
Case 4) $(-15, -15)$ cm, 8°

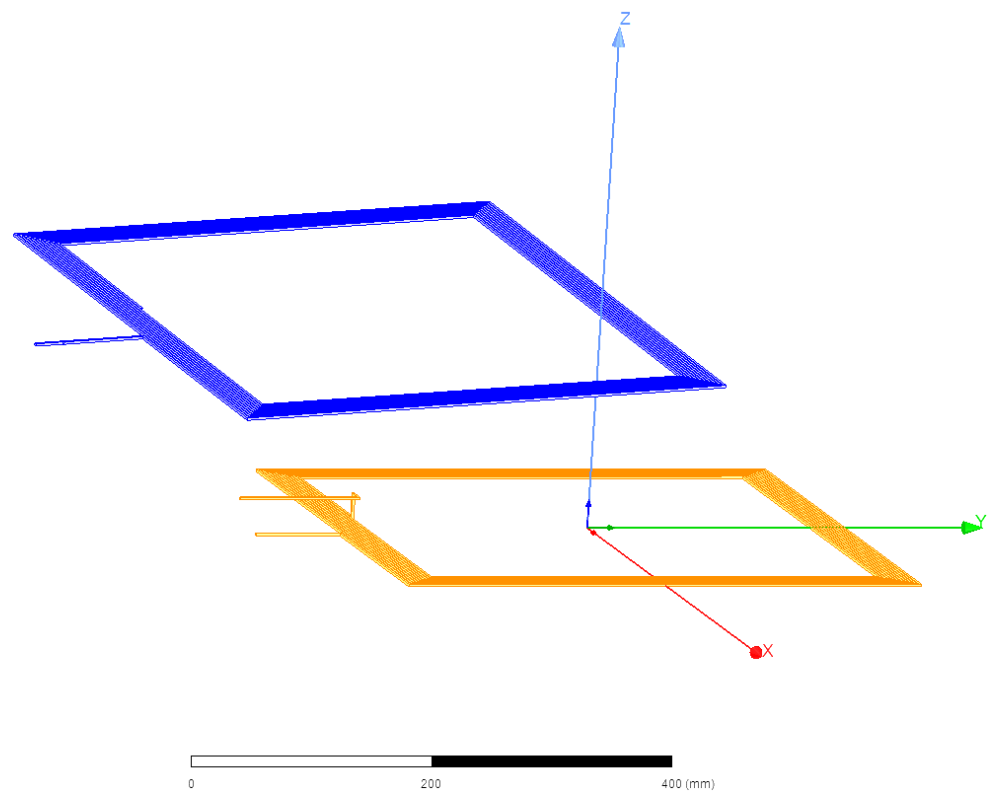
In this final section, we will examine the most extreme cases of multiple misalignments that may occur during the charging of an electric vehicle, particularly when three types of misalignments happen simultaneously.

The final case of multiple misalignment errors is a horizontal misalignment in (-x, -y) direction, combined with a non-uniform air gap in the Z-axis direction with rotation around the Y-axis and Z-axis that identified in table 3-7.

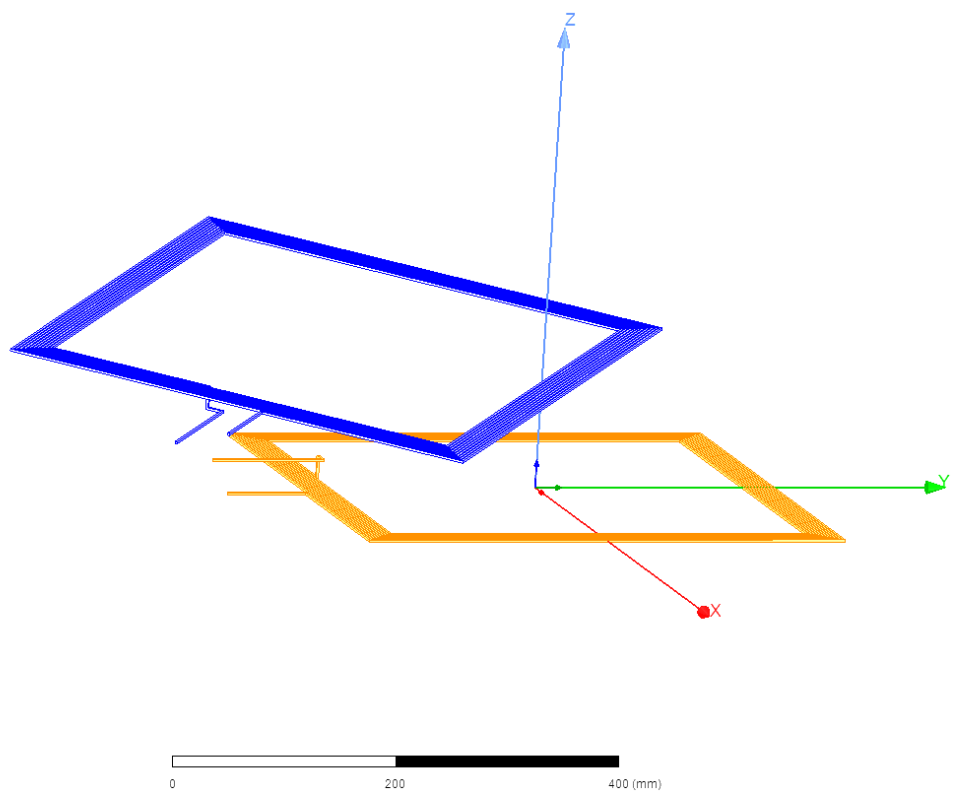
Table 3-7: Multiple misalignment errors is a horizontal misalignment in (-x, -y) direction, combined with a non-uniform air gap in the Z-axis direction with rotation around the Y-axis and Z-axis

Case No.	Non-uniform air gap (D°)	Horizontal (cm)	Around Z-axis (D°)
1	8	(-15, -15)	10
2	8	(-15, -15)	50
3	8	(-10, -12)	10
4	8	(-10, -12)	50
5	5	(-10, -12)	10
6	5	(-15, -15)	10

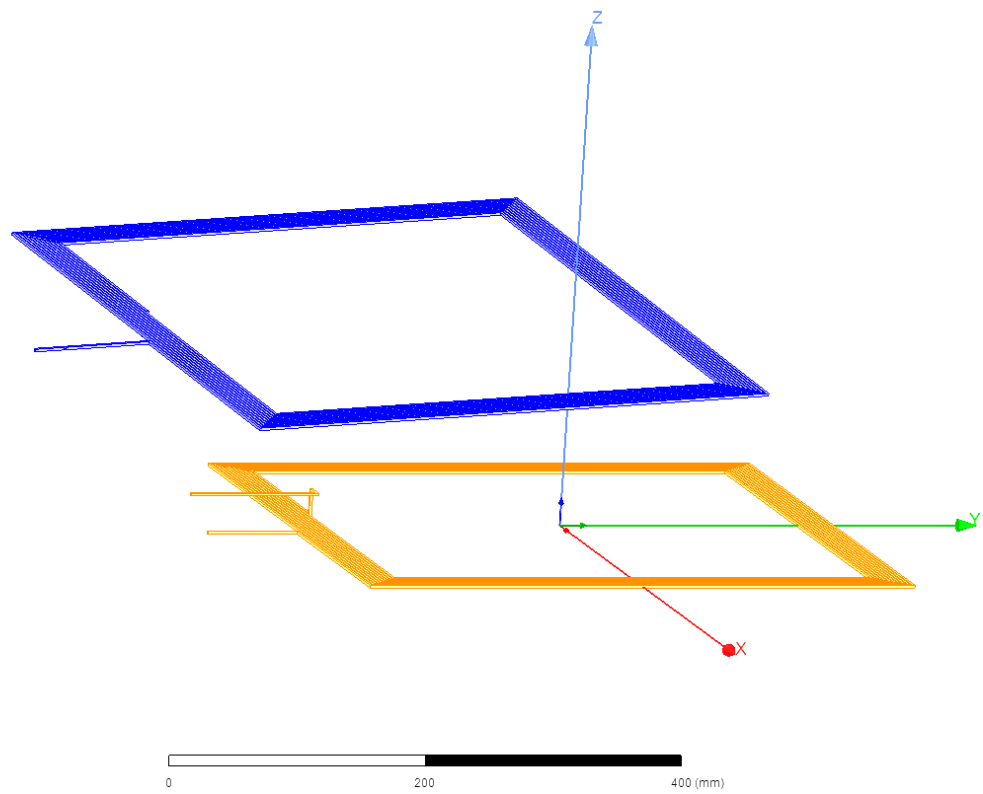
Fig 3-13 shows multiple misalignment error positions since the receiver coil moves horizontally with a non-uniform air gap to the transmitter coil in different positions with an angle existing between them concerning the Y-axis and Z-axis.



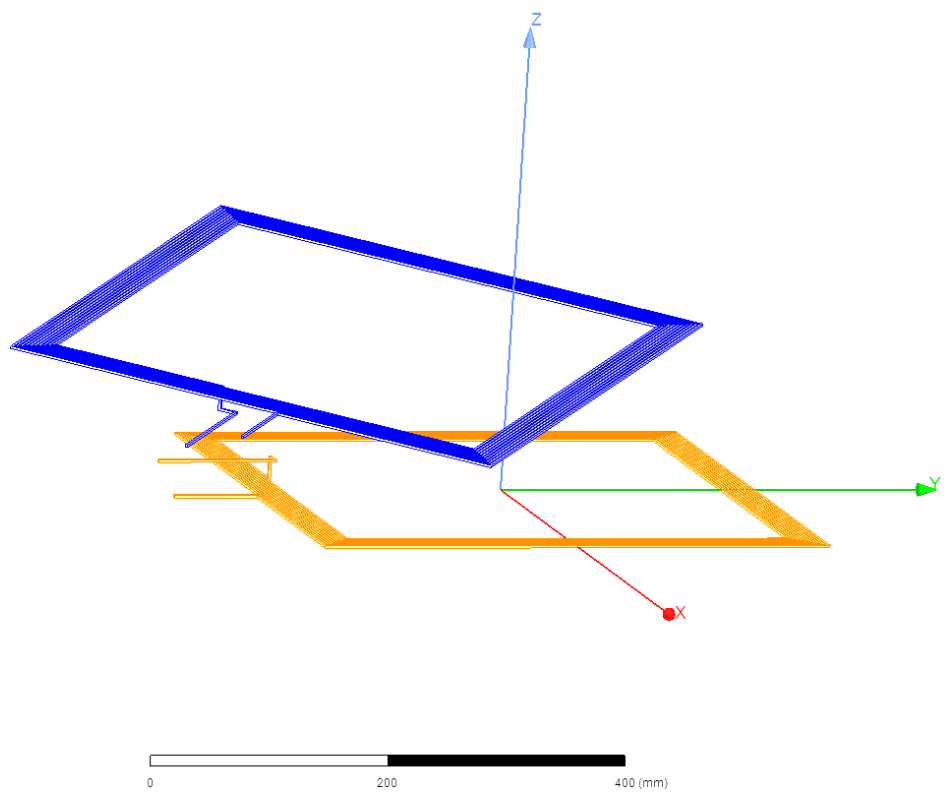
Case 1



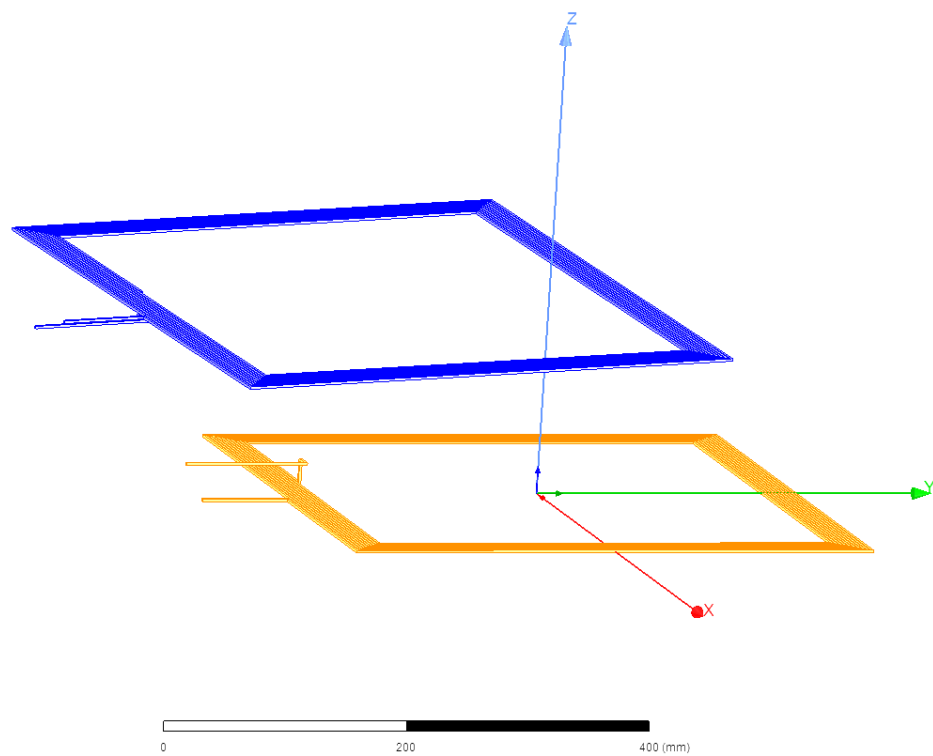
Case 2



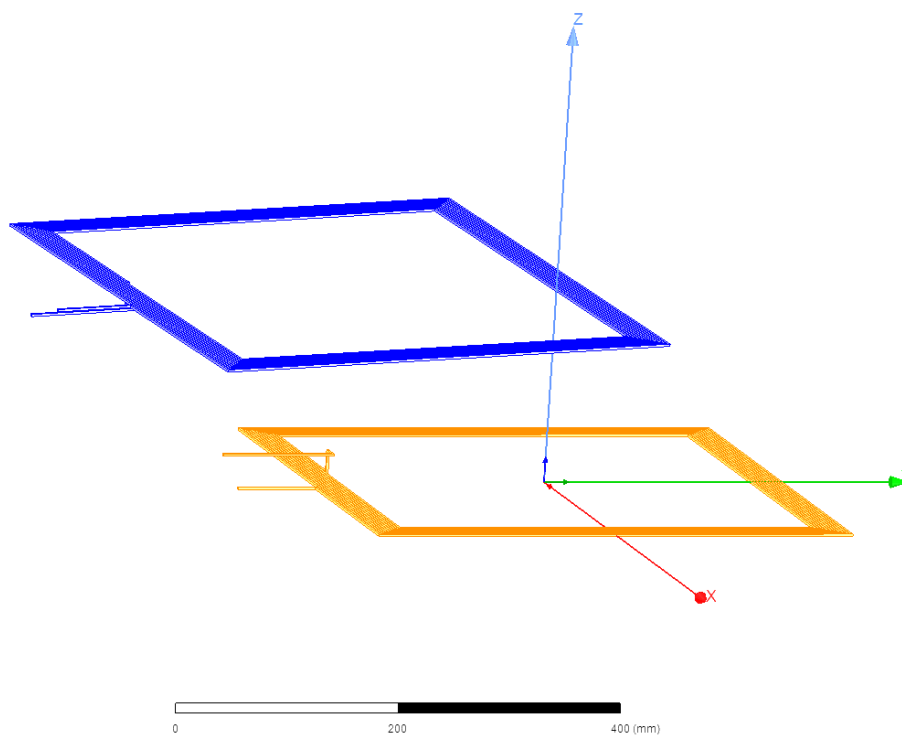
Case 3



Case 4



Case 5



Case 6

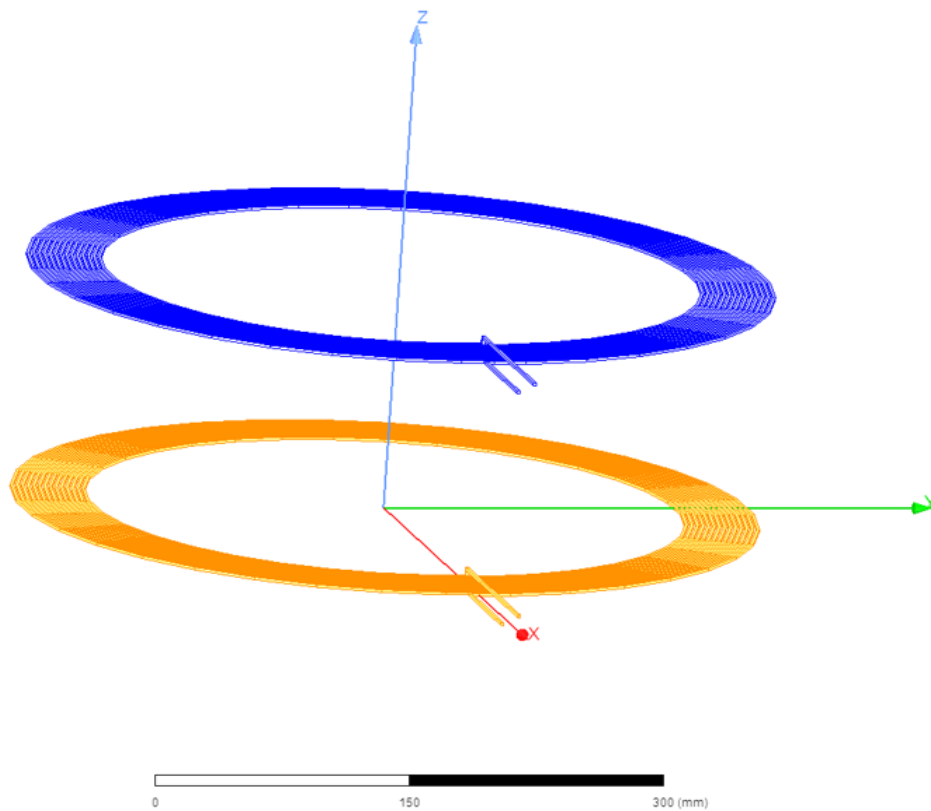
Figure 3-13: Multiple misalignment errors is a horizontal misalignment in (-x, -y) direction, combined with a non-uniform air gap in the Z-axis direction with rotation around the Y-axis and Z-axis

Finally, given that the misalignment issue is the most significant problem, as mentioned earlier, we will conduct an extensive study of the different shapes of the coils. We will compare them in terms of horizontal misalignment scenarios as shown in table 3-8 to identify the most suitable design for the WPT system.

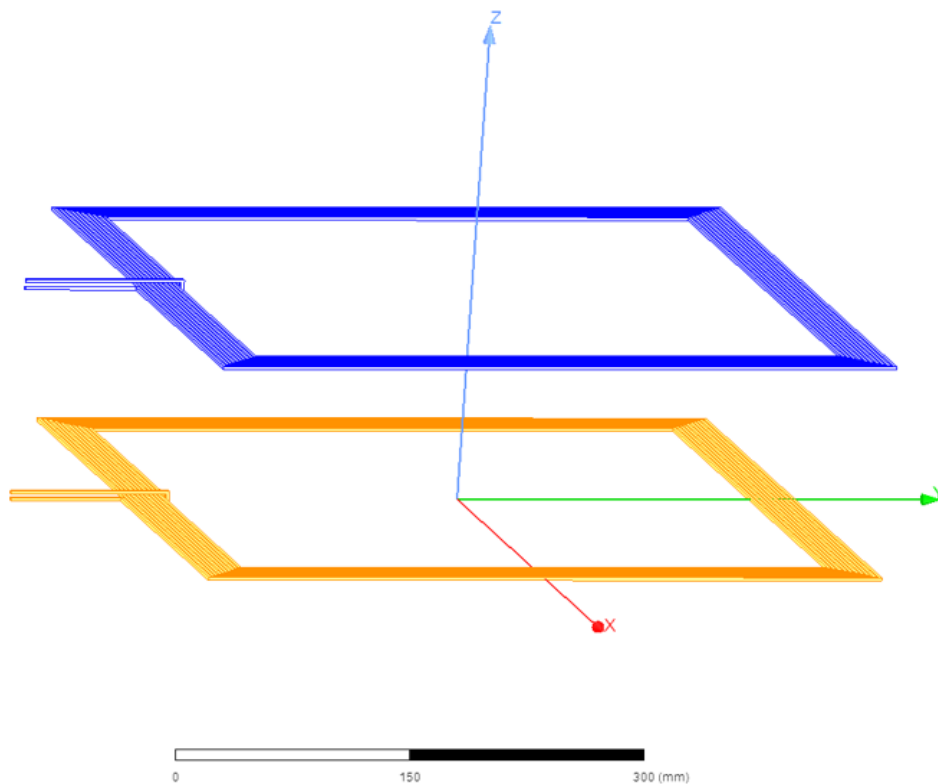
Table 3-8: The models used in design for WPT system

1. Aligned at 15 cm vertical air gap				
	a	b	c	d
Transmitter	Circle	Square	Circle	Square
Receiver	Circle	Square	Square	Circle
2. Horizontal misalignment of 15 cm in X-axis				
	a	b	c	d
Transmitter	Circle	Square	Circle	Square
Receiver	Circle	Square	Square	Circle
3. Horizontal misalignment of 25 cm in X-axis				
	a	b	c	d
Transmitter	Circle	Square	Circle	Square
Receiver	Circle	Square	Square	Circle

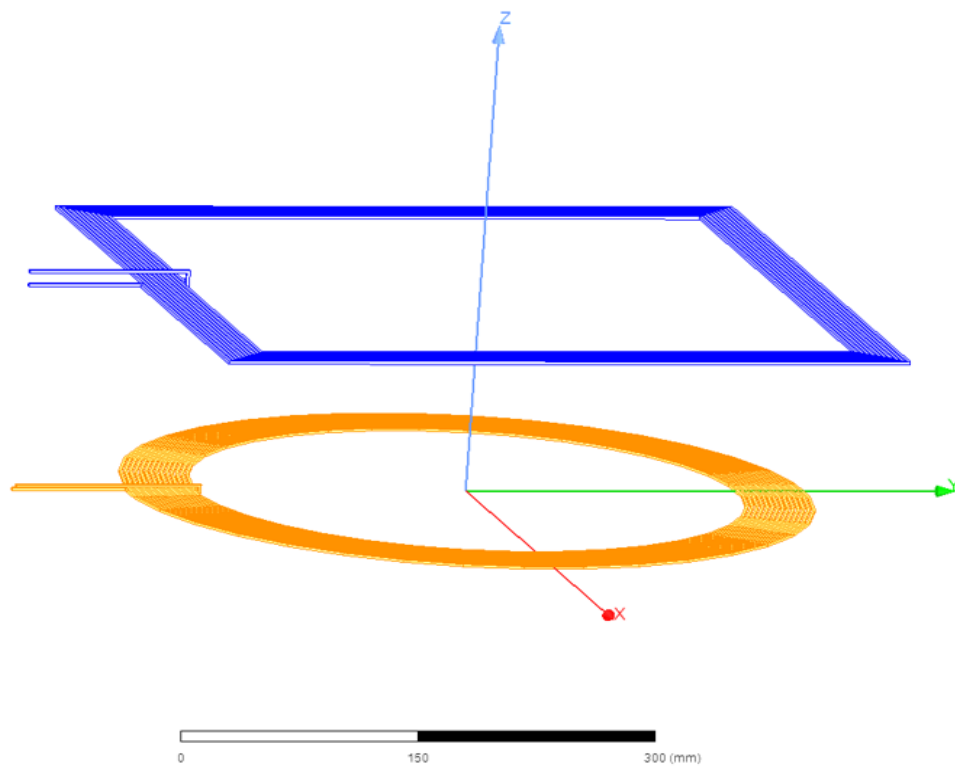
Figure 3-14 Identify the possible different models used in design for WPT system.



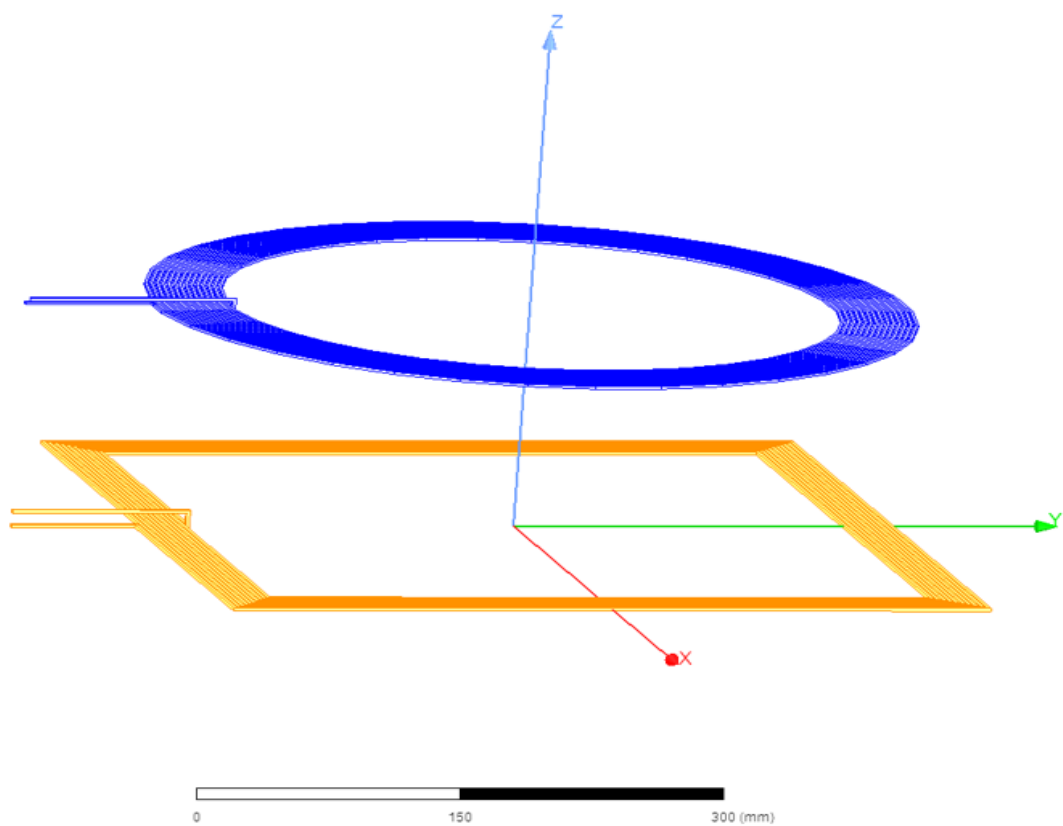
(a)



(b)

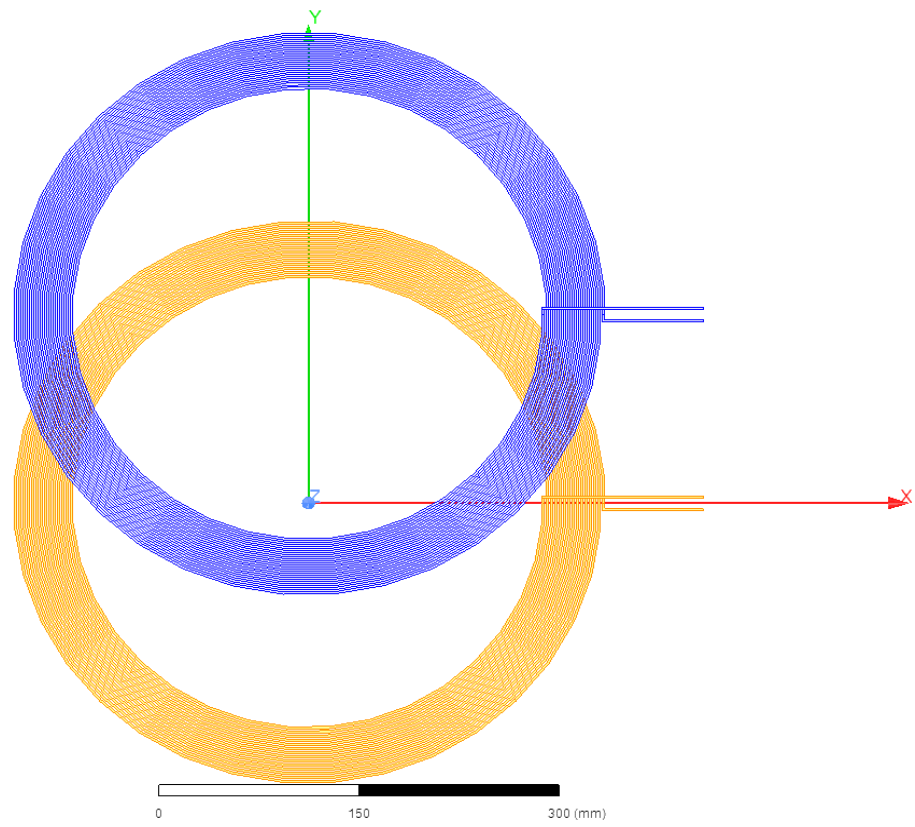


(c)

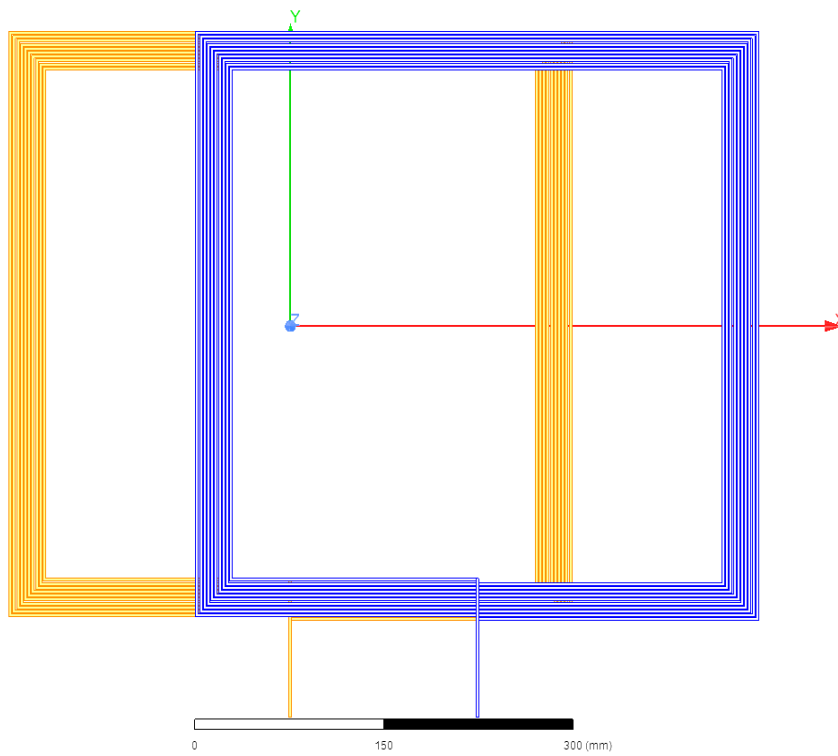


(d)

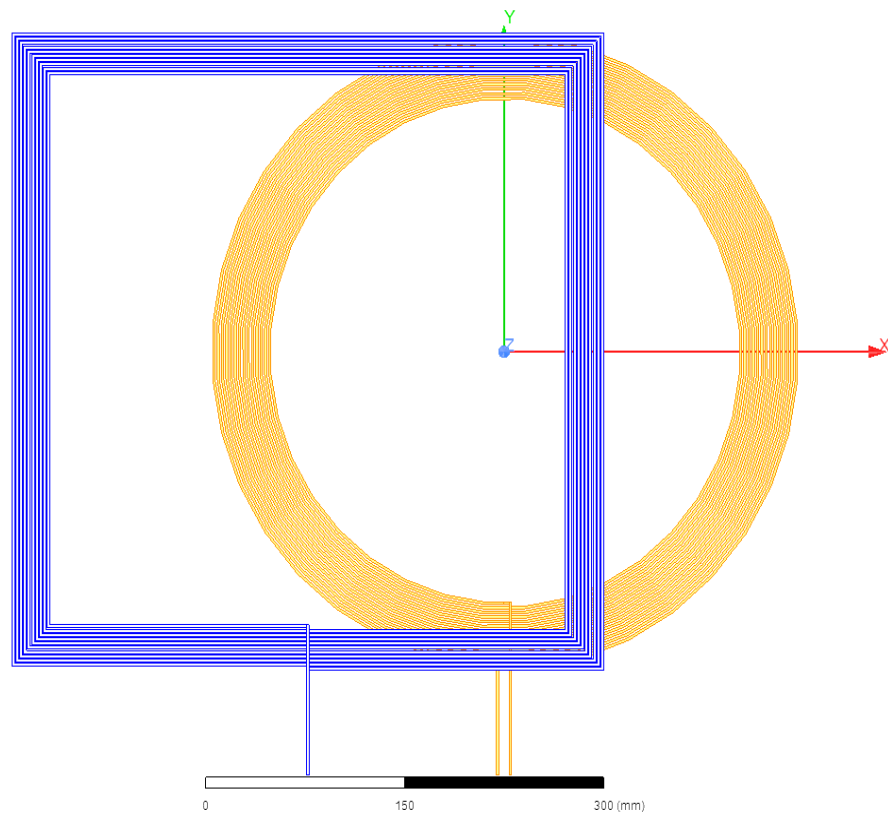
1. Aligned at 15 cm vertical air gap



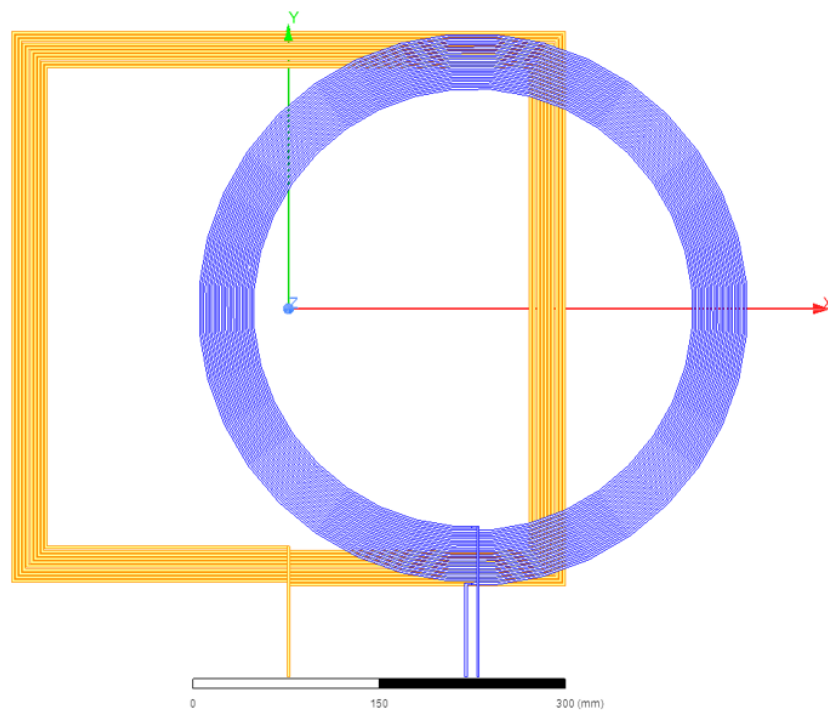
(a)



(b)

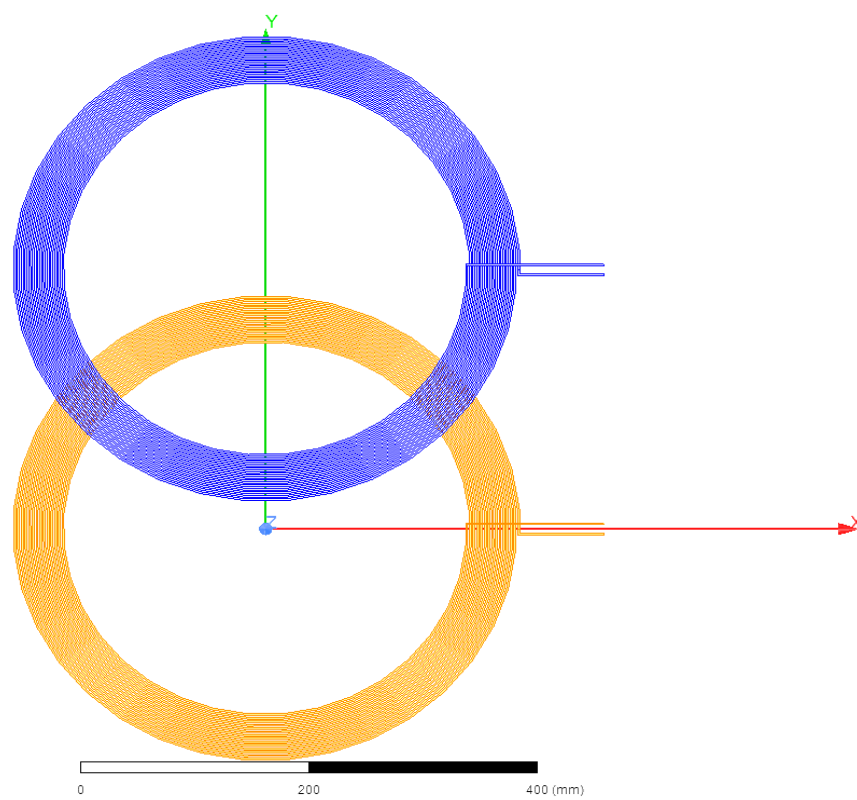


(c)

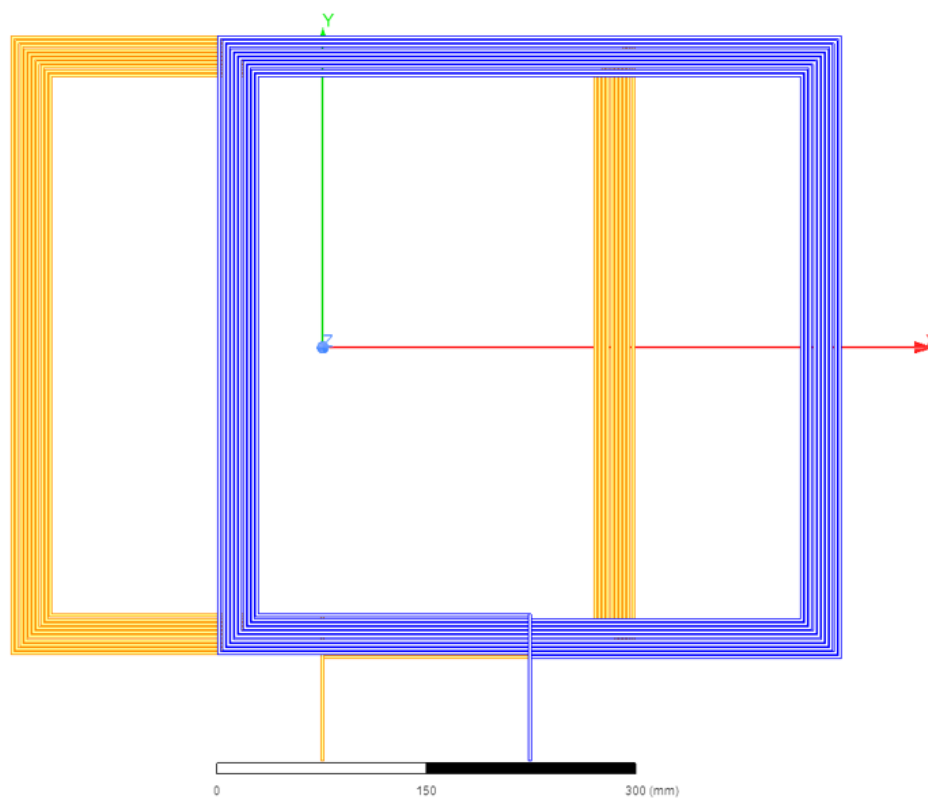


(d)

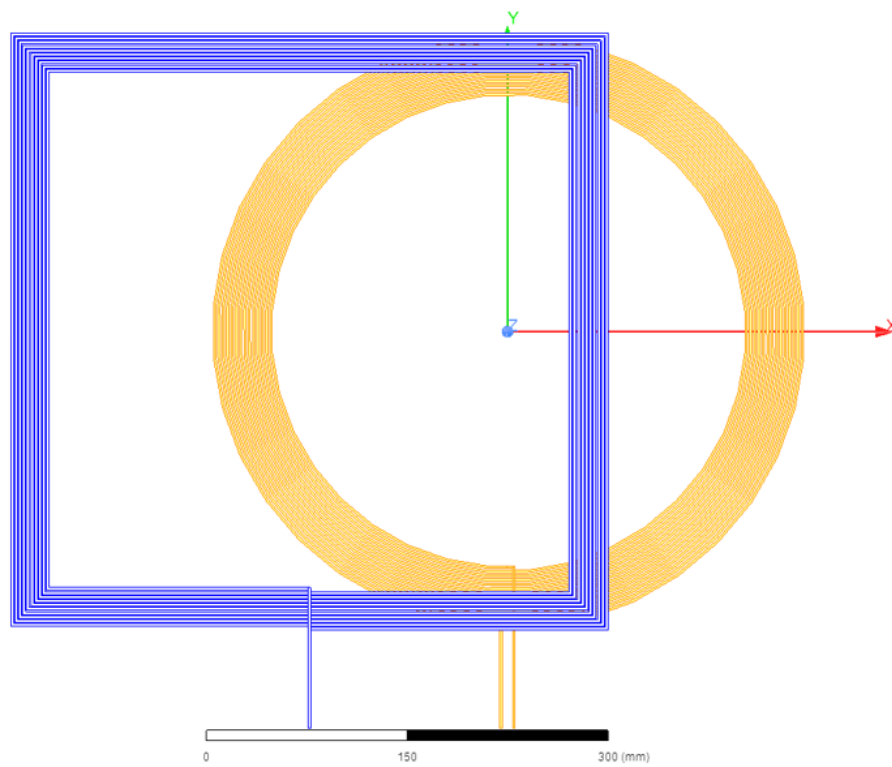
2.Horizontal misalignment of 15 cm in X-axis



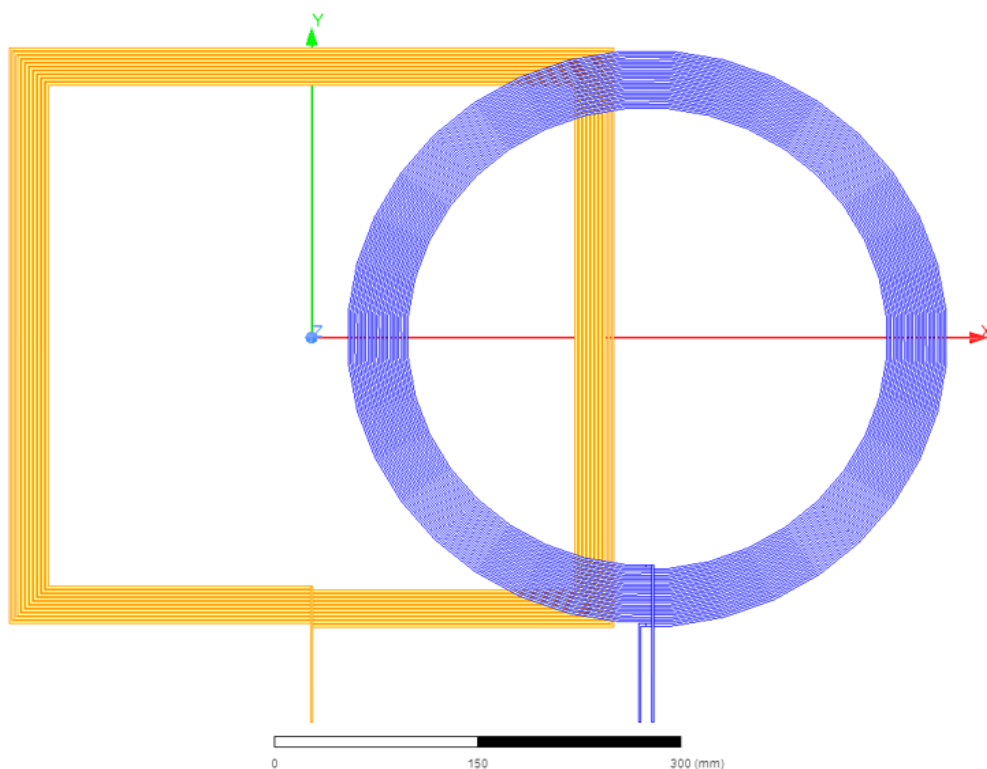
(a)



(b)



(c)



(d)

3.Horizontal misalignment of 25 cm in X-axis

Figure 3-14: Different models used in design for WPT system

3.2.2 Simulation with MATLAB

The values of self-inductances for both the transmitting and receiving coils (L_t and L_r), mutual inductance (M) and coupling coefficient (K) were obtained using ANSYS Maxwell 3D.

Depending on the number of turns and coil dimensions, equations were used in MATLAB code to obtain the values of both resistances and self-inductances of both the transmitting and receiving coils, and depending on the number of turns, coil dimensions and the air gap between the coils the mutual inductance was calculated, and using the values of both the self-inductances and mutual inductance, the value of coupling coefficient was calculated.

After confirming the design values and comparing them with ANSYS Maxwell 3D, MATLAB code was used also to calculate the compensation values, output power and the efficiency of the system and calculate the values of the mutual inductance, coupling coefficient and the output power also when there is a misalignment between the transmitting and receiving coils.

MATLAB®/Simulink is used to simulate the wireless power transfer system and to verify the output power and the efficiency of the system, the simulation model is shown in Figure 3-15.

First the AC voltage is converted to dc voltage by using a rectifier. After that the dc voltage is converted to high frequency (HF) ac voltage using an inverter and the block called mutual inductance then is used in MATLAB as an alternative to the two coils.

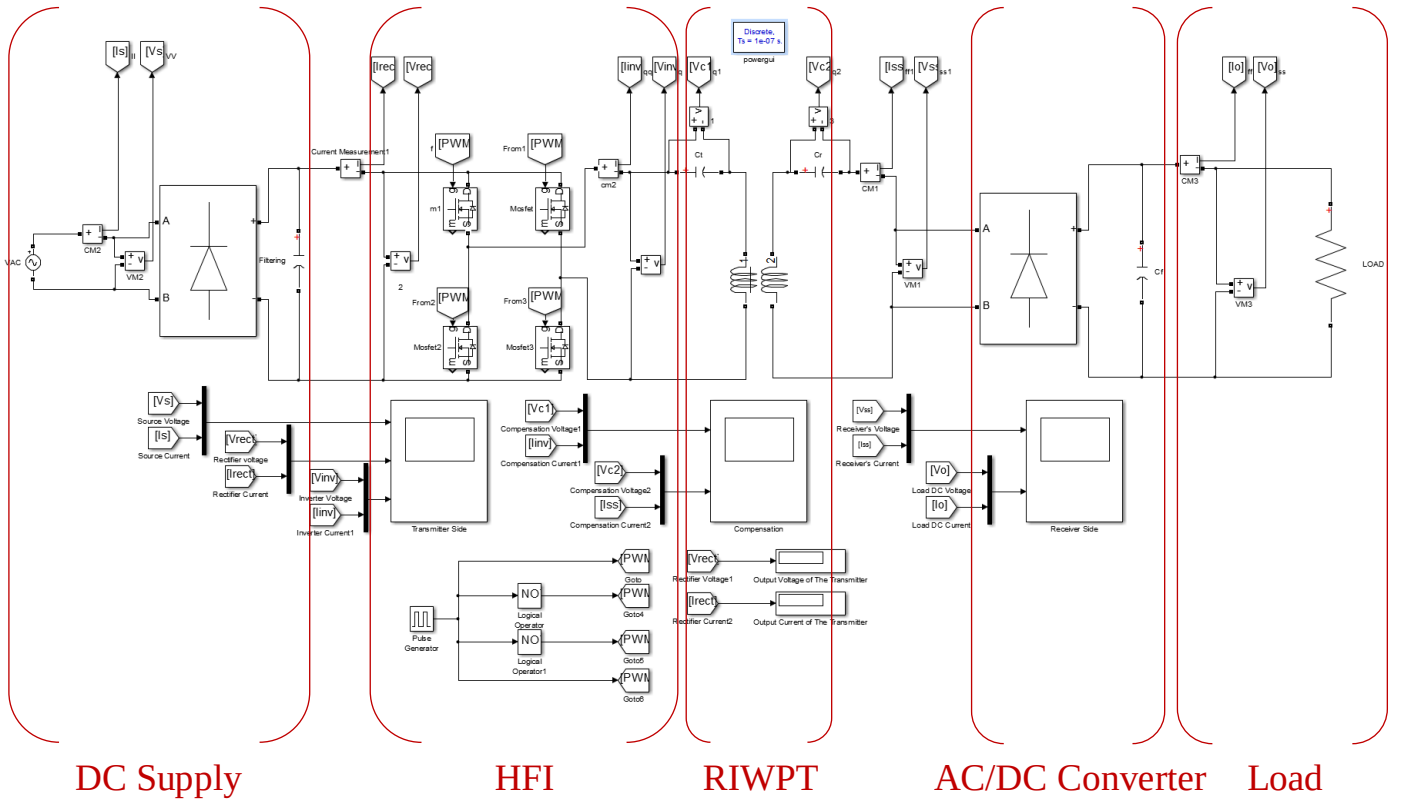


Figure 3-15: MATLAB Simulink RIWPT system

The specifications of the components used in MATLAB are shown in table 3-9.

Table 3-9: Parameters values in MATLAB

Diodes	Value	MOSFETs	Value	Pulse generator	Value
R_{on}	0.001 Ω	R_{on}	0.1 Ω	Amplitude	1
V_f	0.8 V	R_d	0.01 Ω	Period	1/83500
C_s	250 nF	C_s	∞	Pulse Width - Phase delay	50% - 0

Figure 3-16 shows the m-file and the function file for calculating the half of side lengths of both coils.

```
clear all;
clc;

Tt=input('No. of turns of the transmitting coil = ');
Tr=input('No. of turns of the receiving coil = ');
So=input('The surface area of each turn in mm^2= ');

[rt,rr]=HalfSide(Tt,Tr,So);

fprintf('.....\n');
fprintf('The half of side length of the transmitting coil = %0.2f mm\n',rt);
fprintf('The half of side length of the receiving coil = %0.2f mm\n',rr);
```

```
function [rt,rr]=HalfSide(Tt,Tr,So)
rt=Tt*sqrt(So/pi);
rr=Tr*sqrt(So/pi);
end
```

Figure 3-16: The m-file and the function file for calculating the half of side lengths of both coils

Figure 3-17 shows the m-file and the function file for calculating the DC resistances of both coils.

```
clear all;
clc;

Tt=input('No. of turns of the transmitting coil = ');
Tr=input('No. of turns of the receiving coil = ');
Si=input('The surface copper area of each turn in mm^2 = ');
wt=input('The outer width of the transmitting coil in mm = ');
wr=input('The outer width of the receiving coil in mm = ');
lt=input('The outer length of the transmitting coil in mm = ');
lr=input('The outer length of the receiving coil in mm = ');

[Rt,Rr]=Resistance(Tt,Tr,Si,wt,wr,lt,lr);

fprintf('.....\n');
fprintf('The DC resistance of the transmitting coil = %0.3f Ohm\n',Rt);
fprintf('The DC resistance of the receiving coil = %0.3f Ohm\n',Rr);
```

```
function [Rt,Rr]=Resistance(Tt,Tr,Si,wt,wr,lt,lr)
Rt=((1.724e-8)*Tt^2*(wt+lt)/Si)*1000;
Rr=((1.724e-8)*Tr^2*(wr+lr)/Si)*1000;
end
```

Figure 3-17: The m-file and the function file for calculating the DC resistances of both coils

Figure 3-18 shows the m-file and the function file for calculating the self-inductances of both coils without using ferrite.

```
clear all;
clc;

Tt=input('No. of turns of the transmitting coil = ');
Tr=input('No. of turns of the receiving coil = ');
So=input('The surface area of each turn in mm^2 = ');
wt=input('The outer width of the transmitting coil in mm = ');
wr=input('The outer width of the receiving coil in mm = ');
lt=input('The outer length of the transmitting coil in mm = ');
lr=input('The outer length of the receiving coil in mm = ');

[rt,rr]=HalfSide(Tt,Tr,So);
[Lt,Lr]=Self(Tt,Tr,rt,rr,wt,wr,lt,lr);

fprintf('.....\n');
fprintf('The self-inductance of the transmitting coil = %0.3f microH\n',Lt);
fprintf('The self-inductance of the receiving coil = %0.3f microH\n',Lr);
```

```
function [Lt,Lr]=Self(Tt,Tr,rt,rr,wt,wr,lt,lr)
at=wt*log( (2*wt*lt) / (rt*(wt+sqrt(((wt)^2)+((lt)^2)))) );
bt=lt*log( (2*wt*lt) / (rt*(lt+sqrt(((wt)^2)+((lt)^2)))) );
ct=wt+lt-sqrt(((wt)^2)+((lt)^2));

ar=wr*log( (2*wr*lr) / (rr*(wr+sqrt(((wr)^2)+((lr)^2)))) );
br=lr*log( (2*wr*lr) / (rr*(lr+sqrt(((wr)^2)+((lr)^2)))) );
cr=wr+lr-sqrt(((wr)^2)+((lr)^2));

Lt=((4*(10^(-7)))*(Tt^2)*(at+bt-2*ct+0.25*(wt+lt))))*1000;
Lr=((4*(10^(-7)))*(Tr^2)*(ar+br-2*cr+0.25*(wr+lr))))*1000;
end
```

Figure 3-18: The m-file and the function file for calculating the self-inductances of both coils without using ferrite

Figure 3-19 shows the m-file and the function file for calculating the mutual inductance between both coils without using ferrite.

```
clear all;
clc;

Tt=input('No. of turns of the transmitting coil = ');
Tr=input('No. of turns of the receiving coil = ');
u=input('The square side in mm = ');
h=input('The gap distance between the two coils in mm = ');

M=Mutual(Tt,Tr,u,h);

fprintf('.....\n');
fprintf('The mutual inductance = %0.2f microH\n',M);
```

```
function M=Mutual(Tt,Tr,u,h)
term1=u*sqrt((h^2)+(u^2));
term2=(u^2)+(h^2);
term3=(u*h)+h*sqrt((h^2)+2*(u^2));
term4=h-2*sqrt((h^2)+(u^2))+sqrt((h^2)+2*(u^2));

M=(8*(10^-7)*Tt*Tr*(u*log((term1+term2)/term3)+term4))*1000;
end
```

Figure 3-19: The m-file and the function file for calculating the mutual inductance between both coils without using ferrite

Figure 3-20 shows the m-file and the function file for calculating the coupling coefficient between both coils without using ferrite.

```
clear all;
clc;

Lt=input('The self-inductance of the transmitting coil in microH = ');
M=input('The Mutual inductance in microH = ');
Lr=input('The self-inductance of the receiving coil in microH = ');

K=Coupling(Lt,M,Lr);

fprintf('.....\n');
fprintf('The coupling coefficient = %0.3f \n',K);
```

```
function K=Coupling(Lt,M,Lr)
K=M/sqrt(Lt*Lr);
end
```

Figure 3-20: The m-file and the function file for calculating the coupling coefficient between both coils without using ferrite

Figure 3-21 shows the m-file and the function file for calculating the capacitances of the transmitting and the receiving circuits without using ferrite.

```
clear all;
clc;

Lt=input('The self-inductance of the transmitting coil in microH = ');
Lr=input('The self-inductance of the receiving coil in microH = ');
f=input('The resonance frequency in Hz = ');

[Ct,Cr]=Compensation(Lt,Lr,f);

fprintf('.....\n');
fprintf('The capacitance of the transmitting circuit = %0.3f nF\n',Ct);
fprintf('The capacitance of the receiving circuit = %0.3f nF\n',Cr);
```

```
function [Ct,Cr]=Compensation(Lt,Lr,f)
Ct=((1/(4*(pi^2)*(f^2)*Lt)))*(10^15);
Cr=((1/(4*(pi^2)*(f^2)*Lr)))*(10^15);
end
```

Figure 3-21: The m-file and the function file for calculating the capacitances of the transmitting and the receiving circuits without using ferrite

Figure 3-22 shows the m-file and the function file for calculating the RMS transmitting current with using ferrite.

```
clear all;
clc;

V1o=input(' The output HF AC voltage at resonance in V = ');
RLeq=input(' The equivalent AC resistive load in Ohm = ');
Rr=input(' The DC resistance of the receiving coil in Ohm = ');
Rt=input(' The DC resistance of the transmitting coil in Ohm = ');
f=input(' The resonance frequency in Hz = ');
M=input(' The mutual inductance in H = ');

It=TransmitterCurrent(V1o,RLeq,Rr,Rt,f,M);

fprintf(' ..... \n');
fprintf('The RMS transmitting current = %0.3f A\n',It);
```

```
function It=TransmitterCurrent(V1o,RLeq,Rr,Rt,f,M)
w=2*pi*f;
It=(V1o*(RLeq+Rr))/((Rt*(RLeq+Rr))+(w*M)^2);
end
```

Figure 3-22: The m-file and the function file for calculating the RMS transmitting current with using ferrite

Figure 3-23 shows the m-file and the function file for calculating the RMS receiving current with using ferrite.

```
clear all;
clc;

f=input(' The resonance frequency in Hz = ');
M=input(' The mutual inductance in H = ');
It=input(' The RMS transmitting current in A = ');
RLeq=input(' The equivalent AC resistive load in Ohm = ');
Rr=input(' The DC resistance of the receiving coil in Ohm = ');

Ir=ReceiverCurrent(It,f,M,RLeq,Rr);

fprintf('.....\n');
fprintf('The RMS receiving current = %0.3f A\n',Ir);
```

```
function Ir=ReceiverCurrent(It,f,M,RLeq,Rr)
w=2*pi*f;
Ir=(w*M*It)/(RLeq+Rr);
end
```

Figure 3-23: The m-file and the function file for calculating the RMS receiving current with using ferrite

Figure 3-24 shows the m-file and the function file for calculating the RMS power delivered to the equivalent AC resistive load with using ferrite.

```
clear all;
clc;

It=input('The RMS transmitting current in A = ');
f=input(' The resonance frequency in Hz = ');
M=input(' The mutual inductance in microH = ');
Rr=input(' The DC resistance of the receiving coil in Ohm = ');
RLeq=input(' The equivalent AC resistive load in Ohm = ');

P=Power(It,f,M,Rr,RLeq);

fprintf('.....\n');
fprintf('The RMS power delivered to the equivalent AC resistive load = %0.2f W\n',P);
```

```
function P=Power(It,f,M,Rr,RLeq)
w=2*pi*f;
P=(((It*w*M)^2)/(Rr+RLeq))*(10^-12);
end
```

Figure 3-24: The m-file and the function file for calculating the RMS power delivered to the equivalent AC resistive load with using ferrite

Figure 3-25 shows the m-file and the function file for calculating the efficiency while using the SS compensation with using ferrite.

```
clear all;
clc;

f=input('The resonant frequency in Hz = ');
M=input('The mutual inductance in microH = ');
Rt=input('The DC resistance of the transmitting coil in Ohm = ');
Rr=input('The DC resistance of the receiving coil in Ohm = ');
RLeq=input('The equivalent AC resistive load in Ohm = ');

E=Efficiency(f,M,Rt,Rr,RLeq);

fprintf('.....\n');
fprintf('The efficiency while using SS compensation = %0.2f %% \n',E);
```

```
function E=Efficiency(f,M,Rt,Rr,RLeq)
w=2*pi*f;
term1=Rt*((Rr+RLeq)^2);
term2=(w*M)^2;
E=(RLeq/(RLeq+Rr+(term1/term2)))*100;
end
```

Figure 3-25: The m-file and the function file for calculating the efficiency while using the SS compensation with using ferrite

3.2.3 WPT system's costs

Table 3-10: Cost of the HFI

Component	Quantity	Price (each)	Taxes (L.E)	Shipping (L.E)	Total price (L.E)
G3R40MT12D SiC Mosfet [20]	4	131.91	75	290.19	1025
Aluminum Heatsink 100*150*20 [21]	1	260	—	70	330
PCB Prototype Board [22]	1	18	—	—	18
PCB for Inverter [23]	1	800	—	100	900
STM32F401 microcontroller [24]	1	320	—	—	320
ST-LINK V2 Emulator [25]	1	220	—	—	220
3.3V voltage adapter [26]	1	86	—	—	86
IR2110 Gate driver [27]	2	95	—	70	260
UF4007 Fast recovery diode [28]	6	1	—	—	6
Carbon Resistor 13Ω 0.25W [29]	10	0.15	—	—	1.5
22uF 160V Capacitor [30]	2	1.5	—	—	3
0.22uF 50V Capacitor [31]	3	0.75	—	—	2.25
490nF 275V Ceramic Capacitor [32]	4	2	—	—	8
Breadboard [33]	1	40	—	—	40
5.6k 0.25W ohm Resistor [34]	10	0.15	—	—	1.5
6N135 Optocoupler [35]	2	30	—	—	60
220V to 12Vdc converter [36]	3	65	—	—	195
Terminal block 2 pins [37]	5	3.25	—	—	16.25
Red Led [38]	1	0.5	—	—	0.5
Jumper Wire [39]	30	0.75	—	—	22.5
Total Price:					3515.5

Table 3-11: Cost of the transmitter's compensation

Component	Quantity	Price (each)	Taxes (L.E)	Shipping (L.E)	Total price (L.E)
0.33uF Resonance Film Capacitor [40]	14	40	—	70	630
PCB prototype board [41]	1	32	—	—	32
Total Price:					662

Table 3-12: Cost of the induction coils

Component	Quantity	Price (each)	Taxes (L.E)	Shipping (L.E)	Total price (L.E)
Litz Wire 0.1*600 strands (20m) [42]	2	2100	4477.94	7000	15677.94
Iclyric Board 65*65cm	4	822.5	—	—	3290
Iclyric Rod 20cm	4	167.5	—	—	670
PC40 Mn-Zn Soft Ferrite core 4 pieces 60*16*8 [43]	6	425	1580	Free	4130
Total Price:					23767.94

Table 3-13: Cost of the receiver's compensation

Component	Quantity	Price (each)	Taxes (L.E)	Shipping (L.E)	Total price (L.E)
0.33uF Resonance Film Capacitor [40]	14	40	—	70	630
PCB prototype board [41]	1	32	—	—	32
Total Price:					662

Table 3-14: Cost of the AC/DC converter

Component	Quantity	Price (each)	Taxes (L.E)	Shipping (L.E)	Total price (L.E)
220uF 450V Capacitor [44]	3	35	—	70	175
SEI60-10A Diode 1000V 60A [45]	4	90	—	—	360
Secondary Rectifier PCB [46]	1	200	—	—	200
5.8ohm 5KW Load	1	2000	—	20	2020
Total Price:					2755

Table 3-15: Cost of other components

Component	Quantity	Price (each)	Taxes (L.E)	Shipping (L.E)	Total price (L.E)
Electrical Box PI65 (30*40*17) cm	1	1280	—	—	1280
Electrical Box PI65 (25*35*15) cm	1	680	—	—	680
DC Fan 12V 0.22A [47]	3	55	—	—	165
Male Blade crimp Terminal	100	1	—	—	100
Female Blade crimp Terminal	100	1	—	—	100
AC Power Socket 3 Pin [48]	1	15	—	—	15
PCB Male Terminals [49]	20	1	—	—	20
Stranded 6mm copper wire (5m)	2	50	—	—	100
Total Price:					2460

Table 3-16: Total production price for the project

Part	Total price (L.E)
HFI	3515.5
Transmitter's compensation	662
Induction coils	23767.94
Receiver's compensation	662
AC/DC converter	2755
Other components	2460
Total Price:	33822.44

Figure 3-26 shows the distribution of the production price for the project.

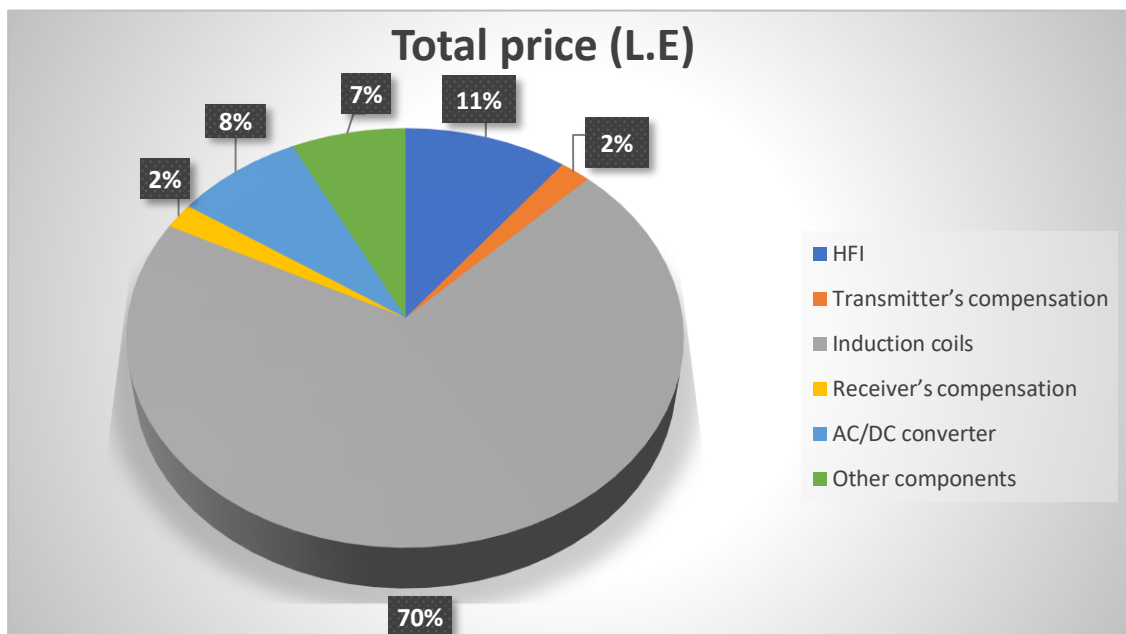


Figure 3-26: Pie chart showing the distribution of the production price for the project

3.3 Practical implementation

The real-life implementation of the proposed system differs from actual charging systems designed for EVs in that it's not designed to charge a battery but rather to provide the power capable of charging the battery in the receiving side, in normal conditions, this power would then be converted to the volts and amps necessary for the battery via a BMS and an on-board charging controller, for the sake of simplicity this step has been skipped and an AC/DC converter and a high power load will take place of the on-board charging system to demonstrate the power transferred. The components of the proposed system are demonstrated in 7 parts as shown in Figure 3-27 and will be discussed in the following steps.

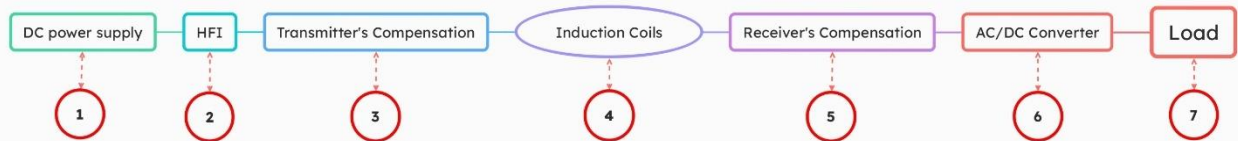


Figure 3-27: The components of the proposed system

1. DC Supply: the start of the system, where it provides the necessary power to be transferred from grid by converting 220v 50Hz AC to DC, rating for this project a 1kVA power supply has been selected as the DC source due to its safety of operation as well as flexibility in varying the output voltage for testing purposes.

The DC supply used in our prototype is shown in Figure 3- 28



Figure 3-28: The laboratory power supply

2. HFI: in order to obtain a high frequency signal for the transmission process, a high frequency inverter is developed, The DC current is turned into a square wave AC current via an H-Bridge MOSFET connection, the gate pulses provided to the MOSFET are through a half bridge drive (IR2110) which takes input from a separate micro-controller (STM32F401RCT6, Black Pill) which can be modified to output a high frequency pulse with the specified duty cycle.

Table 3-17: Gate driver specifications

Gate driver specifications	
Turn-on propagation delay	120 ns
Turn-off propagation delay	94 ns
Shutdown propagation delay	110 ns
Turn-on rise time	25 ns
Turn-off fall time	17 ns
Delay matching	10 ns
High side floating supply offset voltage	500 V
Turn-on propagation delay	120 ns

Table 3-18: STM specifications

STM specifications	
Operating voltage	1.7 to 3.6 V
Maximum CPU frequency	84 MHz
Flash memory	256 Kbytes
SRAM	64 Kbytes
Operating temperatures	Ambient temperatures: -40 to $+85$ °C/ -40 to $+105$ °C/ -40 to $+125$ °C Junction temperature: -40 to $+130$ °C

The MOSFETs are picked up on the basis of their switching speed and current rating. Ultra-fast switching (G3R40MT12D SiC) MOSFETs with ratings of 1200 V were picked as they work perfectly. Table 3-19 shows SiC MOSFET specifications.

Table 3-19: SiC MOSFET specifications

SiC MOSFET specifications	
Drain-Source Voltage $V_{ds(max)}$	1200 V
Continuous Forward Current	T=25°C $I_d=63A$ T=100°C $I_d=44A$ T=135°C $I_d=32A$
Turn on delay time	45 ns
Rise time	16 ns
Turn off delay time	19 ns
Fall time	11 ns
Drain-Source On resistance	40 mΩ
Gate Threshold Voltage	2.7 V

The optocoupler (6N135) is a vital component that ensures safe transmission of the pulses from the microcontroller to the gate driver, the STM's ground pin should be completely isolated from the rest of the circuit otherwise a fault will occur, given the very high frequency this inverter operates at, it is necessary that the optocoupler is capable of handling that frequency, the 6N135 has a rise time and fall time of 1.5 μ s and can handle switching of up to 300 kHz, that's why it was chosen for this operation.

Table 3-20: Optocoupler specifications

Optocoupler specifications	
Rise time	1.5 μ s
Fall time	1.5 μ s
Switching frequency	Up to 300 KHz

The HFI components are shown in Figure 3-29

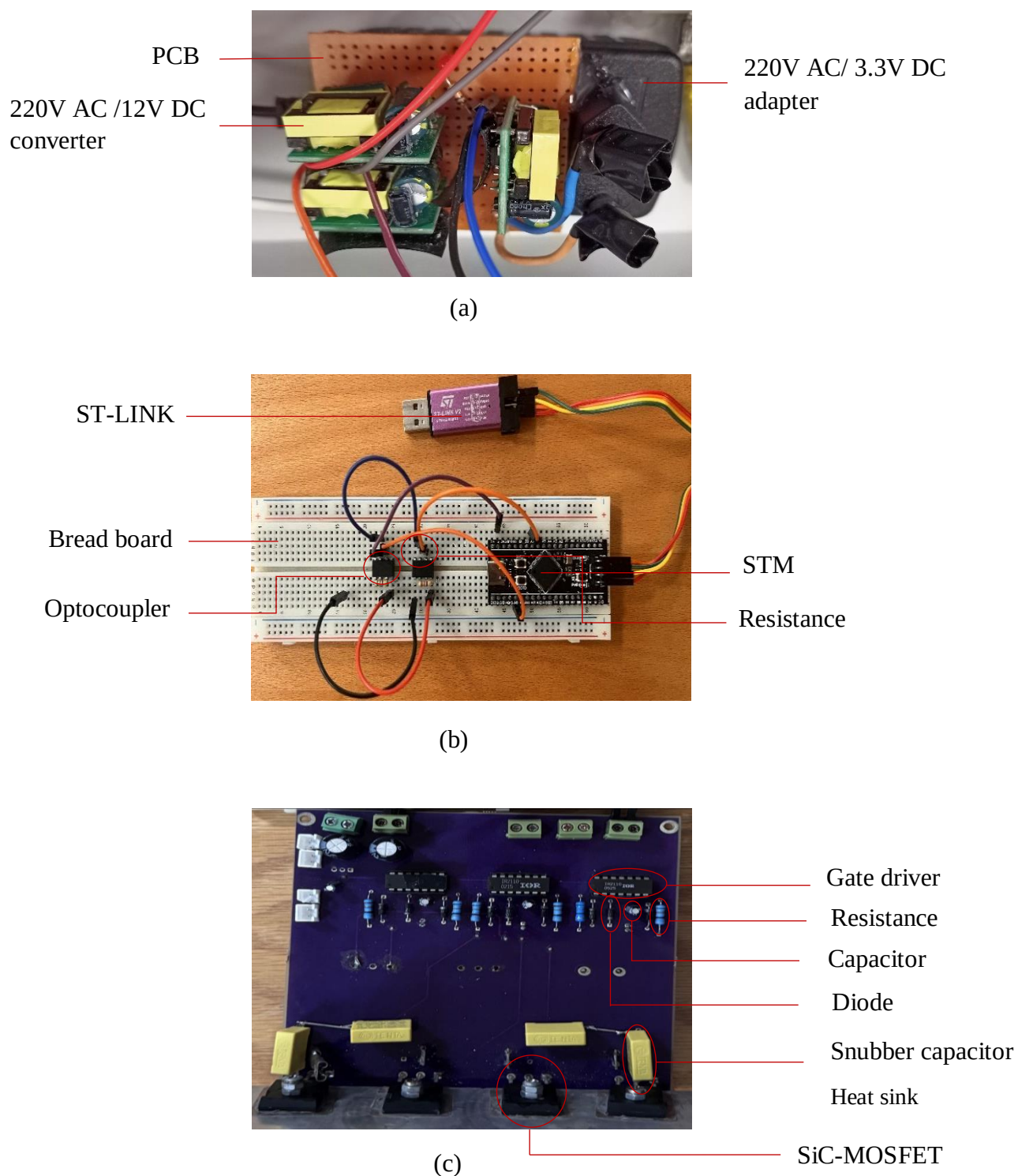


Figure 3-29: HFI components

(a) Feeding of gate driver and fan cooling

(b) STM and optocoupler circuits

(c) MOSFETs and gate driver circuits

3. Transmitter's Compensation Topology: an important part which is responsible for the resonance in the transmitter's circuit, the topology was picked to be series as discussed in [6] and the capacitors used to fit that purpose have characteristics which are presented in table 3-21.

Table 3-21: Compensation characteristics

Metallized Polypropylene Film Capacity	
Rated DC voltage	1200 V
Capacitance	0.33 μ F
DC Voltage proof	2*UR/ 10s
Insulation resistance	$\geq 30000 \Omega$ (100 DC V, 60s)
Tangent of loss angle (20°C)	≤ 0.0007 (10 kHz)
Pulse rise time (dv/dt)	500V/ μ s Max

The series transmitter's compensation with its capacitance read by the LCR meter is shown in Figure 3-30



Figure 3-30: Transmitter's practical compensation

4. Inductive Coupling: the design of the two coils was identical in order to achieve optimal operation in the SS topology state [1]. As well as it was easier to predict the behavior of equation-wise. Two designs were discussed, circular and square, the material used for the construction of the coils was Litz wire 0.1x600 strands, and that's due to skin effect of current passing through the conductor, each strand of Litz wire is insulated from one another. The number of the strands is high to reduce conduction losses as well as heating losses within the wire.

The coils were placed on a platform made out of Iclrylic for transparency as well as not to interfere with the electromagnetic field radiated from the coil [19]. The design and construction of the coils is shown in Figure 3-31.

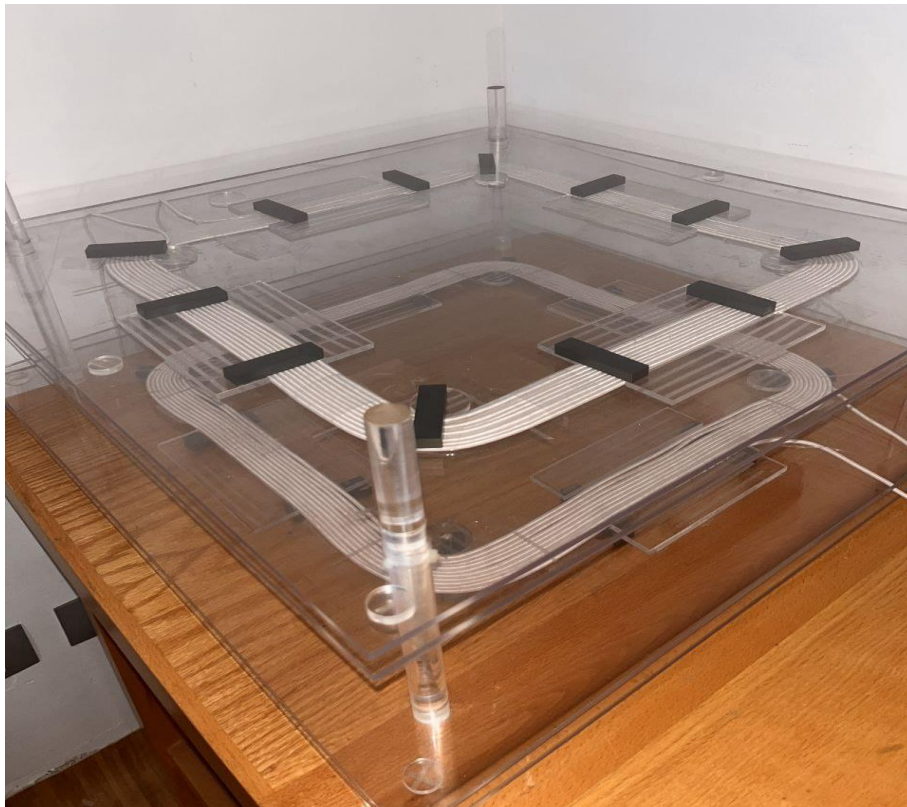


Figure 3-31: The coil design

The parameters of the coils were studied to reach optimal efficiency. The model followed by constructing such a design is presented in the flowchart shown in Figure 3-32. This flowchart determines the number of turns and dimensions of coils, assuming a desired air gap distance. The relation will also determine the approximate wire length needed for the construction of the system.

Table 3-22: WPT Coil Properties

Parameter	Value	Parameter	Value
D_i	2.45 mm	r_t	16.25 mm
D_o	3.25 mm	S_i	4.17 mm ²
D_s	0.1 mm	S_o	8.3 mm ²
h	150 mm	T_r	10 turns
l_{Litz}	40 m	T_t	10 turns
l_r	450 mm	u	450 mm
l_{r_i}	385 mm	w_r	450 mm
l_t	450 mm	w_{r_i}	385 mm
l_{t_i}	385 mm	w_t	450 mm
N	600 strands	w_{t_i}	385 mm
R_{DC_t}	0.006 Ω /m	μ_o	$4 \pi \times 10^{-7}$ H/m
r_r	16.25 mm	ρ_{cu}	1.724×10^{-8} Ω m

Table 3-23: The DC resistances of the Litz wire used

DC resistance	Calculated value by Eq.4 and Eq.5	Measured Value by LCR meter device
R_r	0.0744 Ω	0.1 Ω
R_t	0.0744 Ω	0.1 Ω

After the coil parameters are obtained, the values of self and mutual inductances are calculated based on the m-file script provided in [Figure 3-18 and Figure 3-19].

Once these values have been calculated a coupling coefficient is calculated to determine whether the system produced is viable or not. According to SAE J2954, K should not reach below 0.2 or the WPT system will not be efficient.

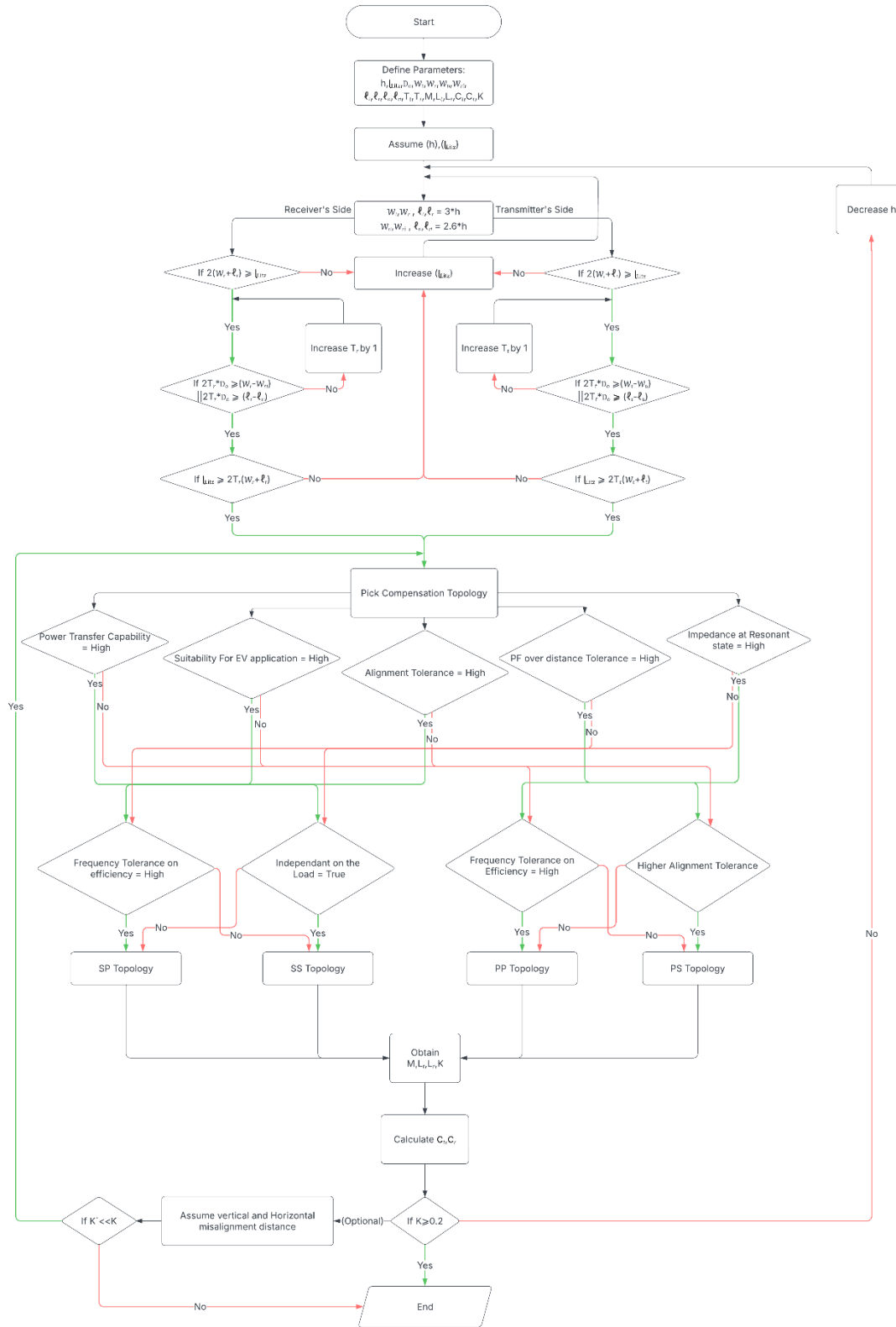


Figure 3-32: Flow chart of the design

The general design procedure for implementing WPT system, as outlined in the flowchart:

- a) Define system parameters such as the coil's inner and outer dimensions from height and width, length of the wire used as well as the number of turns, air gap distance between the two coils and special parameters such as coils' self and mutual inductances, transmitter and receiver capacitances as well as coupling coefficient.
 - b) We obtain the required Litz wire length per air gap distance by assuming their desired values for start, these values will change until a proper solution is achieved based on the equations presented in the flowchart.
 - c) Coils' inner and outer dimensions are obtained with an equation that involves the air gap distance between them, the first stage involves ensuring that the wire could complete one full turn of the coil based on the obtained dimensions.
 - d) The possible number of turns is calculated based on the difference between the inner and outer dimensions; the second stage ensures that the wire length could cover the coil dimensions per turn of the obtained number of turns value.
 - e) Picking the compensation topology depends on the type of application needed for your system, for example if a high alignment tolerance is needed as well as frequency tolerance on efficiency then SP compensation is most fitting.
 - f) Once a compensation topology has been picked, the rest of the coil's parameters from self and mutual inductances as well as compensation capacitors' values are calculated based on the perspective equations of picked compensation topology.
 - g) To ensure the system is reliable, based on the SAE J2954 standard, if
-

the coupling coefficient of 0.2 or above is not met, the assumed air gap distance at the start will have to be reconsidered and the process will repeat.

- h) Optionally, if alignment considerations are not met as desired, the compensation topology will have to be reconsidered for one that works better for misalignment conditions.

5. Receiver's Compensation Topology: the connection is series to obtain the SS topology, the capacitor characteristics used is given in table 3-21.

The series receiver's compensation with its capacitance read by the LCR meter is shown in Figure 3-33.

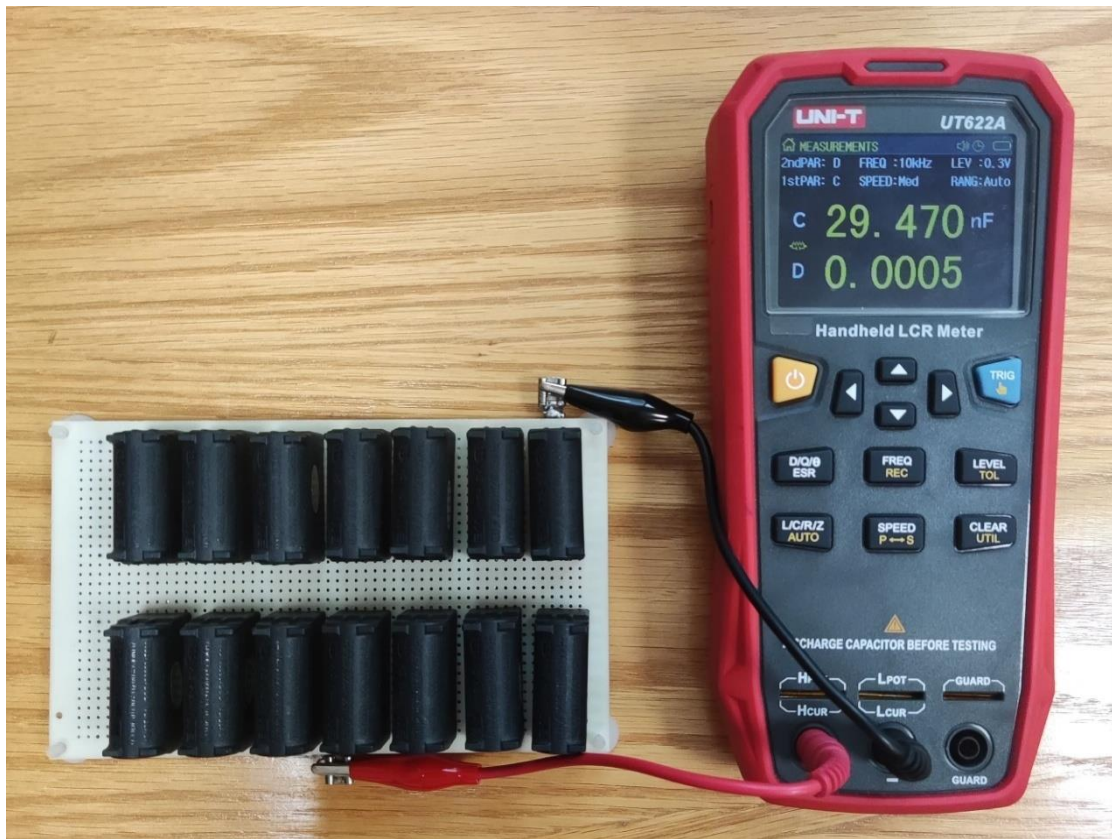


Figure 3-33: Receiver's practical compensation

6. AC/DC Converter: to turn the received power from the coil into DC, the use of ultra-high switching speed diodes is required, this minimizes the amount of power lost in rectification thereby increasing efficiency for this prototype a (DSEI 60) diode was used. Table 3-24 shows Diode specifications.

Table 3-24: Diode specifications

Diodes specifications	
Reverse Recovery Time	50 ns
Peak Inverse Voltage	1000 V
Forward Current Rating	60 A

Figure 3-34 shows the full bridge AC/DC converter before the load which consists of four ultra-high switching, 1000V-60A rating diodes and $3 \times 220 \mu\text{F}$ electrolytic capacitors in order to filter the DC output voltage across the load and ensure small ripples in it.

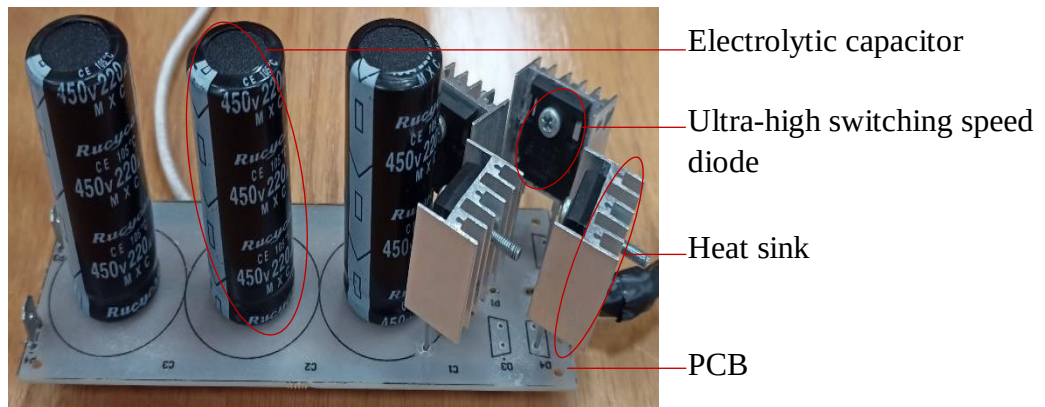


Figure 3-34: Practical AC/DC converter

7. Load: the load used is 5 Ω power resistance rated at 5 KW, which can handle the high power coming from the coil.

Figure 3-35 shows the 5KW rated load which followed the AC/DC converter, and we will measure the DC current through it and also the DC voltage across it.

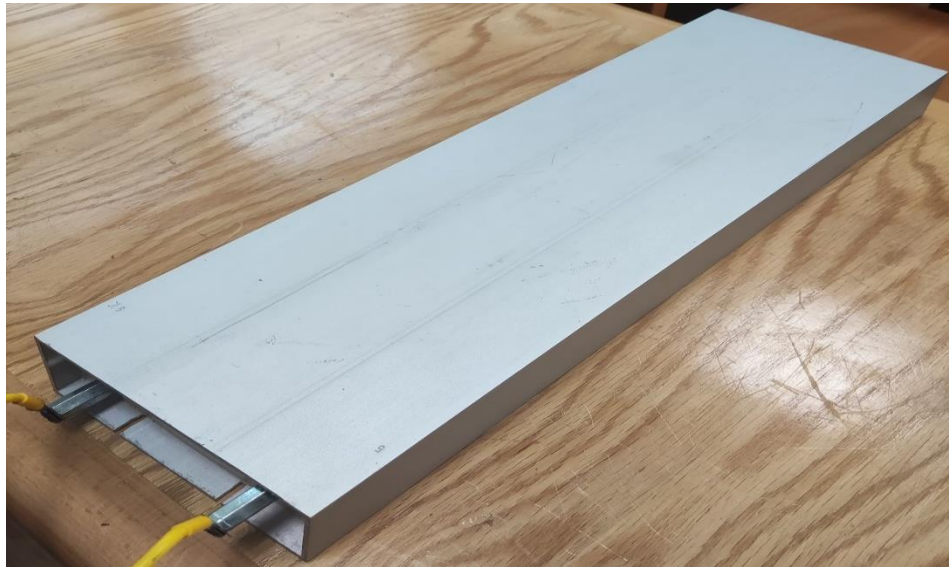


Figure 3-35: The practical resistive load

4 Chapter (4): Results and Discussion

4.1 Simulation results

Table 4-1: The simulated, calculated and measured values of the self-inductances for both coils, without using ferrite

Self-inductance	Obtained value from ANSYS Maxwell 3D	Calculated value from Eq.6 and Eq.7	Measured value by LCR meter device
L_t	113.187 μH	100.688 μH	101 μH
L_r	113.182 μH	100.688 μH	101.5 μH

Table 4-2: The simulated, calculated and practical values of the coupling coefficient and the mutual inductance, without using ferrite

Coupling coefficient and mutual inductance	Obtained value from ANSYS Maxwell 3D	Calculated value from Eq.8 and Eq.9	Practical Values
M	19.343 μH	23.47 μH	17.5 μH
K	0.171	0.233	0.173

Table 4-3: The simulated, calculated and measured values of the self-inductances for both coils, with using ferrite

Self-inductance	Obtained value from ANSYS Maxwell 3D	Calculated value from Eq.6 and Eq.7	Measured value by LCR meter device
L_t	125.817 μH	—	116.7 μH
L_r	125.83 μH	—	117.13 μH

Table 4-4: The simulated, calculated and practical values of the coupling coefficient and the mutual inductance, with using ferrite

Coupling coefficient and mutual inductance	Obtained value from ANSYS Maxwell 3D	Calculated value from Eq.8 and Eq.9	Practical Values
M	21.9 μH	—	20 μH
K	0.174	—	0.171

Table 4-5: The simulated, calculated and measured values of the capacitances of the receiving and the transmitting circuits without using ferrite

Capacitances	Simulated value used in ANSYS Simplorer	Calculated value from Eq.10 and Eq.11	Measured value by LCR meter device
C_t	32.097 nF	36.082 nF	35.964 nF
C_r	32.099 nF	36.082 nF	35.691 nF

Table 4-6: The simulated, calculated and measured values of the capacitances of the receiving and the transmitting circuits with using ferrite

Capacitances	Simulated value used in ANSYS Simplorer	Calculated value from Eq.10 and Eq. 11	Measured value by LCR meter device
C_t	28.875 nF	—	29.555 nF
C_r	28.872 nF	—	29.47 nF

Table 4-7: ANSYS Maxwell 3D results of the M, K and the magnitude of the magnetic flux of the two coils and ANSYS Simpler results of the PAC and η_{s-s} for different cases

Cases	M (μ H)	K	Magnetic flux of transmitter (mWb)	Magnetic flux of receiving (mWb)	P _{AC} (KW)	η_{s-s}
<i>Figure 3-6: The different air gaps between the transmitter and receiver coils</i>						
10	28.28	0.253	1.12	1.12	1.6	97.63
15	19.34	0.171	1.06	1.06	3.346	97.31
20	13.57	0.12	1.011	1.01	6.178	96.5
25	9.737	0.085	0.993	0.998	10.83	95.44
<i>Figure 3-7: The different positions of horizontal misalignment for each scenario</i>						
<i>1. (-x, -y)</i>						
(-10, -12)	13.07	0.115	1.013	1.01	6.667	96.6
(-15, -15)	9.51	0.084	0.982	0.979	10.4	95.38
<i>2. (-x)</i>						
-10	16.52	0.146	1.032	1.032	4.353	97.07
-15	13.75	0.12	1.02	1.021	5.92	96.6
-30	4.745	0.042	0.94	0.941	12.1	87.29
<i>3. (-y)</i>						
-12	15.38	0.135	1.031	1.029	5.1	96.92
-15	13.67	0.119	1.024	1.02	6.1	96.65
-30	4.97	0.042	0.94	0.941	13.67	87.9
<i>4. (-x, +y)</i>						
(-10, +12)	13.07	0.115	1.014	1.017	6.468	96.45
(-15, +15)	9.64	0.085	0.979	0.982	9.65	95.3

Figure 3-8: Angles exist between the transmitting and receiving coil concerning the Z-axis

5°	19.63	0.17	1.074	1.072	3.22	97.28
10°	19.65	0.173	1.063	1.063	3.222	97.35
50°	19.2	0.168	1.066	1.067	3.333	97.27

Figure 3-9: Angles exist between the transmitting and receiving coil concerning the Y-axis

5°	19.58	0.172	1.066	1.065	3.258	97.36
8°	19.81	0.174	1.065	1.066	3.175	97.36
20°	21.63	0.19	1.081	1.081	2.684	97.26

Figure 3-10: Multiple misalignment error positions in case of horizontal misalignment with rotation around the Z-axis

(-5,10°)	18.85	0.164	1.06	1.07	3.47	97.18
(-5,50°)	18.55	0.162	1.064	1.064	3.57	97.18
(-15,10°)	14	0.122	1.026	1.026	5.783	96.66
(-15,50°)	14.1	0.123	1.027	1.027	5.731	96.68
(-30,10°)	5	0.043	0.953	0.952	15.19	88.57
(-30,50°)	4.95	0.043	0.952	0.953	14.5	88.53

Figure 3-11: Multiple misalignment error positions in case of horizontal misalignment in (-x, -y) with rotation around the Z-axis

((-5, -5), 10°)	18.06	0.157	1.06	1.06	3.72	97.1
((-5, -5), 50°)	17.91	0.156	1.057	1.057	3.77	97.15
((-15, -15), 10°)	9.82	0.086	0.986	0.985	9.7	95.47
((-15, -15), 50°)	10.24	0.089	0.996	0.995	8.9	95.51

Figure 3-12: Multiple misalignment error positions in case of horizontal misalignment in (-x, -y) with a non-uniform air gap in the Z-axis direction with rotation around the Y- axis

(5°, (-5, -5))	17.93	0.158	1.046	1.046	3.774	96.98
(8°, (-5, -5))	18.03	0.159	1.047	1.046	3.736	97.01
(5°, (-15, -15))	9.63	0.084	0.985	0.982	10.28	95.44
(8°, (-15, -15))	9.64	0.084	0.988	0.985	10.202	95.43

Figure 3-13: Multiple misalignment errors is a horizontal misalignment in (-x, -y) direction, combined with a non-uniform air gap in the Z-axis direction with rotation around the Y-axis and Z-axis

(-15, -15) cm, 10° around Z, 8° around Y	9.55	0.084	0.985	0.980	12.809	95.9
(-15, -15) cm, 50° around Z, 8° around Y	10.25	0.091	0.988	0.984	10.58	96
(-10, -12) cm, 10° around Z, 8° around Y	13.18	0.117	1.012	1.008	7.212	96.85
(-10, -12) cm, 50° around Z, 8° around Y	13.77	0.122	1.015	1.014	6.378	96.85
(-10, -12) cm, 10° around Z, 5° around Y	13.13	0.116	1.015	1.011	7.249	96.83
(-15, -15) cm, 10° around Z, 5° around Y	9.53	0.085	0.979	0.975	12.833	95.87

Figure 3-14: Different models used in design for WPT system

1. Aligned at 15cm vertical air gap

(circular, circular)	27.37	0.181	1.429	1.428	1.713	97.61
(square, square)	19.33	0.171	1.058	1.06	3.508	97.37
(circular, square)	22.148	0.17	1.384	1.077	2.571	97.5
(square, circular)	22.129	0.17	1.076	1.384	2.584	97.46

<i>2. Horizontal misalignment of 15cm in X-axis</i>						
(circular, circular)	17.84	0.117	1.355	1.355	3.76	97.167
(square, square)	13.77	0.122	1.013	1.015	6.112	96.76
(circular, square)	15.573	0.119	1.334	1.028	5.159	97.14
(square, circular)	15.569	0.12	1.022	1.332	5.382	97.15
<i>3. Horizontal misalignment of 25cm in X-axis</i>						
(circular, circular)	7.85	0.052	1.27	1.27	15.724	94.67
(square, square)	7.75	0.068	0.966	0.966	15.342	94.5
(circular, square)	7.73	0.06	1.27	0.961	16.542	94.88
(square, circular)	7.78	0.06	0.962	1.27	16.475	94.6

All ANSYS Simpler results at table 4-7, are obtained at 311 V as the input DC voltage to the HFI with resonance frequency of 83.3 KHz.

The number of turns of each circular coil designed at a ANSYS Maxwell 3D is 14 turns with a 450 mm outer diameter. The number of turns of each square coil designed at a ANSYS Maxwell 3D is 10 turns with a size of 450 mm × 450 mm.

The length of the Litz wire of each circular coil is 17.8 m. and is 17.4 m for each square coil.

The self-inductance of each circular coil is about 150 μH , and it is about 113 μH for each square coil.

The capacitance of the series compensation with each circular coil is about 24 nF at a resonance frequency of 83.3 KHz. And the capacitance of the series compensation with each square coil is about 32 nF at a resonance frequency of 83.3 KHz.

The RMS power delivered to the equivalent AC resistive load is taken at a certain equivalent AC resistive load of a 4.7 Ω .

At a constant output HF AC voltage, and from Eq. (18), the RMS transmitting current is approximately inversely proportional to the square of the mutual inductance, since all the other parameters in Eq. (18) are constants.

From Eq. (20), the RMS power delivered to the equivalent AC resistive load is directly proportional to the square of both the mutual inductance and the RMS transmitting current which as we said, is inversely proportional to the square of the mutual inductance at a constant output HF AC voltage.

So, we can conclude that the RMS power delivered to the equivalent AC resistive load is inversely proportional to the square of the mutual inductance, at a constant output HF AC voltage.

This illustrates why the RMS power delivered to the equivalent AC resistive load increases when the gap distance between the two coils increased where the mutual inductance decreases.

But if the RMS transmitting current is kept constant, then from Eq. (20), the RMS power delivered to the equivalent AC resistive load is directly proportional to the square of the mutual inductance.

Figure 4-1 shows the relationship between horizontal misalignment in X-axis and coupling coefficient.

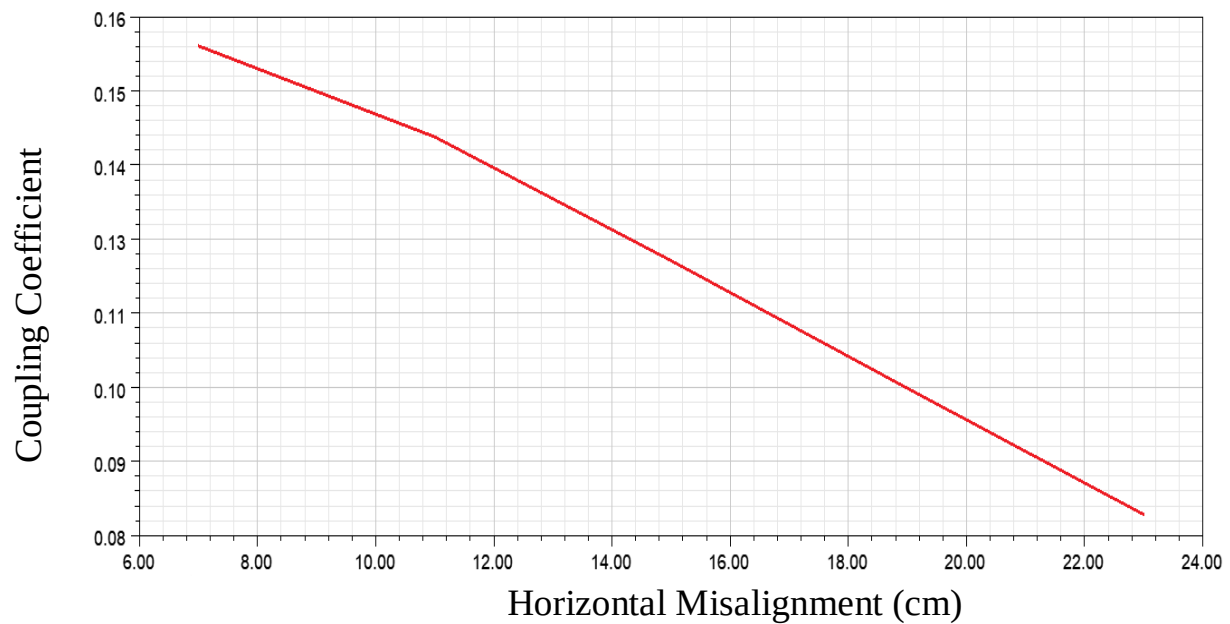


Figure 4-1: the relationship between horizontal misalignment in X-axis and coupling coefficient

Figure 4-2 shows the relationship between horizontal misalignment in X-axis and mutual inductance.

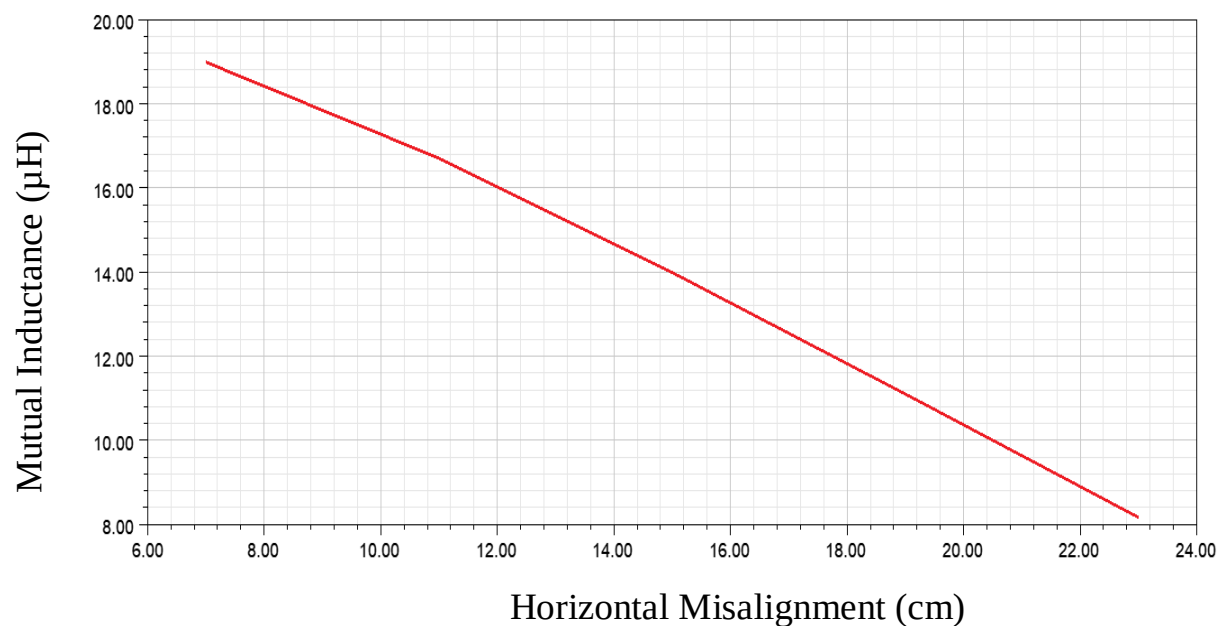


Figure 4-2: The relationship between horizontal misalignment in X-axis and mutual inductance

Figure 4-3 shows the relationship between vertical misalignment in Z-axis and coupling coefficient.

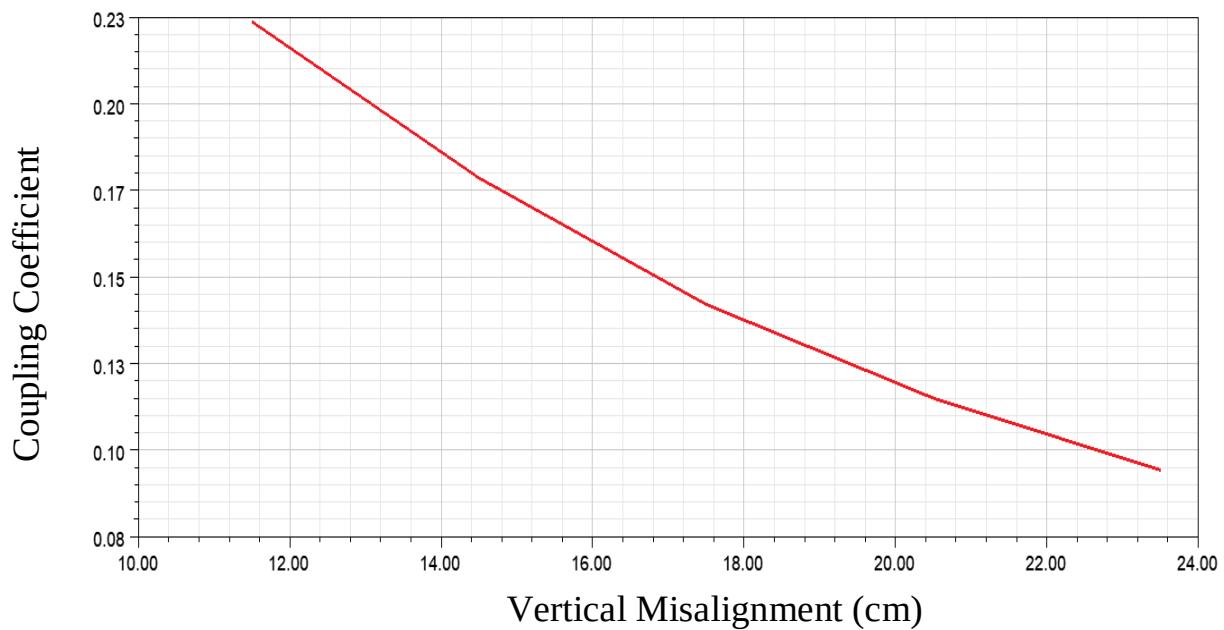


Figure 4-3: The relationship between vertical misalignment in Z-axis and coupling coefficient

Figure 4-4 shows the relationship between vertical misalignment in Z-axis and mutual inductance.

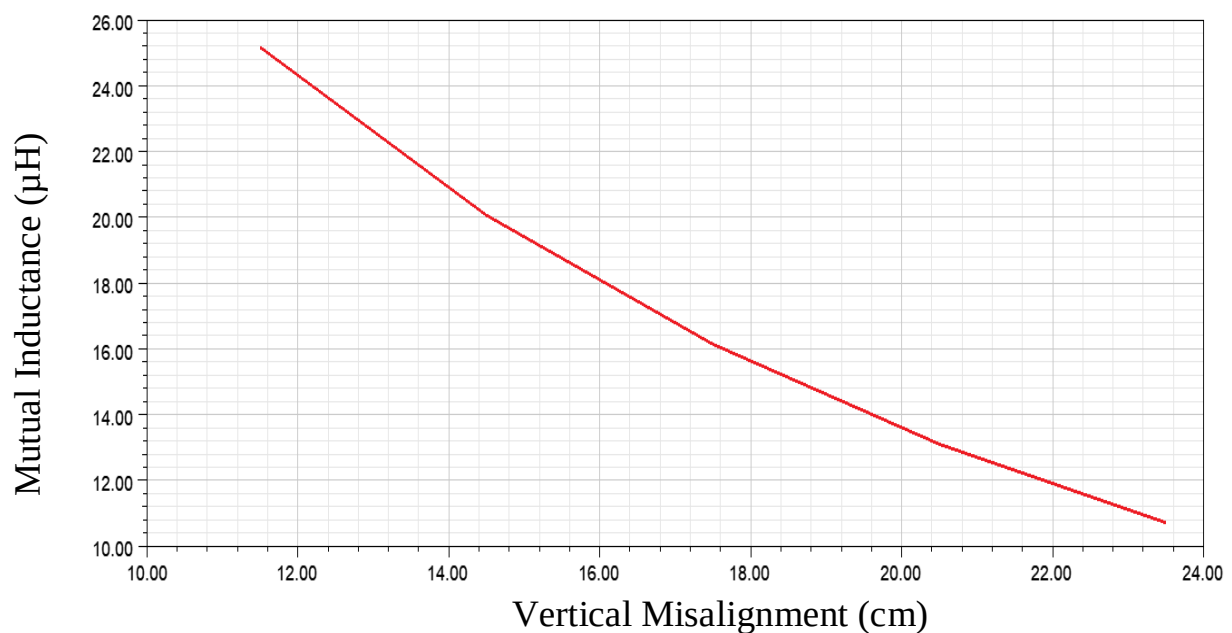


Figure 4-4: The relationship between vertical misalignment in Z-axis and mutual inductance

Figure 4-5 shows the relationship between horizontal misalignment in X-axis, vertical misalignment in Z-axis and coupling coefficient.

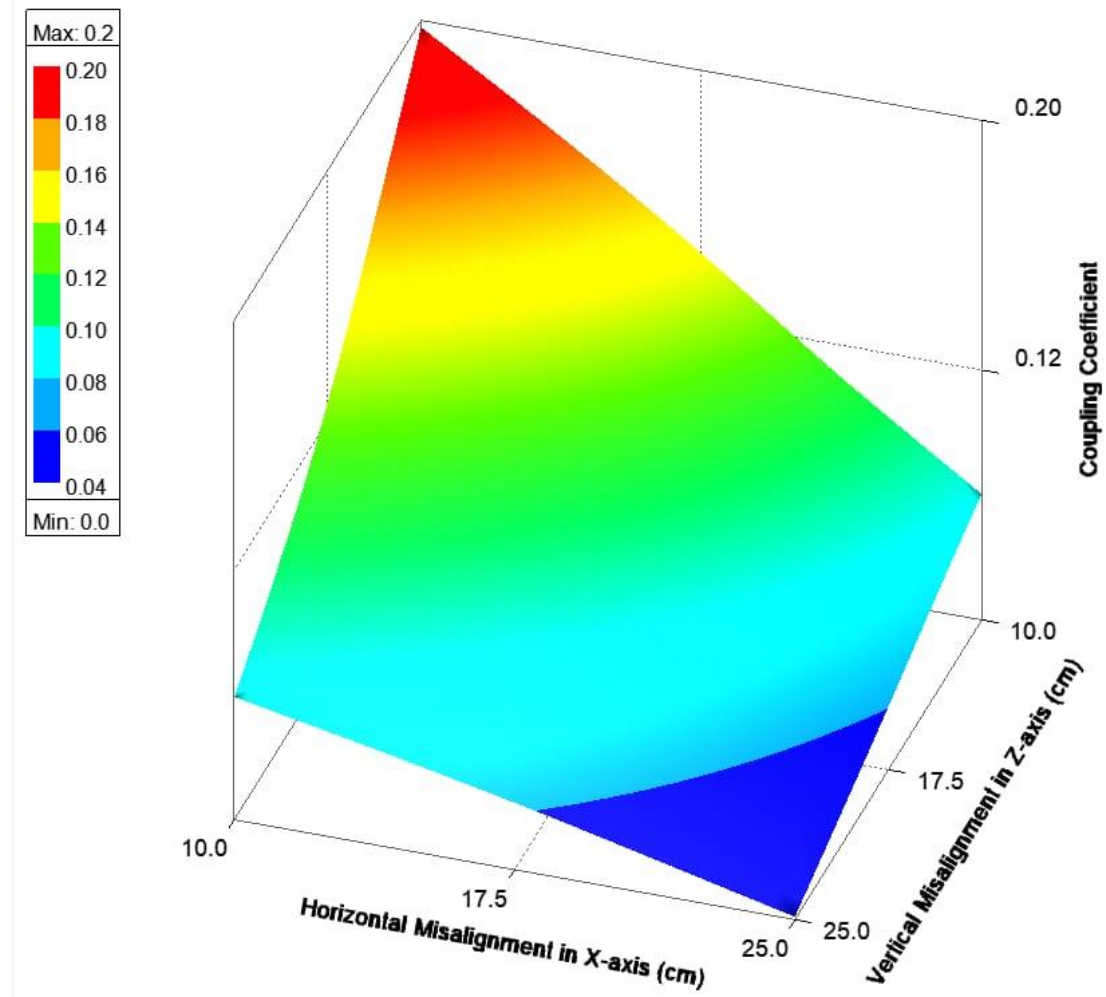


Figure 4-5: The relationship between horizontal misalignment in X-axis, vertical misalignment in Z-axis and coupling coefficient

Figure 4-6 shows the relationship between horizontal misalignment in X-axis, vertical misalignment in Z-axis and mutual inductance.

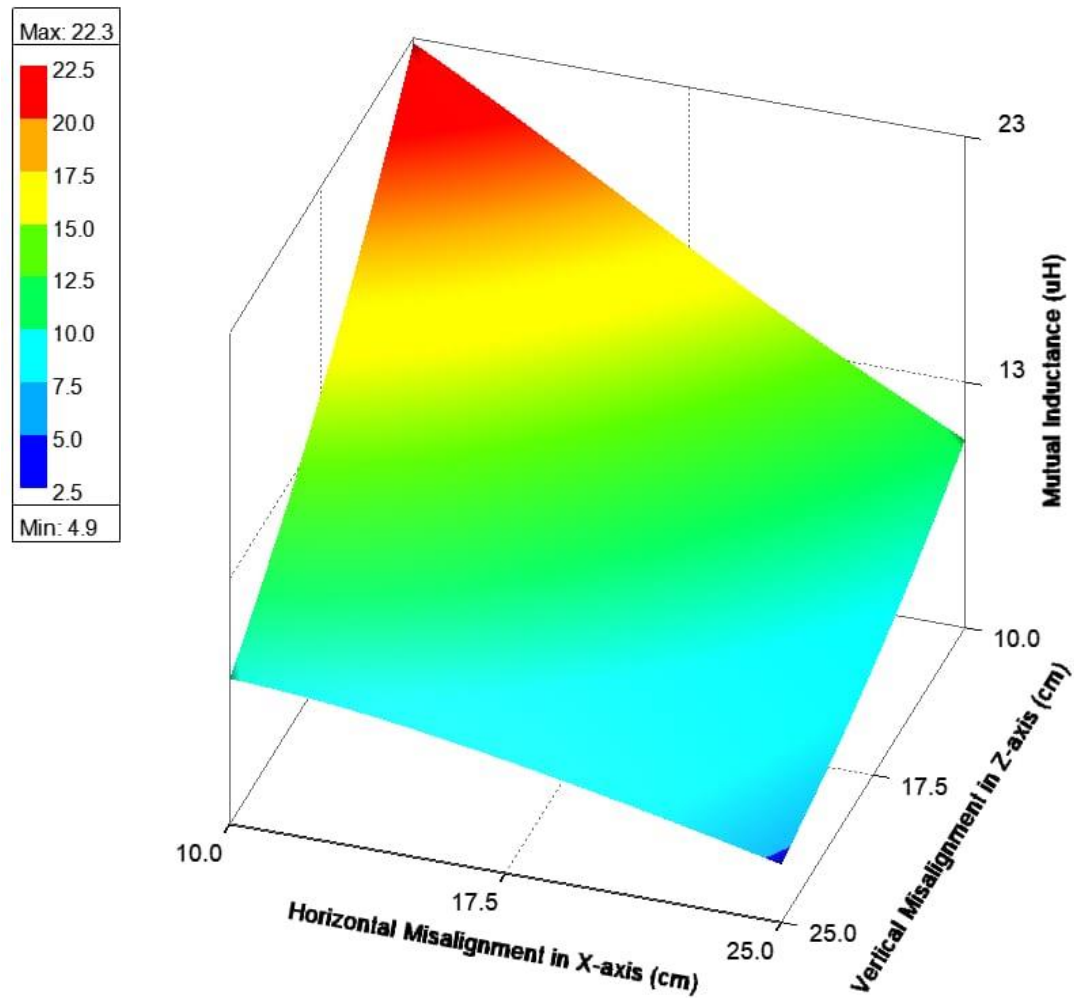
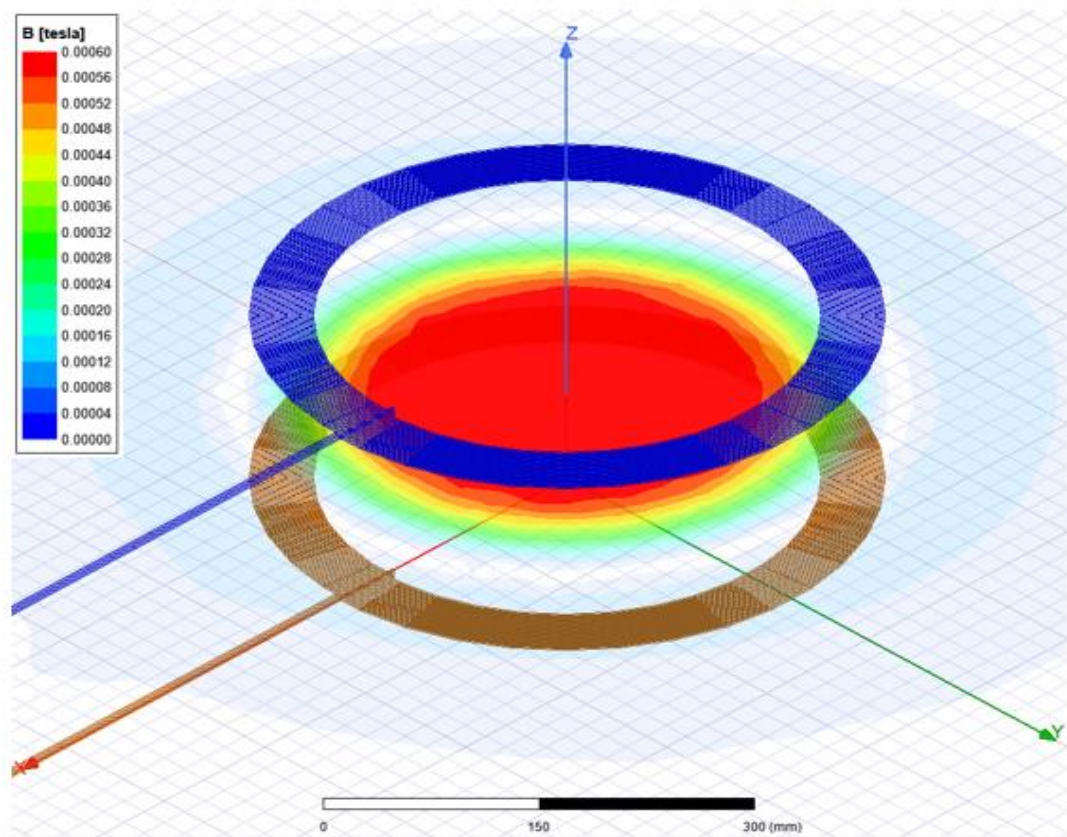
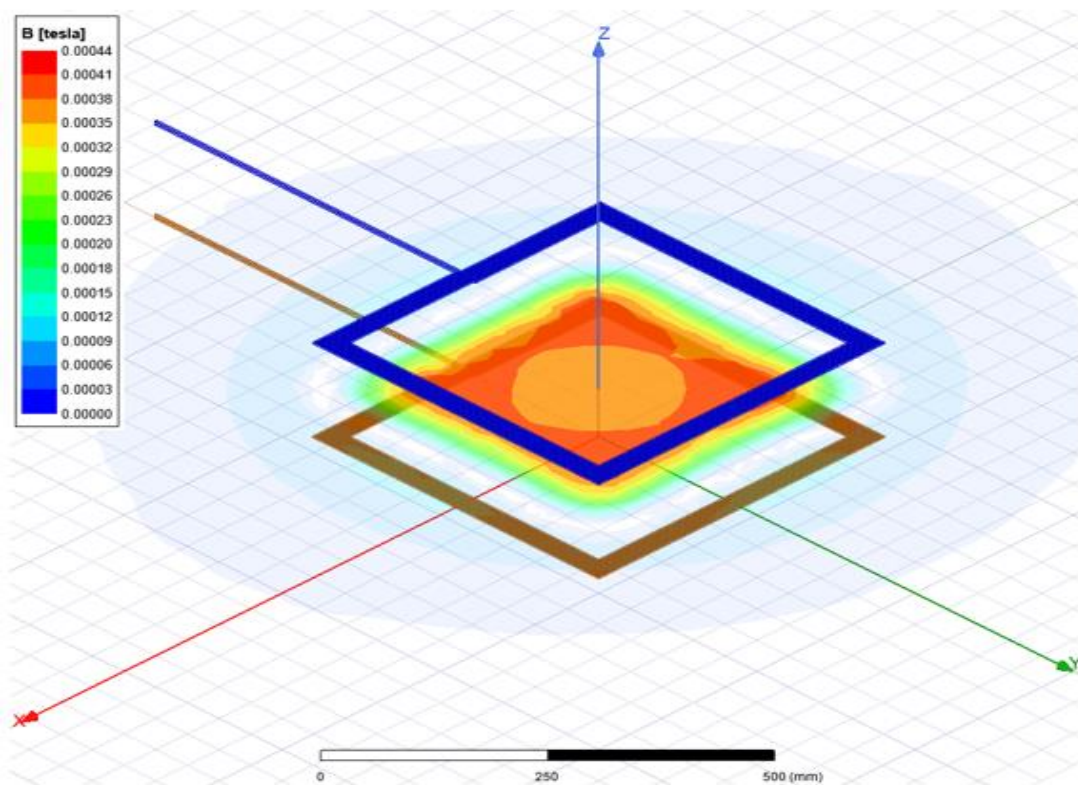


Figure 4-6: The relationship between horizontal misalignment in X-axis, vertical misalignment in Z-axis and mutual inductance

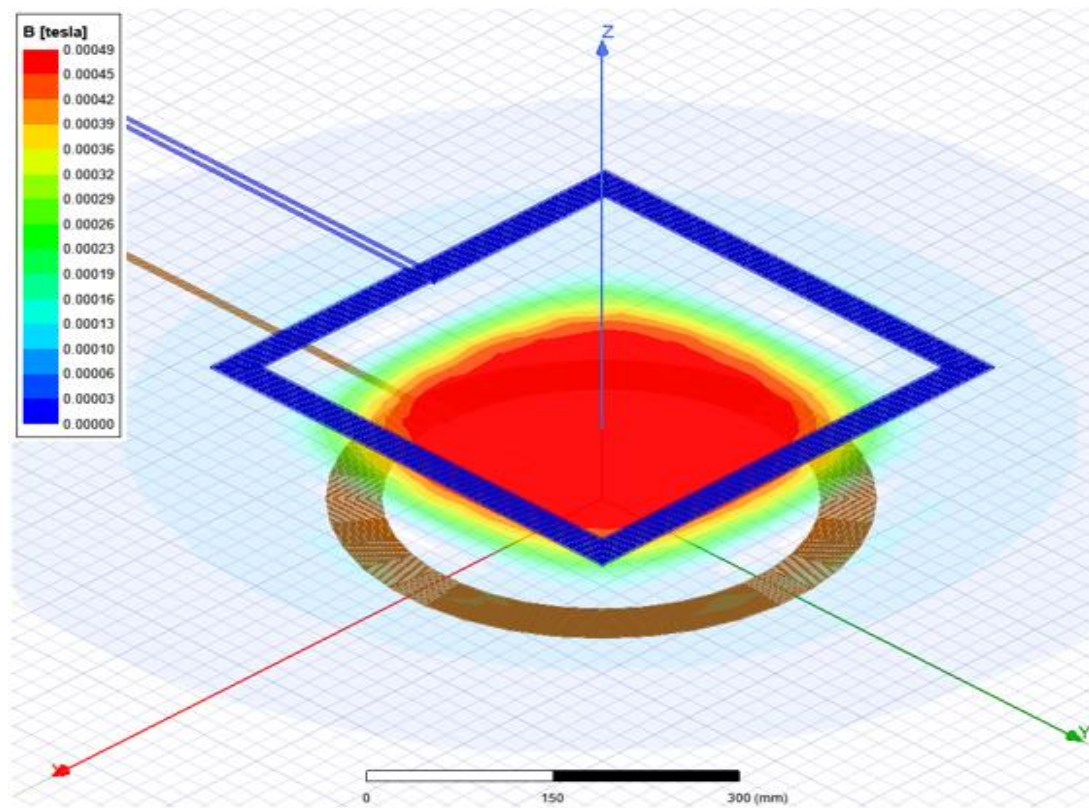
Figure 4-7 shows the magnetic flux animations of shapes of figure 3-13, 1. aligned at 15 cm vertical air gap.



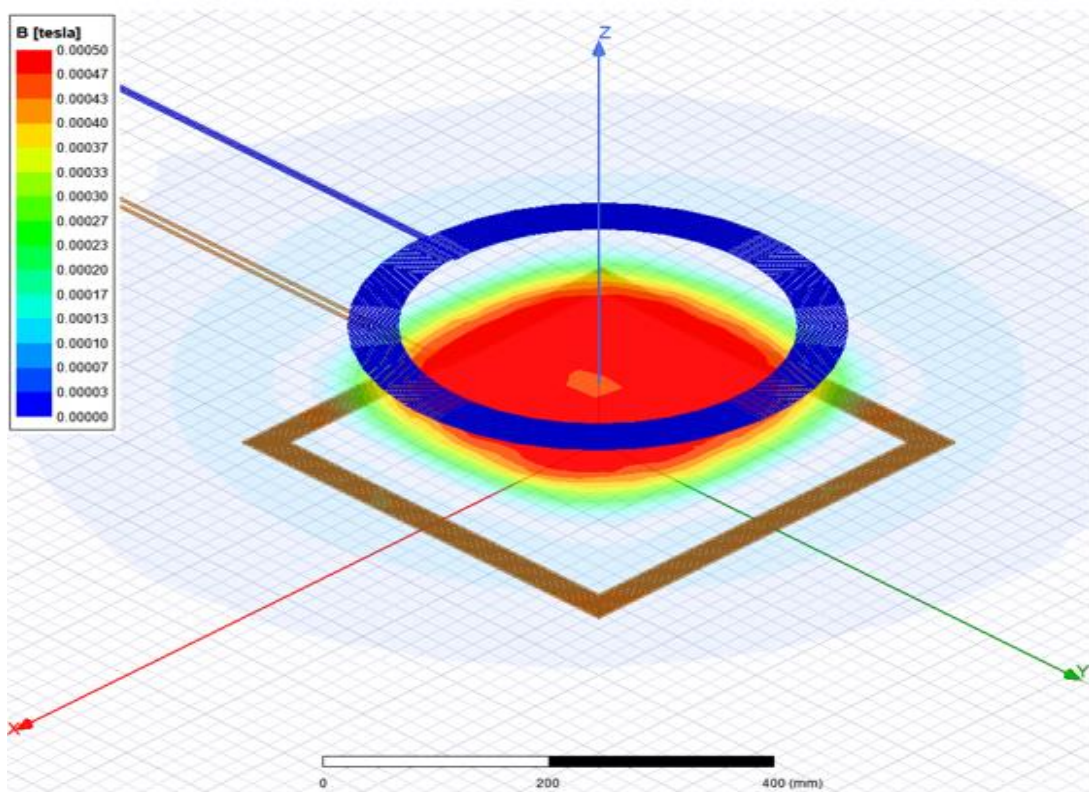
(a)



(b)



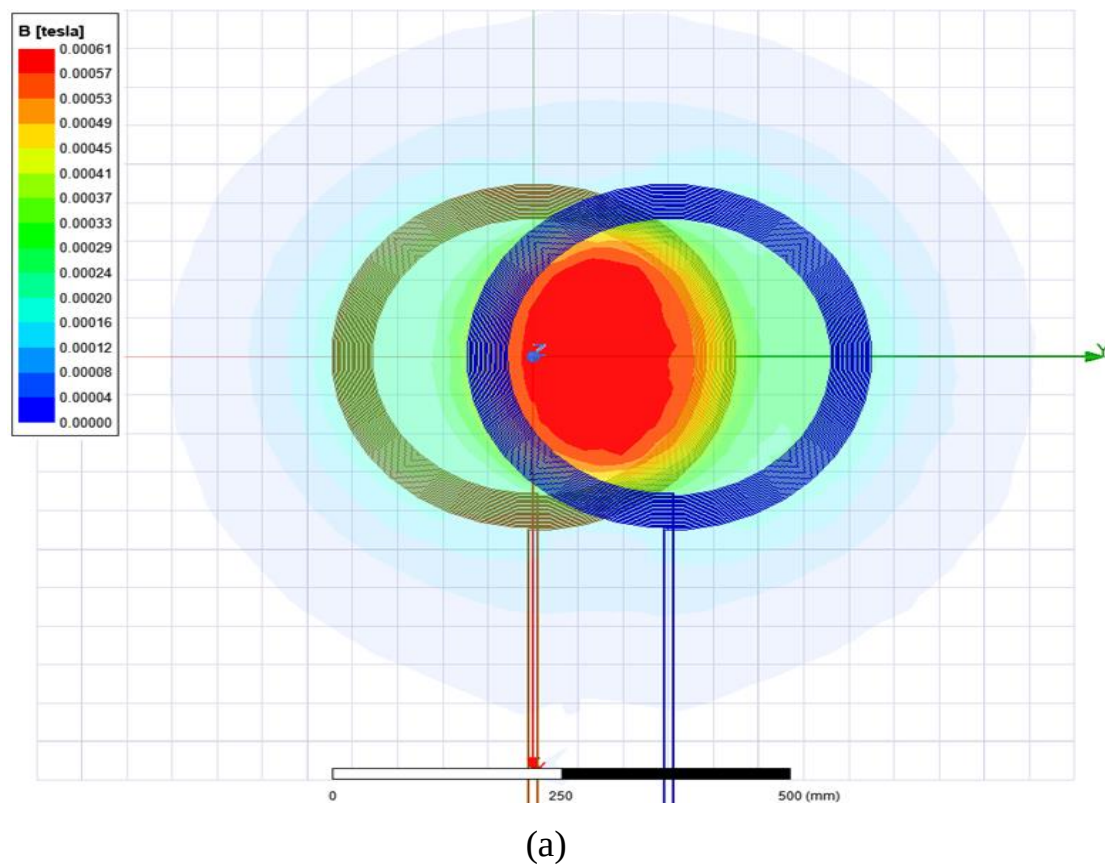
(c)

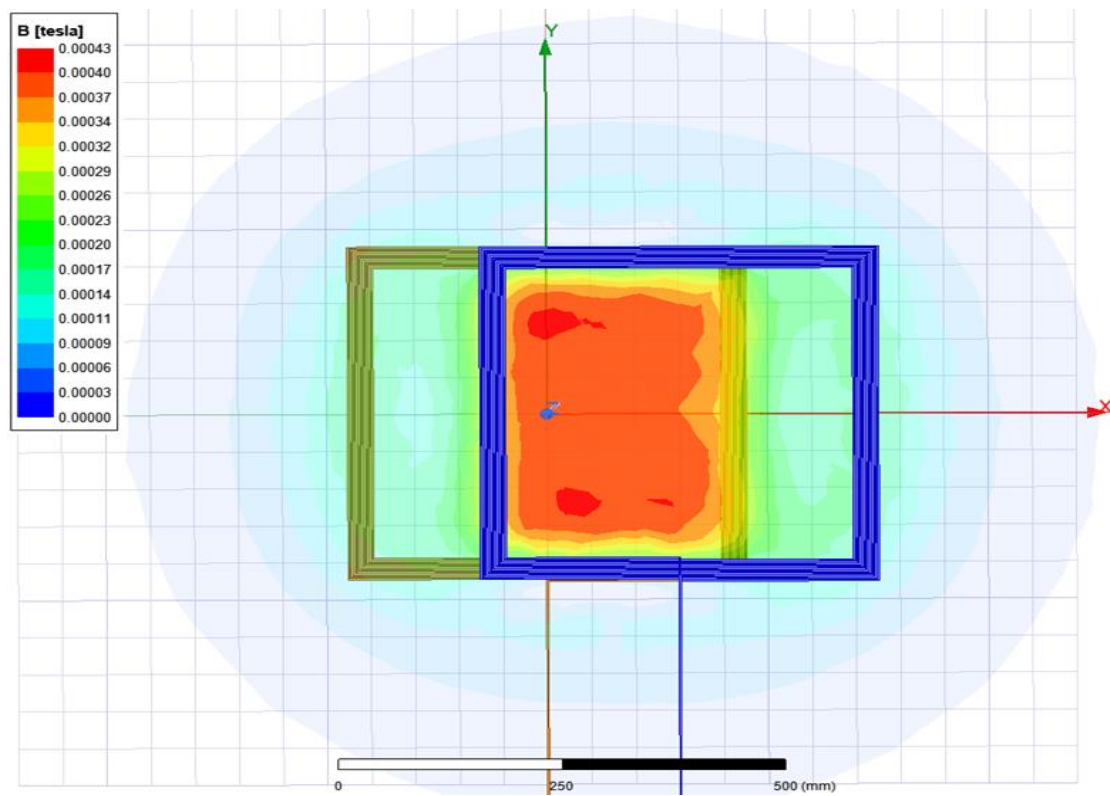


(d)

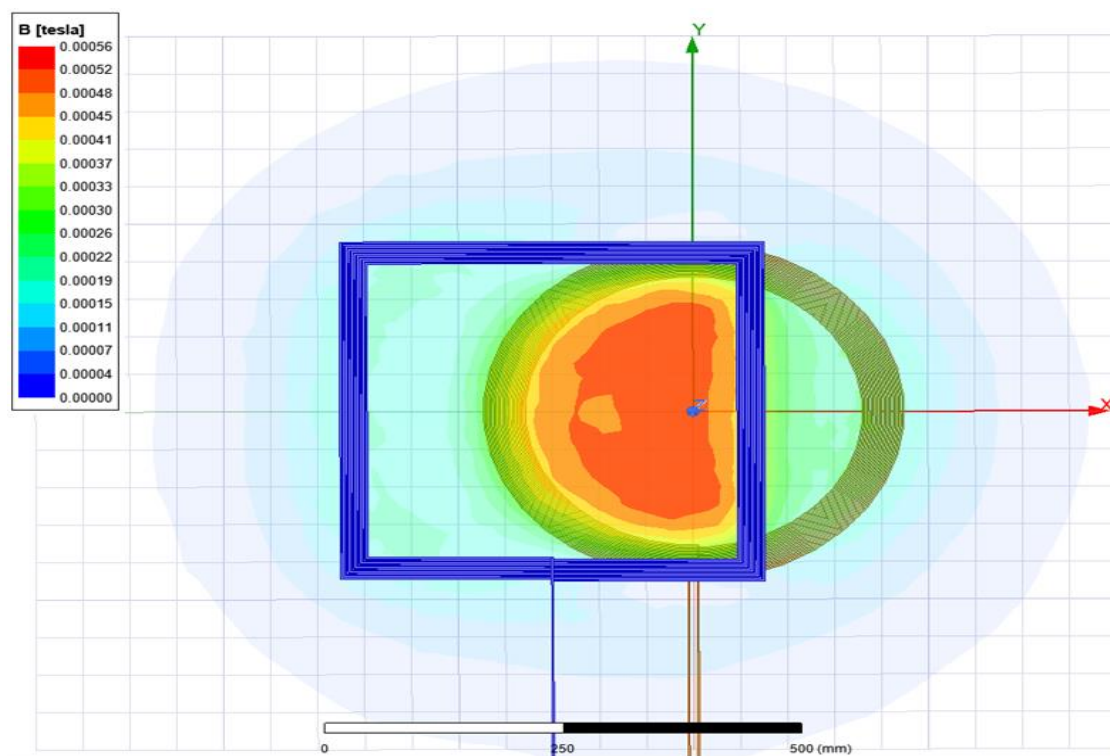
Figure 4-7: The magnetic flux animations of shapes of figure 3-14, 1. aligned at 15 cm vertical air gap

Figure 4-8 shows the magnetic flux animations of shapes of figure 3-13, 2. horizontal misalignment of 15 cm in X-axis.

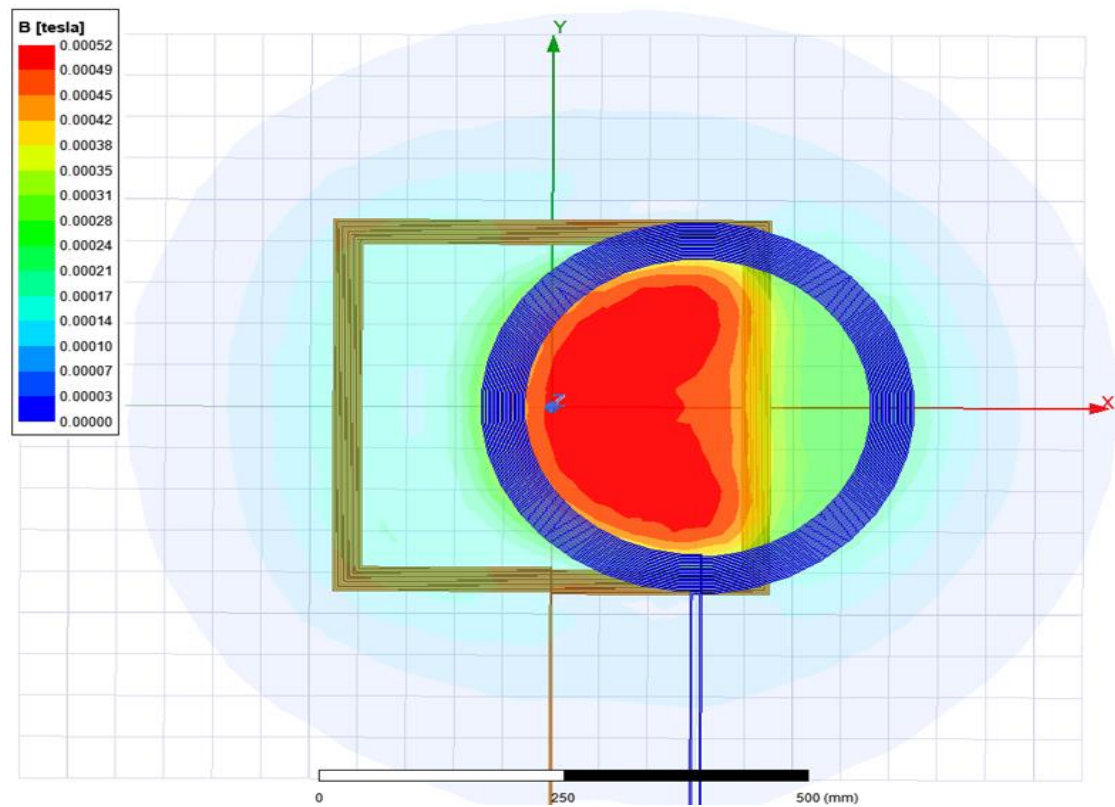




(b)



(c)



(d)

Figure 4-8: The magnetic flux animations of shapes of figure 3-14, 2. horizontal misalignment of 15 cm in X-axis

Figure 4-9 shows the magnetic flux animation of the WPT design using ferrite.

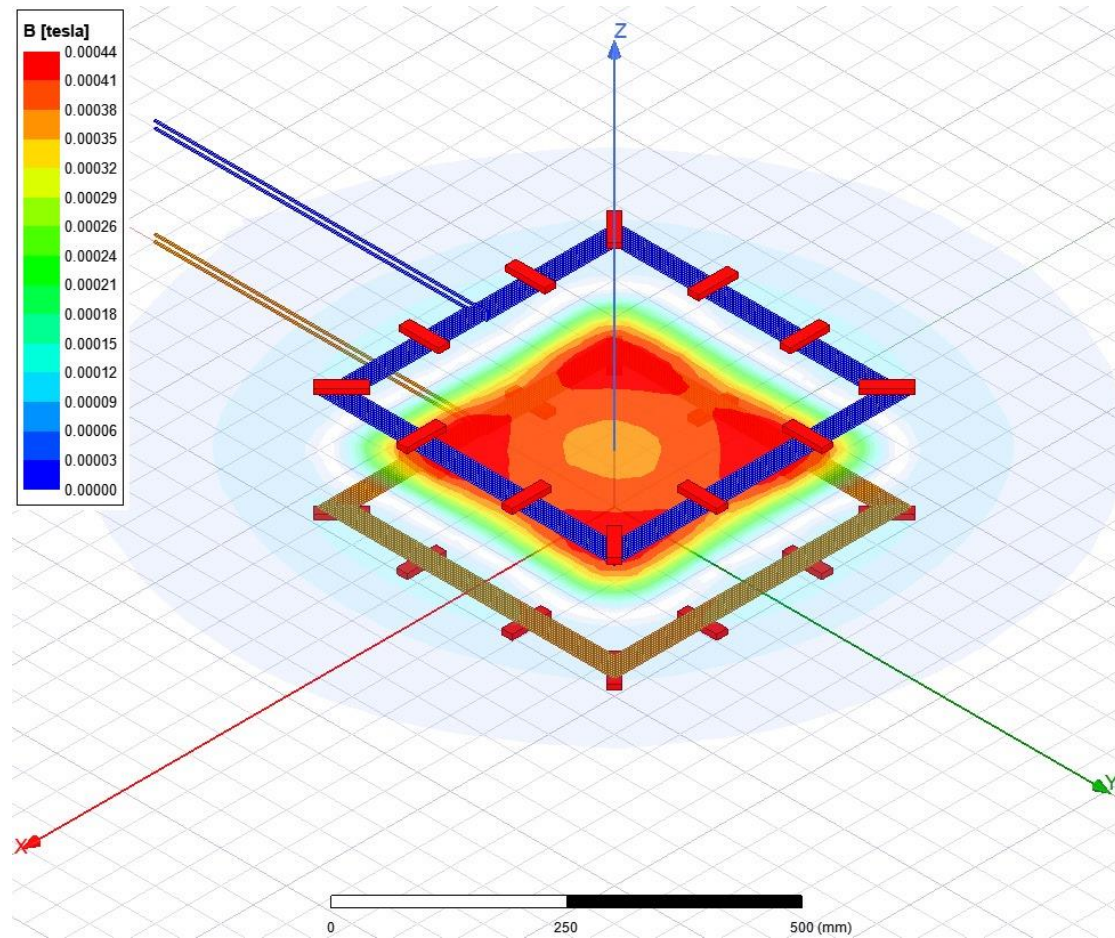
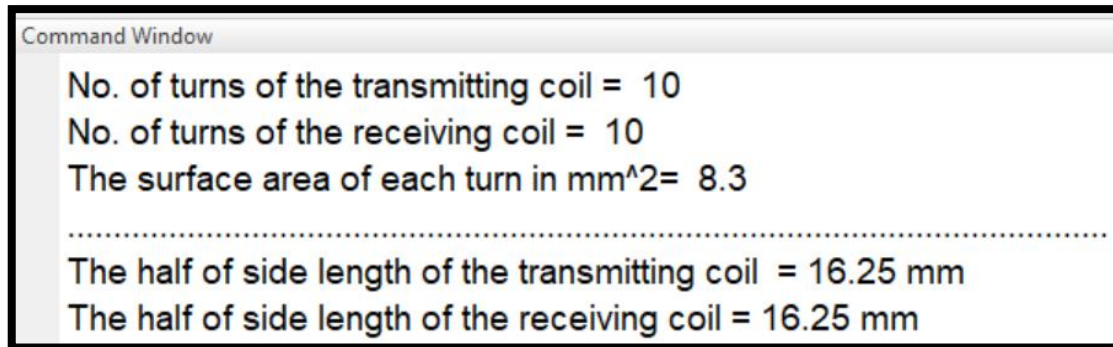


Figure 4-9: The magnetic flux animation of the WPT design using ferrite

Figure 4-10 shows the results of m-file for calculating the half of side lengths of both coils.

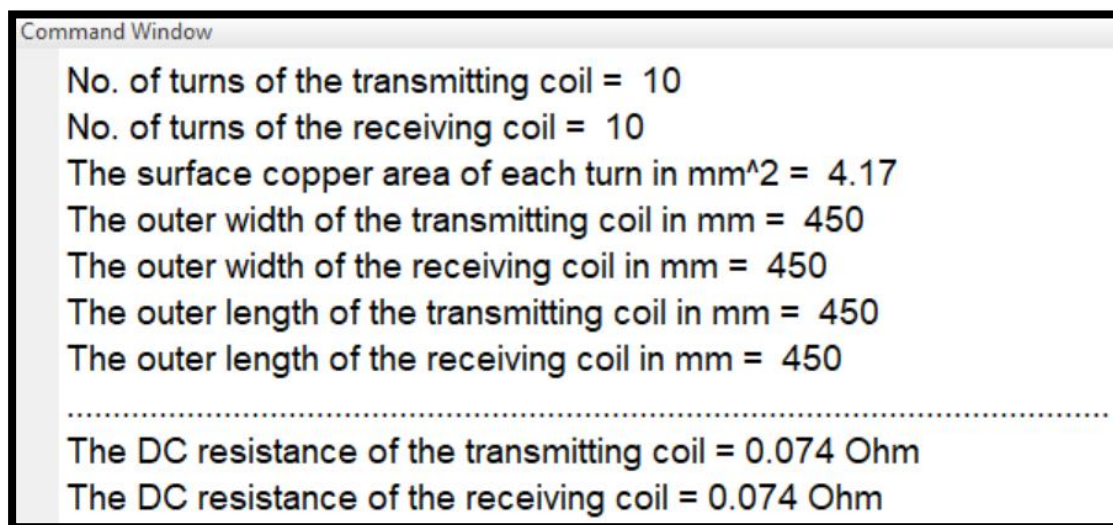


```
Command Window

No. of turns of the transmitting coil = 10
No. of turns of the receiving coil = 10
The surface area of each turn in mm^2= 8.3
.....
The half of side length of the transmitting coil = 16.25 mm
The half of side length of the receiving coil = 16.25 mm
```

Figure 4-10: The results of m-file for calculating the half of side lengths of both coils

Figure 4-11 shows the results of m-file for calculating the DC resistances of both coils.

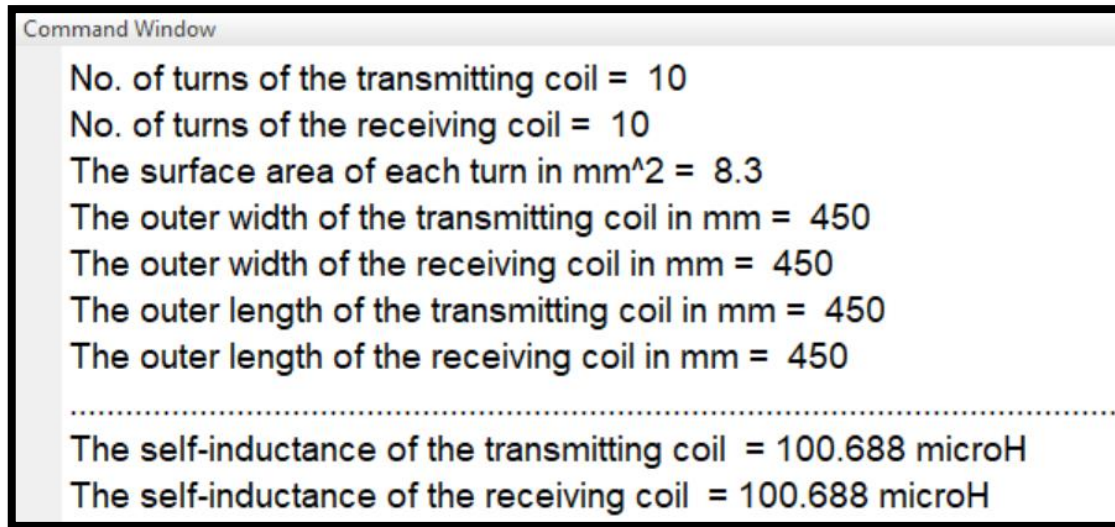


```
Command Window

No. of turns of the transmitting coil = 10
No. of turns of the receiving coil = 10
The surface copper area of each turn in mm^2 = 4.17
The outer width of the transmitting coil in mm = 450
The outer width of the receiving coil in mm = 450
The outer length of the transmitting coil in mm = 450
The outer length of the receiving coil in mm = 450
.....
The DC resistance of the transmitting coil = 0.074 Ohm
The DC resistance of the receiving coil = 0.074 Ohm
```

Figure 4-11: The results of m-file for calculating the DC resistances of both coils

Figure 4-12 shows the results of m-file for calculating the self-inductances of both coils without using ferrite.

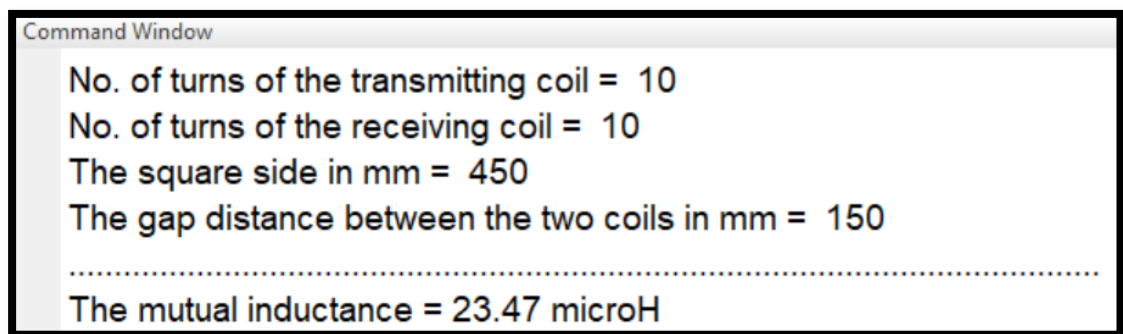


```
Command Window

No. of turns of the transmitting coil = 10
No. of turns of the receiving coil = 10
The surface area of each turn in mm^2 = 8.3
The outer width of the transmitting coil in mm = 450
The outer width of the receiving coil in mm = 450
The outer length of the transmitting coil in mm = 450
The outer length of the receiving coil in mm = 450
.....
The self-inductance of the transmitting coil = 100.688 microH
The self-inductance of the receiving coil = 100.688 microH
```

Figure 4-12: The results of m-file for calculating the self-inductances of both coils without using ferrite

Figure 4-13 shows the results of m-file for calculating the mutual inductance between both coils without using ferrite.

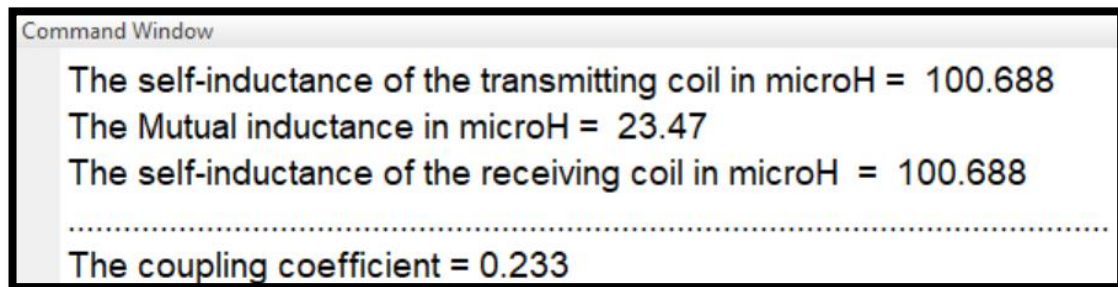


```
Command Window

No. of turns of the transmitting coil = 10
No. of turns of the receiving coil = 10
The square side in mm = 450
The gap distance between the two coils in mm = 150
.....
The mutual inductance = 23.47 microH
```

Figure 4-13: The results of m-file for calculating the mutual inductance between both coils without using ferrite

Figure 4-14 shows the results of m-file for calculating the coupling coefficient between both coils without using ferrite.

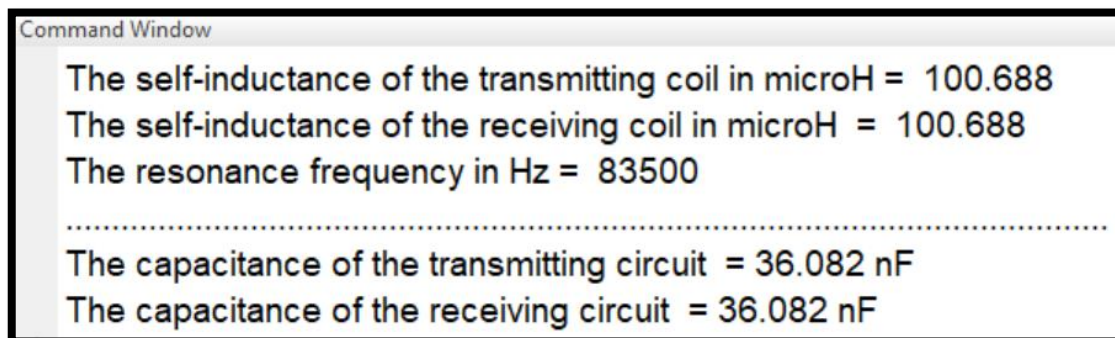


```
Command Window

The self-inductance of the transmitting coil in microH = 100.688
The Mutual inductance in microH = 23.47
The self-inductance of the receiving coil in microH = 100.688
.....
The coupling coefficient = 0.233
```

Figure 4-14: The results of m-file for calculating the coupling coefficient between both coils without using ferrite

Figure 4-15 shows the results of m-file for calculating the capacitances of the transmitting and the receiving circuits without using ferrite.

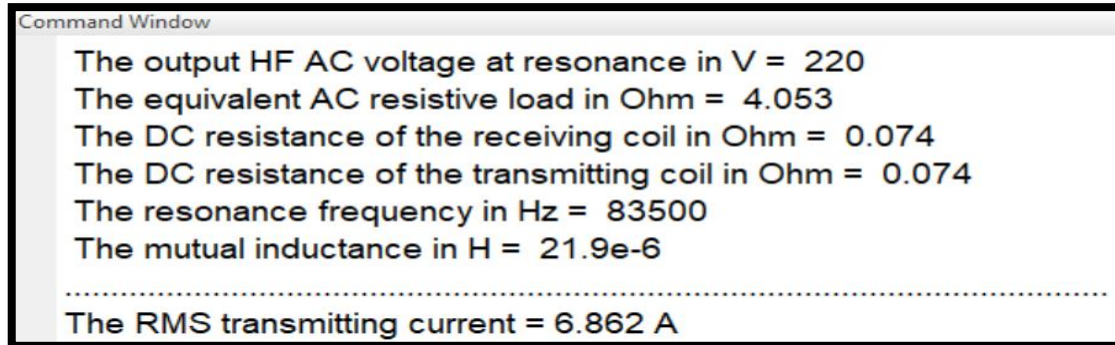


```
Command Window

The self-inductance of the transmitting coil in microH = 100.688
The self-inductance of the receiving coil in microH = 100.688
The resonance frequency in Hz = 83500
.....
The capacitance of the transmitting circuit = 36.082 nF
The capacitance of the receiving circuit = 36.082 nF
```

Figure 4-15: The results of m-file for calculating the capacitances of the transmitting and the receiving circuits without using ferrite

Figure 4-16 shows the results of m-file for calculating the RMS transmitting current with using ferrite.

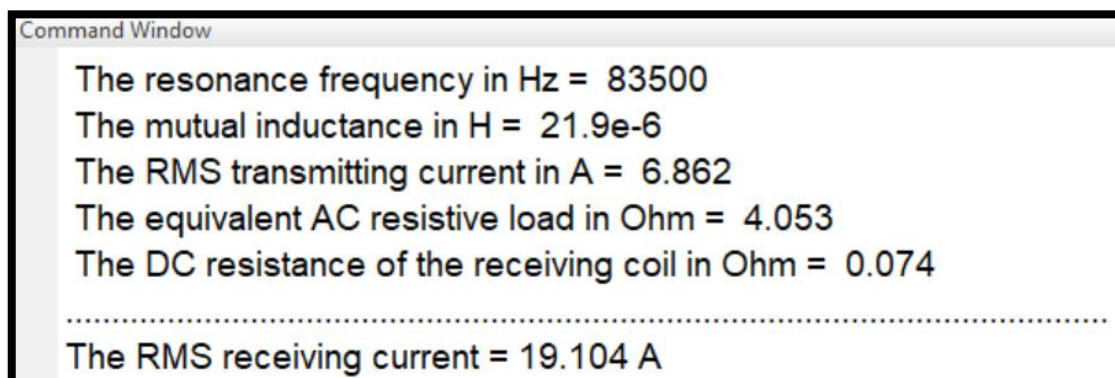


```
Command Window

The output HF AC voltage at resonance in V = 220
The equivalent AC resistive load in Ohm = 4.053
The DC resistance of the receiving coil in Ohm = 0.074
The DC resistance of the transmitting coil in Ohm = 0.074
The resonance frequency in Hz = 83500
The mutual inductance in H = 21.9e-6
.....
The RMS transmitting current = 6.862 A
```

Figure 4-16: The results of m-file for calculating the RMS transmitting current with using ferrite

Figure 4-17 shows the results of m-file for calculating the RMS receiving current with using ferrite.

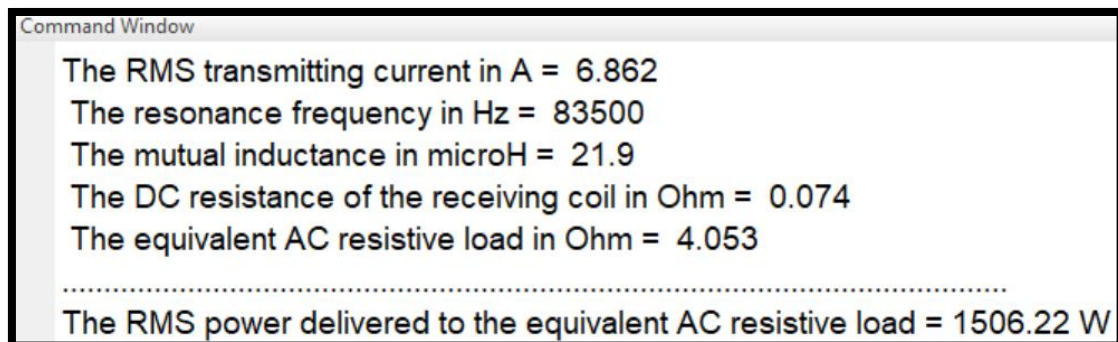


```
Command Window

The resonance frequency in Hz = 83500
The mutual inductance in H = 21.9e-6
The RMS transmitting current in A = 6.862
The equivalent AC resistive load in Ohm = 4.053
The DC resistance of the receiving coil in Ohm = 0.074
.....
The RMS receiving current = 19.104 A
```

Figure 4-17: The results of m-file for calculating the RMS receiving current with using ferrite

Figure 4-18 shows the results of m-file for calculating the RMS power delivered to the equivalent AC resistive load.

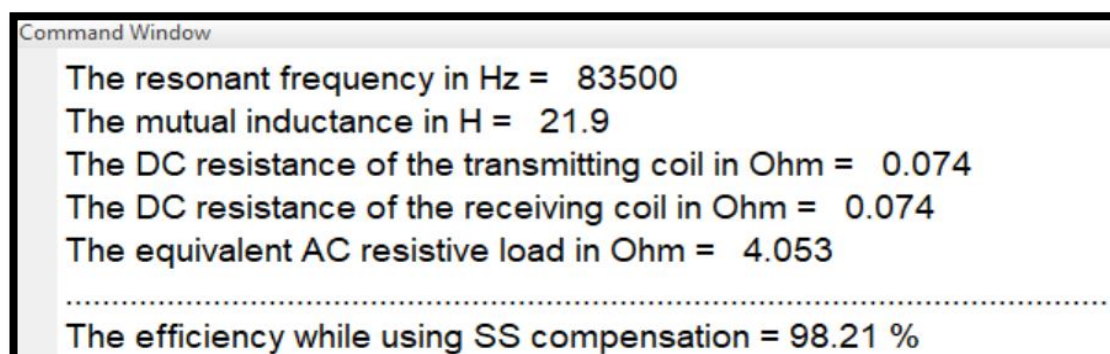


```
Command Window

The RMS transmitting current in A = 6.862
The resonance frequency in Hz = 83500
The mutual inductance in microH = 21.9
The DC resistance of the receiving coil in Ohm = 0.074
The equivalent AC resistive load in Ohm = 4.053
.....
The RMS power delivered to the equivalent AC resistive load = 1506.22 W
```

Figure 4-18: The results of m-file for calculating the RMS power delivered to the equivalent AC resistive load

Figure 4-19 shows the results of m-file for calculating the efficiency while using the SS compensation.



```
Command Window

The resonant frequency in Hz = 83500
The mutual inductance in H = 21.9
The DC resistance of the transmitting coil in Ohm = 0.074
The DC resistance of the receiving coil in Ohm = 0.074
The equivalent AC resistive load in Ohm = 4.053
.....
The efficiency while using SS compensation = 98.21 %
```

Figure 4-19: The results of m-file for calculating the efficiency while using the SS compensation

Figure 4-20 shows the transmitter's AC current wave obtained from ANSYS Simpler simulation in case of 15 cm air gap.

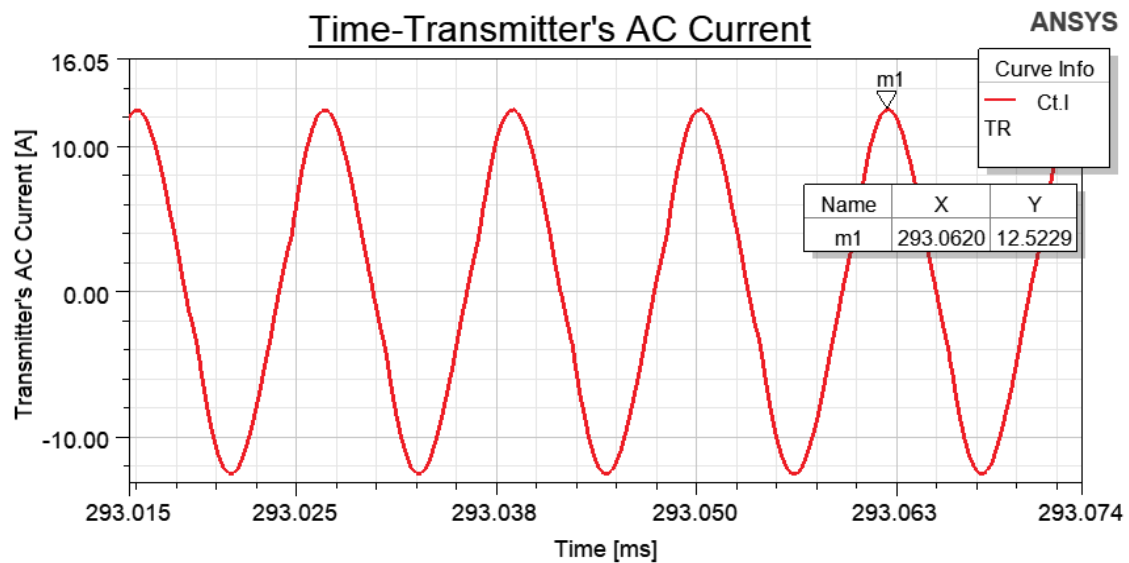


Figure 4-20: The transmitter's AC current wave obtained from ANSYS Simpler simulation in case of 15 cm air gap

Figure 4-21 shows the receiver's AC current wave obtained from ANSYS Simpler simulation in case of 15 cm air gap.

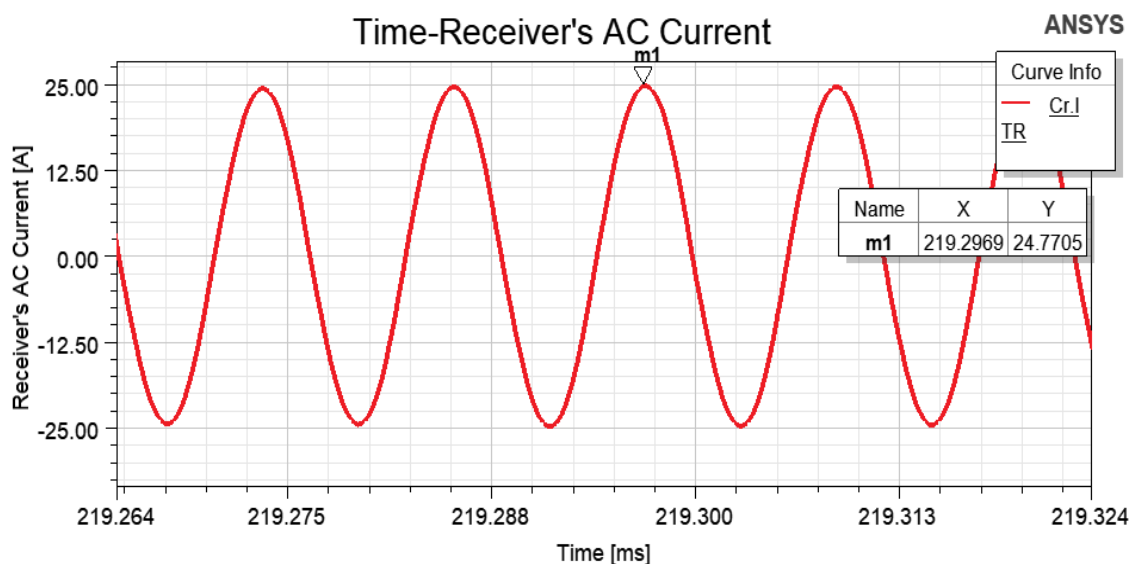


Figure 4-21: The receiver's AC current wave obtained from ANSYS Simpler simulation in case of 15 cm air gap

Figure 4-22 shows the load DC current through the resistive load obtained from ANSYS Simpler simulation in case of 15 cm air gap.

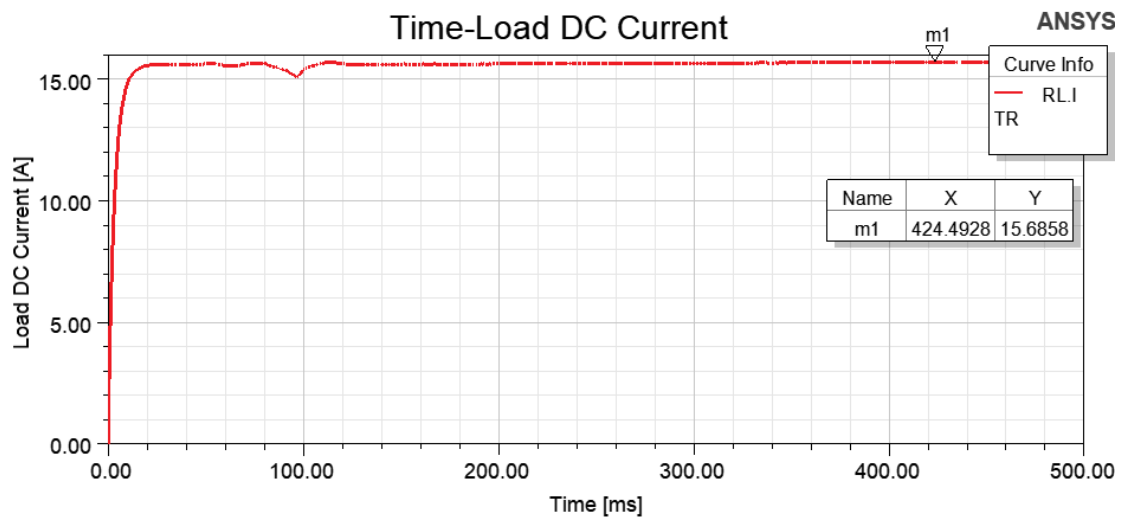


Figure 4-22: The load DC current through the resistive load obtained from ANSYS Simpler simulation in case of 15 cm air gap

Figure 4-23 shows the load DC voltage at the resistive load obtained from ANSYS Simpler simulation in case of 15 cm air gap.

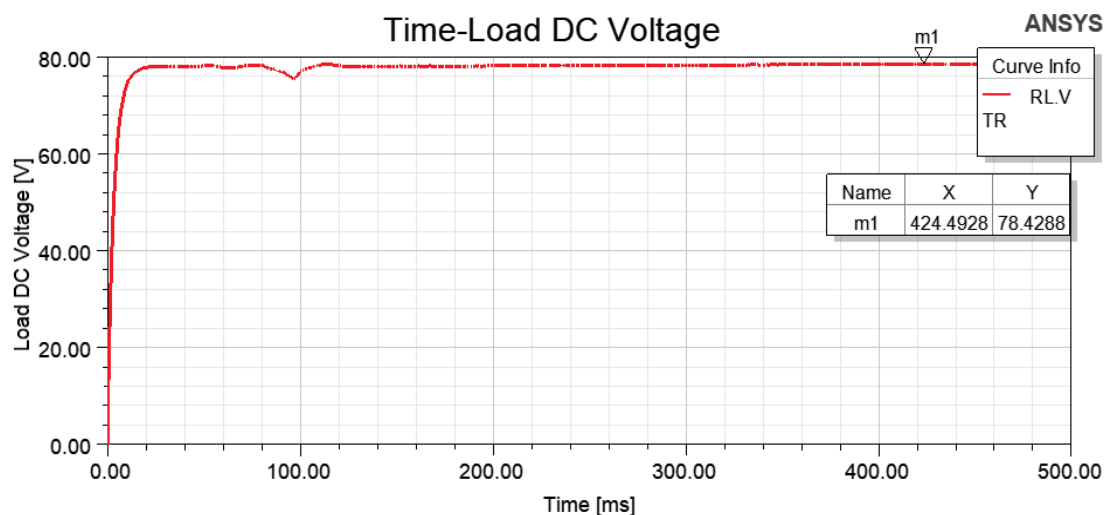


Figure 4-23: The load DC voltage at the resistive load obtained from ANSYS Simpler simulation in case of 15 cm air gap

Figure 4-24 shows the transmitter's AC current wave obtained from MATLAB Simulink in case of 15 cm air gap.

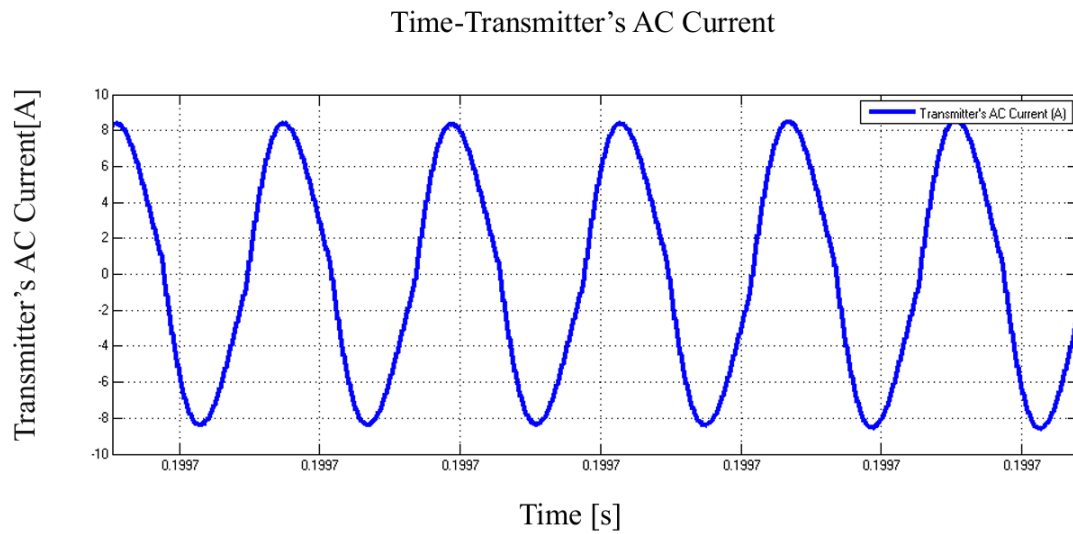


Figure 4-24: The transmitter's AC current wave obtained from MATLAB Simulink in case of 15 cm air gap

Figure 4-25 shows the receiver's AC current wave obtained from MATLAB Simulink in case of 15 cm air gap.

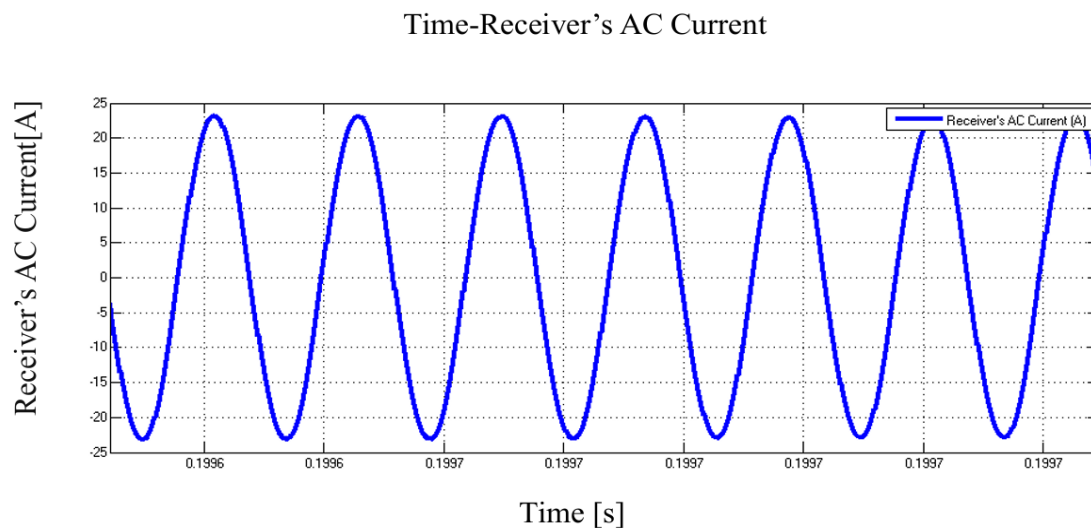


Figure 4-25: The receiver's AC current wave obtained from MATLAB Simulink in case of 15 cm air gap

Figure 4-26 shows the load DC current through the resistive load obtained from MATLAB Simulink in case of 15 cm air gap.

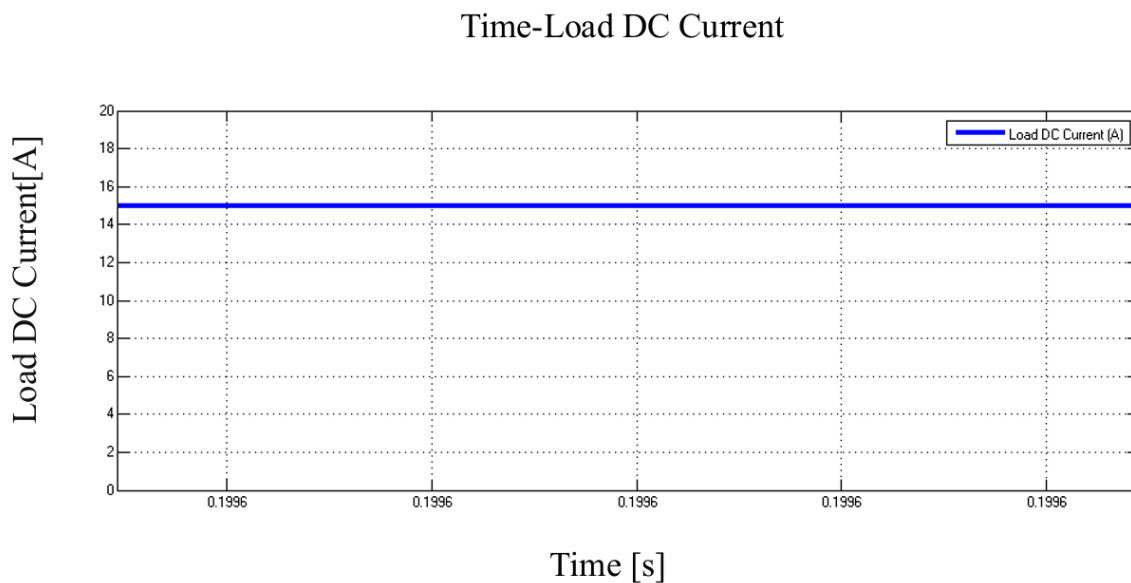


Figure 4-26: The load DC current through the resistive load obtained from MATLAB Simulink in case of 15 cm air gap

Figure 4-27 shows the load DC voltage through the resistive load obtained from MATLAB Simulink in case of 15 cm air gap.

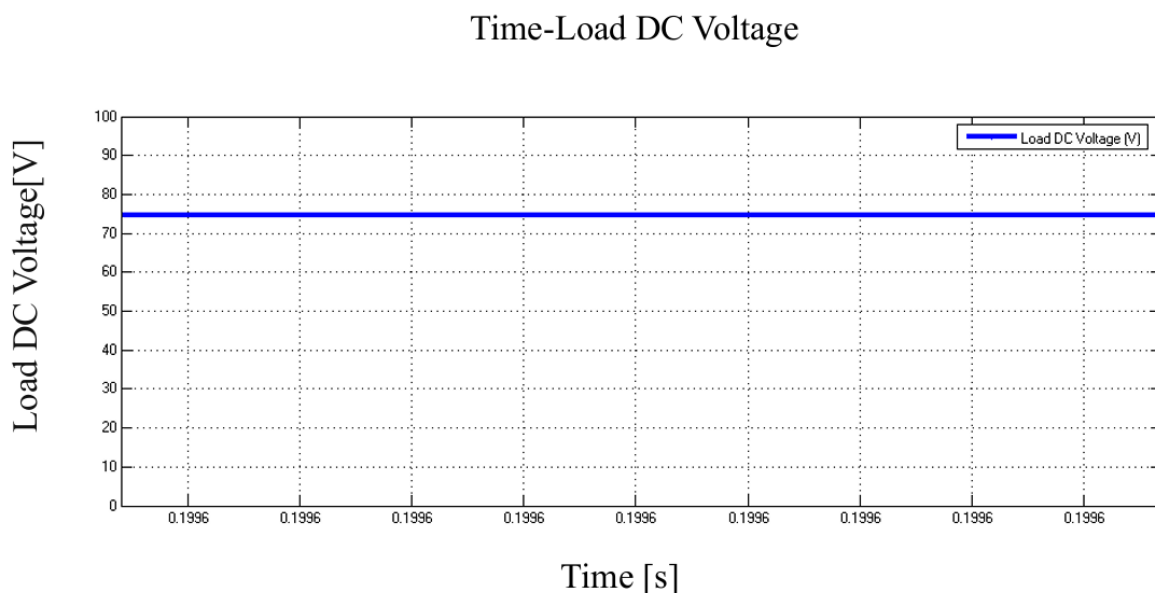


Figure 4-27: The load DC voltage through the resistive load obtained from MATLAB Simulink in case of 15 cm air gap

Table 4-8: The comparison between the ANSYS Simplorer, MATLAB Simulink, calculated and practical values of the different electrical parameters

The different electrical parameters	Obtained value from ANSYS Simplorer	Obtained value from MATLAB Simulink	Calculated Value	Practical value
<i>Transmitter's RMS current</i>	8.855 A	6.01 A	7.935 A	—
<i>Receiver's RMS current</i>	17.51 A	16.79 A	19.095 A	—
<i>The load DC voltage</i>	78.43 V	75 V	—	64.15
<i>The load DC current</i>	15.68 A	15 A	—	12.83

All results obtained from ANSYS Simplorer are according to the electrical circuit shown in Figure 3-5.

This electrical circuit starts with a square wave 83.5 KHz, 220 V RMS (the RMS value for the square wave voltage source is same as the peak value). This voltage source represents the output HF voltage from the HFI.

The reason for not starting in ANSYS Simplorer with DC supply feeds the MOSFETs, is that the solver time of ANSYS Simplorer while using many electronic components is slow.

Other than the simulation done by MATLAB Simulink which according to the electrical circuit shown in Figure 3-15 , the starting of the circuit is the grid feeds the DC supply. This is due to the fast solver time of MATLAB Simulink.

As we see, the results obtained from both programs are close together.

Figure 4-28 shows the relation between the frequency and output power, which indicates that the output power is maximum at the resonant frequency.

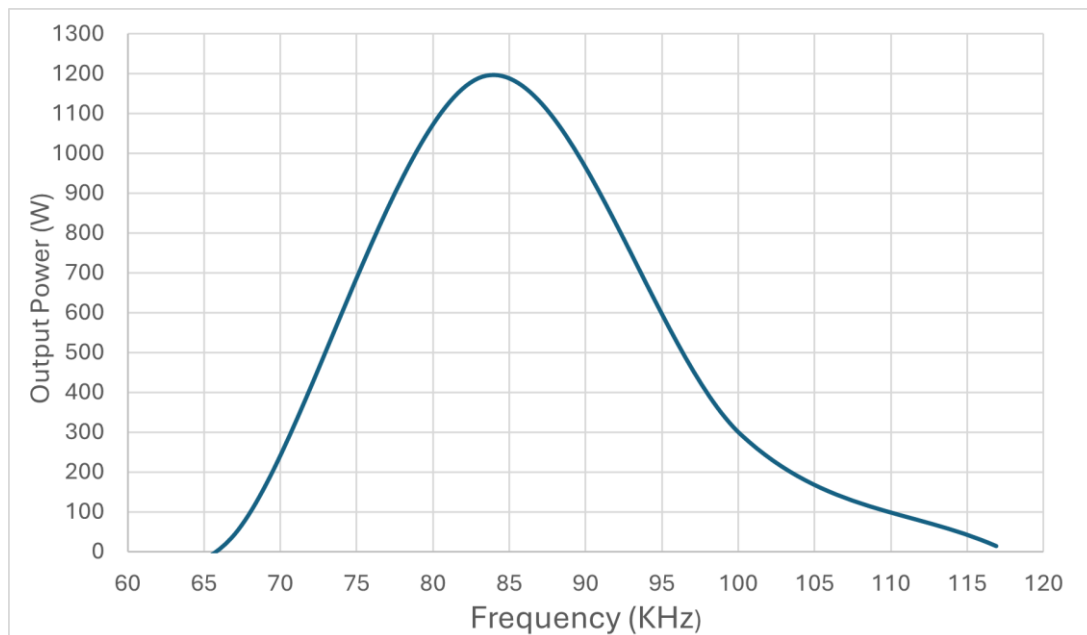


Figure 4-28: The relation between the frequency and output power

Figure 4-29 shows the relation between the frequency and the efficiency, which indicates that the efficiency is maximum at the resonant frequency.

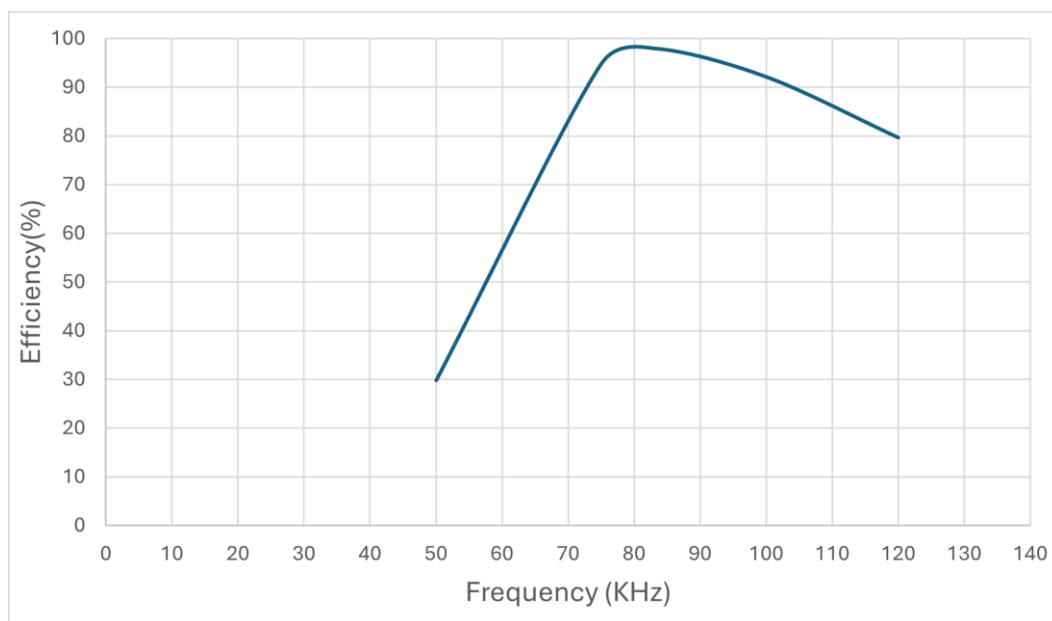


Figure 4-29: The relation between the frequency and the efficiency

The previous two relations are done for the system shown in Figure 4-9, with a 150 mm gap between both coils, with the output HF AC voltage of a 220 V.

There is only one resonant frequency in the X-axis of both previous figures, and each frequency of the other frequencies is not a resonant one.

Meaning that the capacitances of the SS compensation are kept constants for all frequencies, achieving only the resonant frequency of a 83.5 KHz.

The maximum output power and the maximum efficiency are obtained at this only resonant frequency.

4.2 Practical results

Figure 4-30 shows the transmitter electrical box which includes the HFI and the series transmitter's compensation.

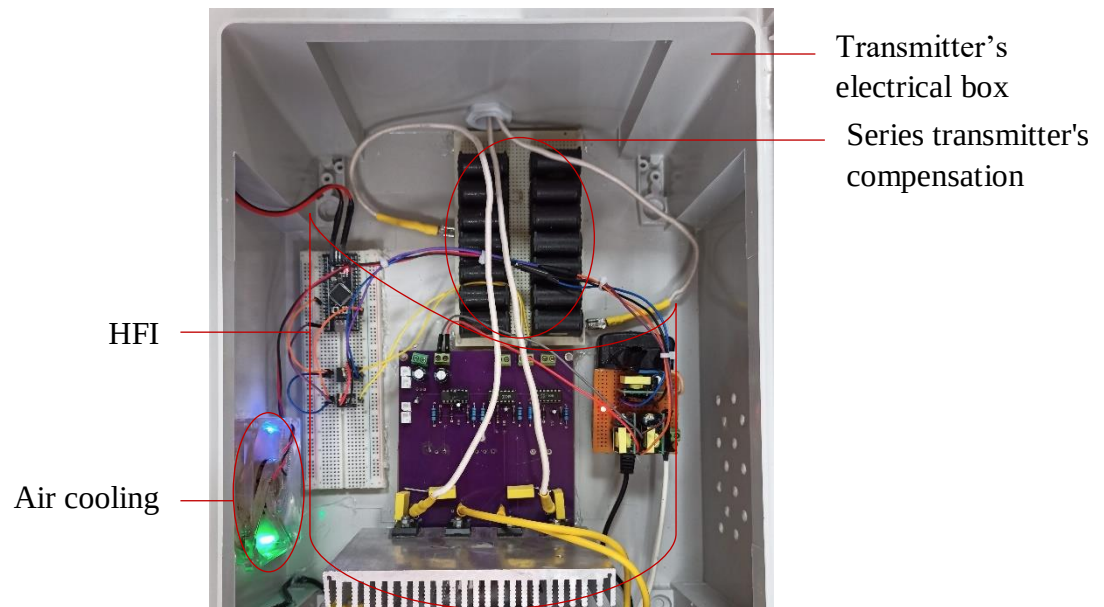


Figure 4-30: The transmitter electrical box which includes the HFI and the series transmitter's compensation

Figure 4-31 shows the receiver electrical box which includes the series receiver's compensation and the AC/DC converter.

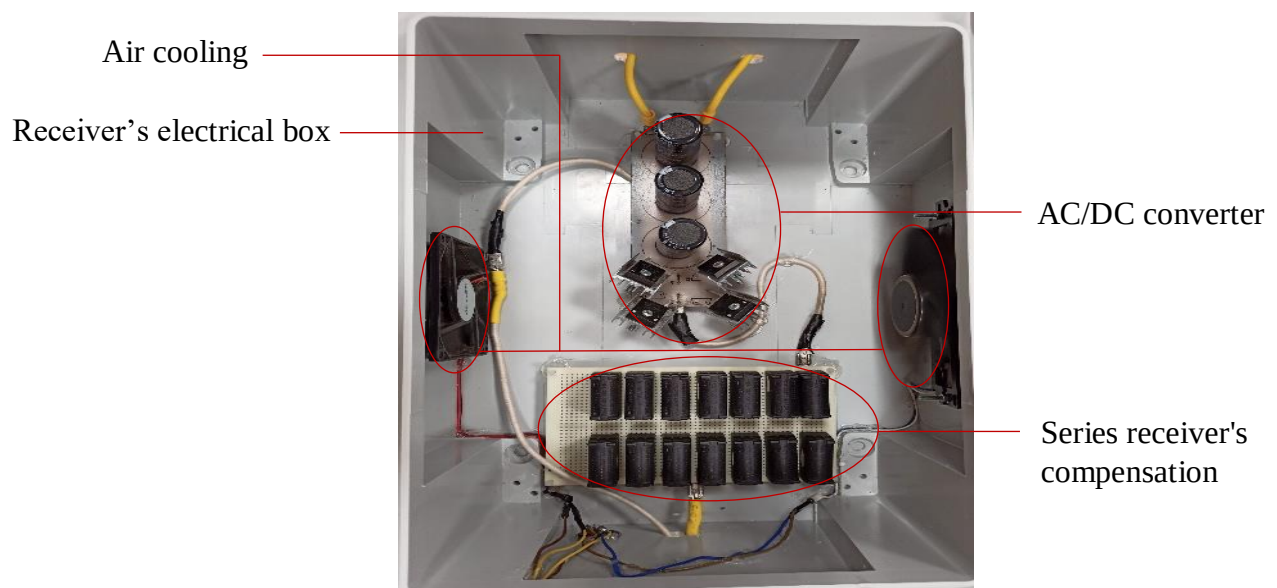


Figure 4-31: The receiver electrical box which includes the series receiver's compensation and the AC/DC converter

Figure 4-32 shows the practical RIWPT system

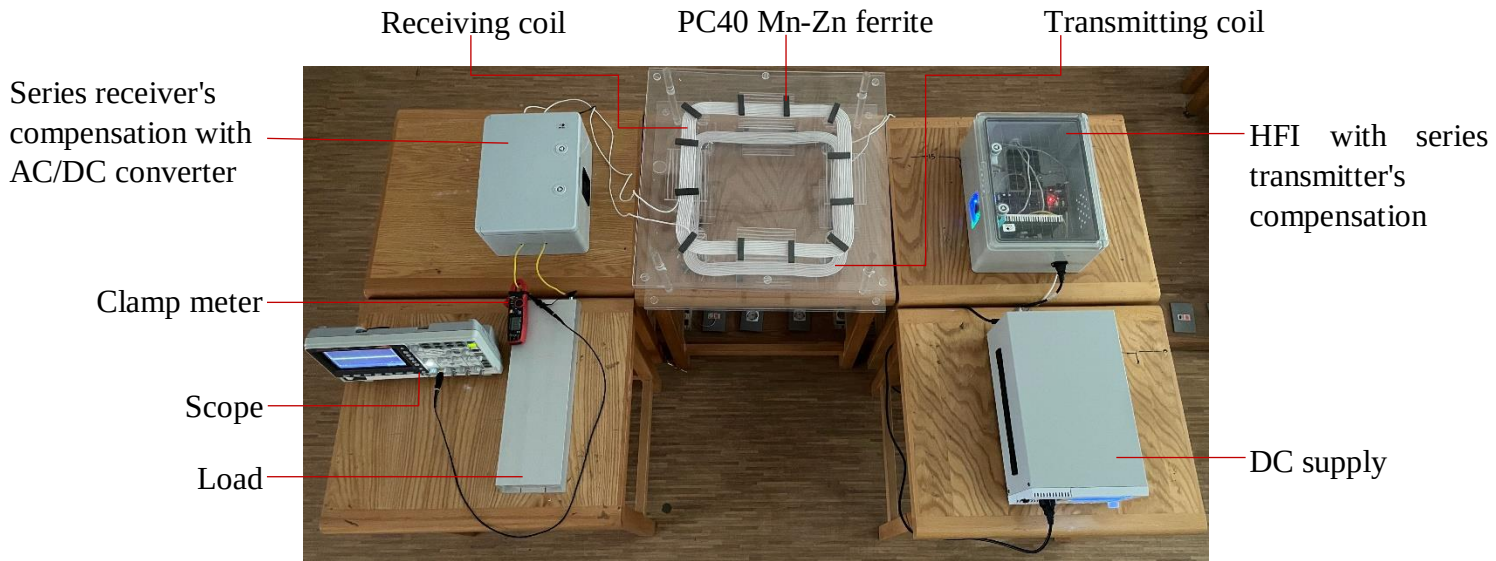


Figure 4-32: The practical RIWPT system

Figure 4-33 shows the reading of the DC voltage through the load by a laboratory scope.

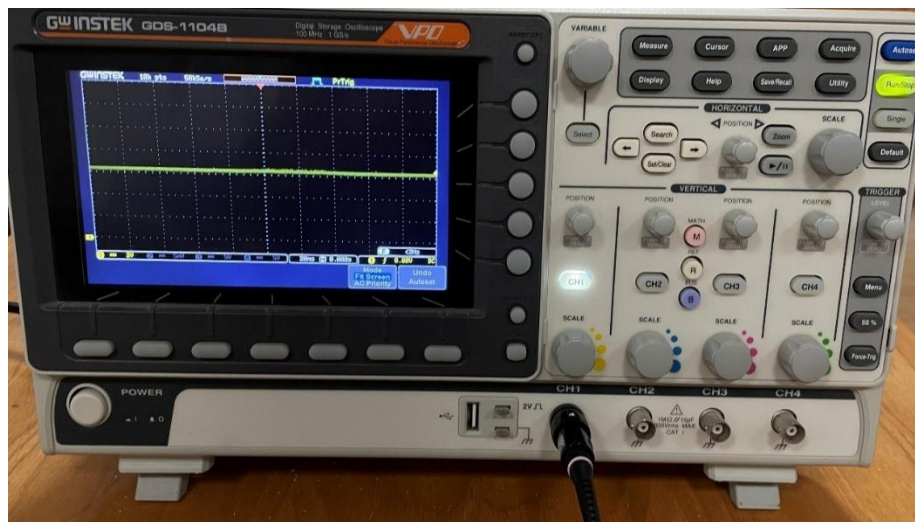


Figure 4-33: The reading of the DC voltage through the load by a laboratory scope

The attenuator in an oscilloscope is set to 10x and the amplitude of the input signal is reduced to 1/10th of its original value, The voltage per division in oscilloscope is set to 2V/div and the oscilloscope reads approximately 6.415V, So the original input signal is about 64.15V.

Figure 4-34 shows the reading of the DC current through the load by a laboratory clamp meter.



Figure 4-34: The reading of the DC current through the load by a laboratory clamp meter

Table 4-9: Designed parameters of the practical system

Symbol/ Abbreviation	Parameter	Value
V_{in}	Input RMS supply voltage	187.1 V
I_{in}	Input RMS supply current	5.74 A
S_{in}	Input apparent power	1074 VA
Q_{in}	Input reactive power	166 Var
P_{in}	Input active power	1061 W
PF	Power factor	0.99
CF	Crest factor	1.33
V_{DC}	DC output voltage from the DC supply (Input DC voltage to HFI)	220 V
u	The square side length of each coil	450 mm
T_t	No. of turns of the transmitting coil	10 turns
T_r	No. of turns of the receiving coil	10 turns
S_i	Surface copper area of each turn, without the insulation	4.17 mm ²
S_o	Surface area of each turn, considering the insulation	8.3 mm ²
h	The gap distance between the two coils	15 cm
L_t	The self-inductance of the transmitting coil	116.7 μ H
L_r	The self-inductance of the receiving coil	117.13 μ H
M	The mutual inductance	20 μ H
K	The coupling coefficient	0.171
f	The resonant frequency	83500 Hz
C_t	The capacitance of the receiving circuit	29.555 nF
C_r	The capacitance of the transmitting circuit	29.47 nF
C_F	The filtering capacitance before the load	660 μ F
R_L	The DC resistive load	5 Ω
V_L	Output load DC voltage	64.15 V
I_L	Output load DC current	12.83 A
P_L	Output load DC power	823 W
η_{s-s}	Practical efficiency	77.5 %

The power supply has a step-down transformer which converts the grid 220V RMS to 187.1 V RMS. The RMS input current taken by the power supply from the grid is 5.74 A. The input apparent power is 187.1V multiplied by 5.74 A, which equals 1074 VA.

The power supply readings show that there is a small value of the reactive power drawn by the proposed system, which equals 166 VAr. The input active power is 1061 W, so the input power factor equals the input active power divided by the input apparent power, which equals to 0.99 approximately.

The measured Crest Factor of 1.33, instead of the ideal 1.414 for a pure sine wave, indicates that the output waveform of the power supply is not a perfect sinusoidal signal. This deviation can be due to several reasons: most laboratory power supplies produce a filtered or modified output waveform. As a result, the waveform has lower peak values which reduces the Crest Factor. Additionally, waveform distortion caused by internal filtering circuits or the power supply design itself can also lead to a lower Crest Factor.

The RMS 187.1 V is converted by the power supply into a DC 220 V, which is the input to the HFI.

The power pads were square shaped coils, which have a 450 mm square side and a 150 mm as an air gap distance between them. Each coil was covered by a 12 piece of PC40 Mn-Zn ferrite for improving the coupling coefficient.

The self-inductances of both coils were measured by the LCR meter. The SS compensation is designed so that a resonant frequency of an 83.5 KHz was achieved, and the capacitances of a SS compensation were measured by LCR meter.

The output load DC voltage measured by the laboratory scope is 64.15 V which is relatively different from values obtained from simulations, and this is because that the magnetic and switching losses were not considered in the simulations. The output load DC current measured by a clamp meter device is 12.83 A.

The output load DC power is the output load DC voltage multiplied by the output load DC current, or it is the square of the output load DC voltage divided by the value of the resistive load ($5\ \Omega$), or it is the square of the output load DC current multiplied by the value of the resistive load ($5\ \Omega$). The output load DC power calculated by any previous method equals 823 W.

By dividing the output load DC power by the input active power, we can calculate the practical efficiency which equals to 77.5%, a difference arises between the efficiency values obtained from the practical and those calculated using the MATLAB code because of a certain factors such as magnetic and switching losses were not accounted for in the MATLAB code by which the efficiency was 98.21%.

5 Chapter (5): Conclusion

5.1 Overview

The project presented an 823 W RIWPT system using SS compensation and square coupling coils. The SS compensation topology was chosen due to its several advantages, including the ability to size the capacitors during the design independently of load or coupling variations, stable voltage source characteristics and built-in short-circuit protection. A resonant frequency of 83.5 kHz was selected for the system's operation.

The system parameters were computed using a MATLAB script, based on the winding dimensions and the selected SS compensation scheme. Using the calculated parameters, the RIWPT system model was developed in ANSYS Maxwell 3D and integrated with ANSYS Simplorer. The square inductive coils were designed using ANSYS Maxwell 3D electromagnetic simulation software, which enables detailed analysis of the magnetic field distribution.

Simulation results indicate that the magnetic field strength and consequently, power transfer efficiency are maximized when the vertical distance between the transmitter and receiver coils is minimized. Using Litz wire significantly enhances efficiency by minimizing coil resistance. Two key parameters influencing the efficiency of a WPT system are the coupling coefficient and the mutual inductance between the coils, which are highly dependent on the relative positioning of the coils. Any vertical misalignment between the transmitting and receiving coils reduces both the coupling coefficient and the mutual inductance, resulting in a decrease in power transfer efficiency.

The efficiency of the proposed system under ideal operating conditions was found to be 77.5%. However, certain factors, such as magnetic and switching losses, were not considered in the MATLAB code, which yielded a theoretical efficiency of 98.21%. This discrepancy highlights the difference between the practical efficiency values and those calculated using the MATLAB code.

5.2 Challenges

The proposed prototype faces several limitations, including high implementation costs, constraints related to weight and size of electronic components, susceptibility to magnetic interference, sensitivity to temperature variations, a limited operating range, potential safety concerns, and relatively low power transfer efficiency.

6 Chapter (6): References

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7 Chapter (7): Appendices

7.1 MATLAB m-file codes

```
Tt=input('No. of turns of the transmitting coil = ');
Tr=input('No. of turns of the receiving coil = ');
So=input('The surface area of each turn in mm^2 = ');

[rt,rr]=HalfSide(Tt,Tr,So);

fprintf('.....\n');
fprintf('The half of side length of the transmitting coil = %0.2f mm\n',rt);
fprintf('The half of side length of the receiving coil = %0.2f mm\n',rr);
```

```
Tt=input('No. of turns of the transmitting coil = ');
Tr=input('No. of turns of the receiving coil = ');
Si=input('The surface copper area of each turn in mm^2 = ');
wt=input('The outer width of the transmitting coil in mm = ');
wr=input('The outer width of the receiving coil in mm = ');
lt=input('The outer length of the transmitting coil in mm = ');
lr=input('The outer length of the receiving coil in mm = ');

[Rt,Rr]=Resistance(Tt,Tr,Si,wt,wr,lt,lr);

fprintf('.....\n');
fprintf('The DC resistance of the transmitting coil = %0.3f Ohm\n',Rt);
fprintf('The DC resistance of the receiving coil = %0.3f Ohm\n',Rr);
```

```

Tt=input('No. of turns of the transmitting coil = ');
Tr=input('No. of turns of the receiving coil = ');
So=input('The surface area of each turn in mm^2 = ');
wt=input('The outer width of the transmitting coil in mm = ');
wr=input('The outer width of the receiving coil in mm = ');
lt=input('The outer length of the transmitting coil in mm = ');
lr=input('The outer length of the receiving coil in mm = ');

[rt,rr]=HalfSide(Tt,Tr,So);
[Lt,Lr]=Self(Tt,Tr,rt,rr,wt,wr,lt,lr);

fprintf('.....\n');
fprintf('The self-inductance of the transmitting coil = %0.3f microH\n',Lt);
fprintf('The self-inductance of the receiving coil = %0.3f microH\n',Lr);

```

```

Tt=input('No. of turns of the transmitting coil = ');
Tr=input('No. of turns of the receiving coil = ');
u=input('The square side in mm = ');
h=input('The gap distance between the two coils in mm = ');

M=Mutual(Tt,Tr,u,h);

fprintf('.....\n');
fprintf('The mutual inductance = %0.2f microH\n',M);

```

```
Lt=input('The self-inductance of the transmitting coil in microH = ');
M=input('The Mutual inductance in microH = ');
Lr=input('The self-inductance of the receiving coil in microH = ');

K=Coupling(Lt,M,Lr);

fprintf('.....\n');
fprintf('The coupling coefficient = %0.3f \n',K);
```

```
Lt=input('The self-inductance of the transmitting coil in microH = ');
Lr=input('The self-inductance of the receiving coil in microH = ');
f=input('The resonance frequency in Hz = ');

[Ct,Cr]=Compensation(Lt,Lr,f);

fprintf('.....\n');
fprintf('The capacitance of the transmitting circuit = %0.3f nF\n',Ct);
fprintf('The capacitance of the receiving circuit = %0.3f nF\n',Cr);
```

```
V1o=input(' The output HF AC voltage at resonance in V = ');
RLeq=input(' The equivalent AC resistive load in Ohm = ');
Rr=input(' The DC resistance of the receiving coil in Ohm = ');
Rt=input(' The DC resistance of the transmitting coil in Ohm = ');
f=input(' The resonance frequency in Hz = ');
M=input(' The mutual inductance in H = ');

It=TransmitterCurrent(V1o,RLeq,Rr,Rt,f,M);

fprintf('.....\n');
fprintf('The RMS transmitting current = %0.3f A\n',It);
```

```
f=input(' The resonance frequency in Hz = ');
M=input(' The mutual inductance in H = ');
It=input(' The RMS transmitting current in A = ');
RLeq=input(' The equivalent AC resistive load in Ohm = ');
Rr=input(' The DC resistance of the receiving coil in Ohm = ');

Ir=ReceiverCurrent(It,f,M,RLeq,Rr);

fprintf('.....\n');
fprintf('The RMS receiving current = %0.3f A\n',Ir);
```

```
It=input('The RMS transmitting current in A = ');
f=input(' The resonance frequency in Hz = ');
M=input(' The mutual inductance in microH = ');
Rr=input(' The DC resistance of the receiving coil in Ohm = ');
RLeq=input(' The equivalent AC resistive load in Ohm = ');

P=Power(It,f,M,Rr,RLeq);

fprintf('.....\n');
fprintf('The RMS power delivered to the equivalent AC resistive load =
%0.2f W\n',P);
```

```
f=input('The resonant frequency in Hz = ');
M=input('The mutual inductance in H = ');
Rt=input('The DC resistance of the transmitting coil in Ohm = ');
Rr=input('The DC resistance of the receiving coil in Ohm = ');
RLeq=input('The equivalent AC resistive load in Ohm = ');

E=Efficiency(f,M,Rt,Rr,RLeq);

fprintf('.....\n');
fprintf('The efficiency while using SS compensation = %0.2f %% \n',E);
```

7.2 MATLAB function codes

```
function [rt,rr]=HalfSide(Tt,Tr,So)
rt=Tt*sqrt(So/pi);
rr=Tr*sqrt(So/pi);
End
```

```
function [Rt,Rr]=Resistance(Tt,Tr,Si,wt,wr,lt,lr)
Rt=((1.724e-8)Tt*2(wt+lt)/Si)*1000;
Rr=((1.724e-8)Tr*2(wr+lr)/Si)*1000;
End
```

```
function [Lt,Lr]=Self(Tt,Tr,rt,rr,wt,wr,lt,lr)
at=wt*log( (2*wt*lt) / (rt*(wt+sqrt(((wt)^2)+((lt)^2)))) );
bt=lt*log( (2*wt*lt) / (rt*(lt+sqrt(((wt)^2)+((lt)^2)))) );
ct=wt+lt-sqrt(((wt)^2)+((lt)^2));
ar=wr*log( (2*wr*lr) / (rr*(wr+sqrt(((wr)^2)+((lr)^2)))) );
br=lr*log( (2*wr*lr) / (rr*(lr+sqrt(((wr)^2)+((lr)^2)))) );
cr=wr+lr-sqrt(((wr)^2)+((lr)^2));
Lt=((4*(10^(-7)))(Tt^2)(at+bt-2*ct+0.25*(wt+lt)))*1000;
Lr=((4*(10^(-7)))(Tr^2)(ar+br-2*cr+0.25*(wr+lr)))*1000;
End
```

```

function M=Mutual(Tt,Tr,u,h)
term1=u*sqrt((h^2)+(u^2));
term2=(u^2)+(h^2);
term3=(u*h)+h*sqrt((h^2)+2*(u^2));
term4=h-2*sqrt((h^2)+(u^2))+sqrt((h^2)+2*(u^2));
M=(8*(10^-7)Tt*Tr(u*log((term1+term2)/term3)+term4))*1000;
End

```

```

function K=Coupling(Lt,M,Lr)
K=M/sqrt(Lt*Lr);
End

```

```

function [Ct,Cr]=Compensation(Lt,Lr,f)
Ct=((1/(4*(pi^2)(f^2)*Lt)))(10^15);
Cr=((1/(4*(pi^2)(f^2)*Lr)))(10^15);
End

```

```

function It=TransmitterCurrent(V1o,RLeq,Rr,Rt,f,M)
w=2*pi*f;
It=(V1o*(RLeq+Rr))/((Rt*(RLeq+Rr))+(w*M)^2);
End

```

```

function Ir=ReceiverCurrent(It,f,M,RLeq,Rr)
w=2*pi*f;
Ir=(w*M*It)/(RLeq+Rr);
End

```

```
function P=Power(It,f,M,Rr,RLeq)
w=2*pi*f;
P=((It*w*M)^2)/(Rr+RLeq)*(10^-12);
End
```

```
function E=Efficiency(f,M,Rt,Rr,RLeq)
w=2*pi*f;
term1=Rt*((Rr+RLeq)^2);
term2=(w*M)^2;
E=(RLeq/(RLeq+Rr+(term1/term2)))*100;
End
```

7.3 Stm32CubeIDE code

```
/* USER CODE BEGIN Header */  
  
/**  
  
*****  
  
 * @file      : main.c  
  
 * @brief     : Main program body  
  
*****  
  
 * @attention  
  
 *  
  
 * Copyright (c) 2025 STMicroelectronics.  
  
 * All rights reserved.  
  
 *  
  
 * This software is licensed under terms that can be found in the LICENSE  
file  
  
 * in the root directory of this software component.  
  
 * If no LICENSE file comes with this software, it is provided AS-IS.  
  
 *  
  
*****  
  
*/  
  
/* USER CODE END Header */  
  
/* Includes -----*/  
  
#include "main.h"  
  
  
/* Private includes -----*/  
  
/* USER CODE BEGIN Includes */  
  
  
/* USER CODE END Includes */
```

```
/* Private typedef -----*/
```

```
/* USER CODE BEGIN PTD */
```

```
/* USER CODE END PTD */
```

```
/* Private define -----*/
```

```
/* USER CODE BEGIN PD */
```

```
/* USER CODE END PD */
```

```
/* Private macro -----*/
```

```
/* USER CODE BEGIN PM */
```

```
/* USER CODE END PM */
```

```
/* Private variables -----*/
```

```
TIM_HandleTypeDef htim1;
```

```
/* USER CODE BEGIN PV */
```

```
/* USER CODE END PV */
```

```
/* Private function prototypes -----*/
```

```
void SystemClock_Config(void);
```

```
static void MX_GPIO_Init(void);
```

```
static void MX_TIM1_Init(void);
```

```
/* USER CODE BEGIN PFP */
```

```
/* USER CODE END PFP */
```

```
/* Private user code -----*/
```

```
/* USER CODE BEGIN 0 */
```

```
/* USER CODE END 0 */
```

```
/**
```

```
 * @brief The application entry point.
```

```
 * @retval int
```

```
 */
```

```
int main(void)
```

```
{
```

```
/* USER CODE BEGIN 1 */
```

```
/* USER CODE END 1 */
```

```
/* MCU Configuration-----*/
```

```
/* Reset of all peripherals, Initializes the Flash interface and the Systick.
 */
```

```
HAL_Init ();
```

```
/* USER CODE BEGIN Init */
```

```
/* USER CODE END Init */
```

```
/* Configure the system clock */
SystemClock_Config ();

/* USER CODE BEGIN SysInit */

/* USER CODE END SysInit */

/* Initialize all configured peripherals */
MX_GPIO_Init ();
MX_TIM1_Init ();
/* USER CODE BEGIN 2 */

HAL_TIM_PWM_Start(&htim1, TIM_CHANNEL_1);
HAL_TIMEx_PWMN_Start(&htim1, TIM_CHANNEL_1);

/* USER CODE END 2 */

/* Infinite loop */
/* USER CODE BEGIN WHILE */
while (1)
{
    /* USER CODE END WHILE */

    /* USER CODE BEGIN 3 */
}
```

```
/* USER CODE END 3 */

}

/**
 * @brief System Clock Configuration
 * @retval None
 */
void SystemClock_Config(void)
{
    RCC_OscInitTypeDef RCC_OscInitStruct = {0};
    RCC_ClkInitTypeDef RCC_ClkInitStruct = {0};

    /** Configure the main internal regulator output voltage
     */
    __HAL_RCC_PWR_CLK_ENABLE ();

    __HAL_PWR_VOLTAGESCALING_CONFIG(PWR_REGULATOR_
    VOLTAGE_SCALE2);

    /** Initializes the RCC Oscillators according to the specified parameters
     * in the RCC_OscInitTypeDef structure.
     */
    RCC_OscInitStruct.OscillatorType = RCC_OSCILLATORTYPE_HSI;
    RCC_OscInitStruct.HSIState = RCC_HSI_ON;
    RCC_OscInitStruct.HSICalibrationValue =
    RCC_HSICALIBRATION_DEFAULT;
    RCC_OscInitStruct.PLL.PLLState = RCC_PLL_NONE;
    if (HAL_RCC_OscConfig(&RCC_OscInitStruct) != HAL_OK)
```

```
{
    Error_Handler ();
}

/** Initializes the CPU, AHB and APB buses clocks
 */

RCC_ClkInitStruct.ClockType=
RCC_CLOCKTYPE_HCLK|RCC_CLOCKTYPE_SYSCLK
|RCC_CLOCKTYPE_PCLK1|RCC_CLOCKTYPE_PCLK2;
RCC_ClkInitStruct.SYSCLKSource = RCC_SYSCLKSOURCE_HSI;
RCC_ClkInitStruct.AHBCLKDivider = RCC_SYSCLK_DIV1;
RCC_ClkInitStruct.APB1CLKDivider = RCC_HCLK_DIV1;
RCC_ClkInitStruct.APB2CLKDivider = RCC_HCLK_DIV1;

If(HAL_RCC_ClockConfig(&RCC_ClkInitStruct,FLASH_LATENCY_0
) != HAL_OK)
{
    Error_Handler();
}
}

/**
 * @brief TIM1 Initialization Function
 * @param None
 * @retval None
 */
static void MX_TIM1_Init(void)
{
```

```
/* USER CODE BEGIN TIM1_Init 0 */

/* USER CODE END TIM1_Init 0 */

TIM_MasterConfigTypeDef sMasterConfig = {0};
TIM_OC_InitTypeDef sConfigOC = {0};
TIM_BreakDeadTimeConfigTypeDef sBreakDeadTimeConfig = {0};

/* USER CODE BEGIN TIM1_Init 1 */

/* USER CODE END TIM1_Init 1 */
htim1.Instance = TIM1;
htim1.Init.Prescaler = 0;
htim1.Init.CounterMode = TIM_COUNTERMODE_UP;
htim1.Init.Period = 193;
htim1.Init.ClockDivision = TIM_CLOCKDIVISION_DIV1;
htim1.Init.RepetitionCounter = 0;
htim1.Init.AutoReloadPreload =
TIM_AUTORELOAD_PRELOAD_DISABLE;
if (HAL_TIM_PWM_Init(&htim1) != HAL_OK)
{
    Error_Handler();
}
sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;
sMasterConfig.MasterSlaveMode =
TIM_MASTERSLAVEMODE_DISABLE;
```

```
if(HAL_TIMEx_MasterConfigSynchronization(&htim1,&sMasterConfig
) != HAL_OK)
{
    Error_Handler();
}
sConfigOC.OCMode = TIM_OCMODE_PWM1;
sConfigOC.Pulse = 96;
sConfigOC.OCpolarity = TIM_OCPOLARITY_LOW;
sConfigOC.OCNPolarity = TIM_OCNPOLARITY_LOW;
sConfigOC.OCFastMode = TIM_OCFAST_DISABLE;
sConfigOC.OCIdleState = TIM_OCIDLESTATE_RESET;
sConfigOC.OCNIdleState = TIM_OCNIDLESTATE_RESET;
if(HAL_TIM_PWM_ConfigChannel(&htim1,&sConfigOC,TIM_CHAN
NEL_1) != HAL_OK)
{
    Error_Handler();
}
sBreakDeadTimeConfig.OffStateRunMode = TIM_OSSR_DISABLE;
sBreakDeadTimeConfig.OffStateIDLEMode = TIM_OSSI_DISABLE;
sBreakDeadTimeConfig.LockLevel = TIM_LOCKLEVEL_OFF;
sBreakDeadTimeConfig.DeadTime = 2;
sBreakDeadTimeConfig.BreakState = TIM_BREAK_DISABLE;
sBreakDeadTimeConfig.BreakPolarity =
TIM_BREAKPOLARITY_HIGH;
sBreakDeadTimeConfigAutomaticOutput =
TIM_AUTOMATICOUTPUT_DISABLE;

if(HAL_TIMEx_ConfigBreakDeadTime(&htim1,&sBreakDeadTimeCon
fig) != HAL_OK)
```

```
{
    Error_Handler();
}

/* USER CODE BEGIN TIM1_Init 2 */

/* USER CODE END TIM1_Init 2 */
HAL_TIM_MspPostInit(&htim1);

}

/**
 * @brief GPIO Initialization Function
 * @param None
 * @retval None
 */
static void MX_GPIO_Init(void)
{
    /* USER CODE BEGIN MX_GPIO_Init_1 */

    /* USER CODE END MX_GPIO_Init_1 */

    /* GPIO Ports Clock Enable */
    __HAL_RCC_GPIOA_CLK_ENABLE();

    /* USER CODE BEGIN MX_GPIO_Init_2 */

    /* USER CODE END MX_GPIO_Init_2 */
```

```
}
```

```
/* USER CODE BEGIN 4 */
```

```
/* USER CODE END 4 */
```

```
/**
```

```
 * @brief This function is executed in case of error occurrence.
```

```
 * @retval None
```

```
 */
```

```
void Error_Handler(void)
```

```
{
```

```
 /* USER CODE BEGIN Error_Handler_Debug */
```

```
 /* User can add his own implementation to report the HAL error return state */
```

```
 __disable_irq();
```

```
 while (1)
```

```
 {
```

```
 }
```

```
 /* USER CODE END Error_Handler_Debug */
```

```
}
```

```
#ifdef USE_FULL_ASSERT
```

```
/**
```

```
 * @brief Reports the name of the source file and the source line number
```

```
 *       where the assert_param error has occurred.
```

```
 * @param file: pointer to the source file name
```

```
 * @param line: assert_param error line source number
```

```
* @retval None
*/
void assert_failed(uint8_t *file, uint32_t line)
{
    /* USER CODE BEGIN 6 */
    /* User can add his own implementation to report the file name and line
    number,
    ex: printf("Wrong parameters value: file %s on line %d\r\n", file, line)
    */
    /* USER CODE END 6 */
}
#endif /* USE_FULL_ASSERT */
```
